ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Fundamentals of Electric Drive

هذا العمل صدقة جارية عن أرواح موتى وشهداء المسلمين عامة وشهداء فلسطين خاصة

اللهم إنهم في ذمتك وحبل جوارك، فَقِهِم من فتنة القبر وعذاب النار، وأنت أهل الوفاء والحق، اللهم اغفر لهم وارحمهم، فإنك أنت الغفور الرحيم

اللَّهُمَّ، اغْفِرْ لهم وَارْحَمْهُم، وَاعْفُ عنْهم وَعَافِهِم، وَأَكْرِمْ نُزُلَهُم، وَوَسِّعْ مُدْخَلَهُم، وَاغْسِلْهُم بِمَاءٍ وَثَلْجٍ وَبَرَدٍ، وَنَقِّهِم مِنَ الخَطَايَا كما يُنَقَّى الثَّوْبُ الأبْيَضُ مِنَ الدَّنَسِ، وَأَبْدِلْهُم دَارًا خَيْرًا مِن دَارِهِم، وَأَهْلًا خَيْرًا مِن أَهْلِهِم، وَزَوْجًا خَيْرًا مِن زَوْجِهِم، وَقِهِم فِتْنَةَ القَبْرِ وَعَذَابَ النَّارِ

اللهم إنهم عبيدك وابناء عبيدك وابناء إماءك كانوا يشهدون أن لا إله إلا أنت، وأن محمداً عبدك ورسولك، وأنت أعلم بهم، اللهم إن كانوا محسنين فزدهم في إحسانهم، وإن كانوا مسيئين فتجاوز عن سيئاتهم، اللهم لا تحرمنا أجرهم ولا تفتنا بعدهم

اللَّهُمَّ اغفِرْ لحَيِّنا ومَيِّتِنا، وصَغيرِنا وكَبيرِنا، وذَكَرِنا وأُنثانا، وشاهِدِنا وغائِبِنا، اللَّهُمَّ مَن أحيَيْتَه مِنَّا فأَحْيِه على الإيمانِ، ومَن توَفَّيتَه مِنَّا فتوَفَّه على الإسلامِ، اللَّهُمَّ لا تَحرِمْنا أجْرَهم ولا تُضِلَّنا بَعْدَهم

اللهم سدد رمي المقاومة اللهم لا ترفع لليهود راية ولا تحقق لهم غاية واجعلهم لمن خلفهم عبرة وآية

قال تعالى:{﴿ سُبْحَانَ الَّذِي أَسْرَى بِعَبْدِهِ لَيلاً مِنَ الْمَسْجِدِ الْحَرَامِ إِلَى الْمَسْجِدِ الأَقْصَا الَّذِي بَارَكْنَا حَوْلَهُ لِنُرِيَهُ مِنْ آيَاتِنَا إِنَّهُ هُوَ السَّمِيعُ الْبَصِيرُ * وَآتَيْنَا مُوسَى الْكِتَابَ وَجَعَلْنَاهُ هُدىً لِبَنِي إِسْرَائِيلَ أَلاَّ تَتَّخِذُوا مِنْ دُونِي وَكِيلاً * ذُرِّيَّةَ مَنْ حَمَلْنَا مَعَ نُوحٍ إِنَّهُ كَانَ عَبْداً شَكُوراً * وَقَضَيْنَا إِلَى بَنِي إِسْرَائِيلَ فِي الْكِتَابِ لَتُفْسِدُنَّ فِي الأَرْضِ مَرَتَيْنِ وَلَتَعْلُنَّ عُلُواً كَبِيراً * فَإِذَا جَاءَ وَعْدُ أُولاهُمَا بَعَثْنَا عَلَيْكُمْ عِبَاداً لَنَا أُولِي بَأْسٍ شَدِيدٍ فَجَاسُوا خِلاَلَ الدِّيارِ وَكَانَ وَعْداً مَفْعُولاً * ثُمَّ رَدَدْنَا لَكُمُ الْكَرَّةَ عَلَيْهِمْ وَأَمْدَدْنَاكُمْ بِأَمْوَالٍ وَبَنِينَ وَجَعَلْنَاكُمْ أَكْثَرَ نَفِيراً * إِنْ أَحْسَنْتُمْ أَحْسَنْتُمْ لِأَنْفُسِكُمْ وَإِنْ أَسَأْتُمْ فَلَهَا فَإِذَا جَاءَ وَعْدُ الآخِرَةِ لِيَسُوؤا وُجُوهَكُمْ وَلِيَدْخُلُوا الْمَسْجِدَ كَمَا دَخَلُوهُ أَوَّلَ مَرَّةٍ وَلِيُتَبِّرُوا مَا عَلَوْا تَتْبِيراً * عَسَى رَبُّكُمْ أَنْ يَرْحَمَكُمْ وَإِنْ عُدْتُمْ عُدْنَا وَجَعَلْنَا جَهَنَّمَ لِلْكَافِرِينَ حَصِيراً ﴾} سورة الإسراء

والكمال لله وحده لهذا من الممكن أن تتواجد بعض الأخطاء في الحلول لذلك نرجو منكم الاستفسار عن الأسئلة التي تريدون السؤال عنها بالتواصل مع مُنشأ هذا العمل

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

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Unit one: Electric drive system

1. The drive system output power is:

a. Thermal power
b. Mechanical power
c. Electromagnetic power
d. Electrical power

Correct answer: Mechanical power

2. Loads which require coupling with the motor shaft through transmission mechanical system (TMS) are:

a. Loads which need low speeds and high torques
b. Loads which need low speeds and low torques
c. Loads which need high speeds and high torques
d. Loads which need high speeds and low torques

Correct answer: Loads which need high speeds and low torques

3. Calculate the starting time of the drive system with the following data:

$$J_\Sigma = 1.0\ kg.m^2,\quad T_m = 15 + 0.5\omega,\quad T_L = 5 + 0.6\omega$$

Consider final speed $\omega_2 = 0.95\ \omega_{ss}$

a. $29.96\ s$
b. $2.22\ s$
c. $299.57\ s$
d. None of them

Correct answer: $29.96\ s$

ω_{ss} is obtained when the electric drive system is stable, and stability of electric drive system occurs when $T_m = T_L$, so:

$$T_m = T_L$$
$$15 + 0.5\omega_{ss} = 5 + 0.6\omega_{ss}$$
$$\omega_{ss} = 100\frac{rad}{s}$$

And $w_2 = 0.95\ \omega_{ss} = 0.95 * 100\frac{rad}{s} = 95\frac{rad}{s}$

As the initial operating motor angular speed ω_o isn't mentioned in the question, so the drive system considered start its rotation from rest $w_o = 0\frac{rad}{s}$

$$t_{st} = \int_{w_1}^{w_2} \frac{J_\Sigma}{T - T_L} d\omega = \int_{w_o}^{w} \frac{J_\Sigma}{T - T_L} d\omega = \int_0^{95} \frac{1.0}{15 + 0.5\omega - (5 + 0.6\omega)} d\omega = \int_0^{95} \frac{1.0}{10 - 0.1\omega} d\omega = \frac{1.0}{-0.1}\ln(10 - 0.1\omega)\Big|_0^{95}$$

$$= 10\ln(10 - 0.1\omega)\Big|_{95}^{0} = 10(\ln(10 - 0.1 * 0) - \ln(10 - 0.1 * 95)) = 29.9573227355\ s \cong 29.96\ s$$

4. Component determines the motor starting time is:

a. motor inertia
b. motor load
c. speed change rate
d. all of these

بسواعدنا نبنيها

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Correct answer: all of these (motor inertia, motor load, speed change rate)

The motor torque equation is given by:

$$T = T_L + T_{dyn}$$

Where:

T: Motor Torque

T_L: Load Torque

T_{dyn}: Dynamic Acceleration, which is given by: $T_{dyn} = J_\Sigma \frac{d\omega}{dt}$

$$T = T_L + J_\Sigma \frac{d\omega}{dt}$$

$$T - T_L = J_\Sigma \frac{d\omega}{dt}$$

Rearrange the last equation and let dt on one side alone

$$dt = \frac{J_\Sigma}{T - T_L} d\omega$$

And now integrate both two sides

$$\int_{t_o}^{t} dt = \int_{w_o}^{w} \frac{J_\Sigma}{T - T_L} d\omega$$

$$t_{st} = \int_{w_o}^{w} \frac{J_\Sigma}{T - T_L} d\omega$$

According to the last equation, the starting time depends on the following:

a. motor moment of Inertia (motor inertia) J_m
b. Motor Torque T
c. Load Torque T_L
d. Motor angular speed rate of change $d\omega$
e. Initial operating motor angular speed ω_o, usually motor rotates from rest $\omega_o = 0 \frac{rad}{s}$
f. Steady-state operating motor angular speed ω
g. Motor efficiency η_m, $\eta_m = J_m \frac{\omega^2}{2}$

5. The transmission unit as a component of electric drive system is used to:

a. Decrease load speed and increase load torque
b. Increase load speed and decrease load torque
c. Decrease load speed and decrease load torque
d. Increase load speed and increase load torque

Correct answer: Decrease load speed and increase load torque

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

∴ Transsimmion unit in electric drive system

is und to Decreas the load speed and increas

load torau .

✓b) Decreas load speed and increas load Tora

6. The component of EDS uses to provide the motor with suitable power to meet the load requirements is:
a. power converter
b. sensing unit
c. controller
d. power supply

Correct answer: controller

7. Referring to the drive systems, Lathe cutting machine is considered:
a. Passive load
b. Active load
c. Inductive load
d. Resistive load

Correct answer: Passive load

8. In the electric drive system, the main function of the controller is:
a. Sensing the feedback variables only
b. Controlling the power converter only
c. Matching the motor and power converter to meet the load requirements
d. Adapting to the changes of the environment only

Correct answer: Matching the motor and power converter to meet the load requirements

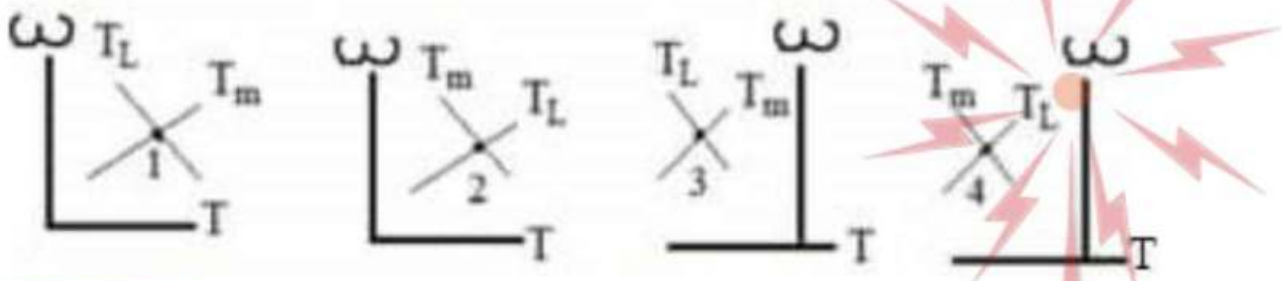
9. Electric drive system is:
a. Electro-mechanical system
b. Electrical system
c. Mechanical system
d. Electro-pneumatic system

Correct answer: Electro-mechanical system

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Unit two: Electric drive system characteristics

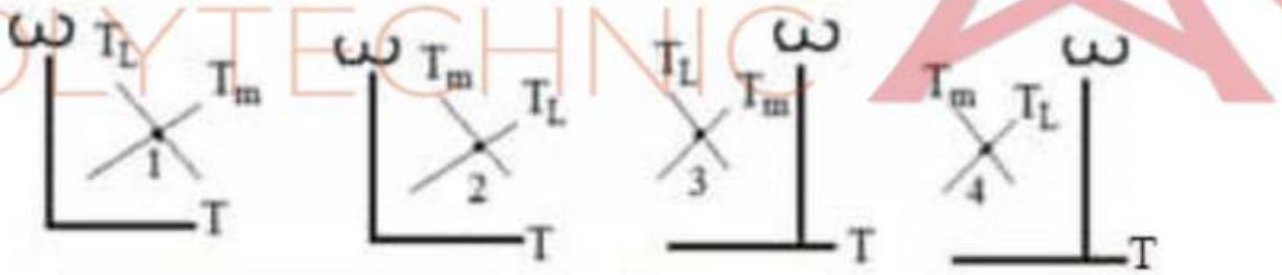
10. For the drive system characteristics below, unstable points of equilibrium are:



a. 1 & 4
b. 2 & 3
c. 1 &3
d. 2 & 4

Correct answer: 1 & 3
Note that: 1 and 3 are unstable point of equilibrium as slope of T_L is lesser than slope of T_m. Hence they are unstable

11. Stable points of equilibrium are:

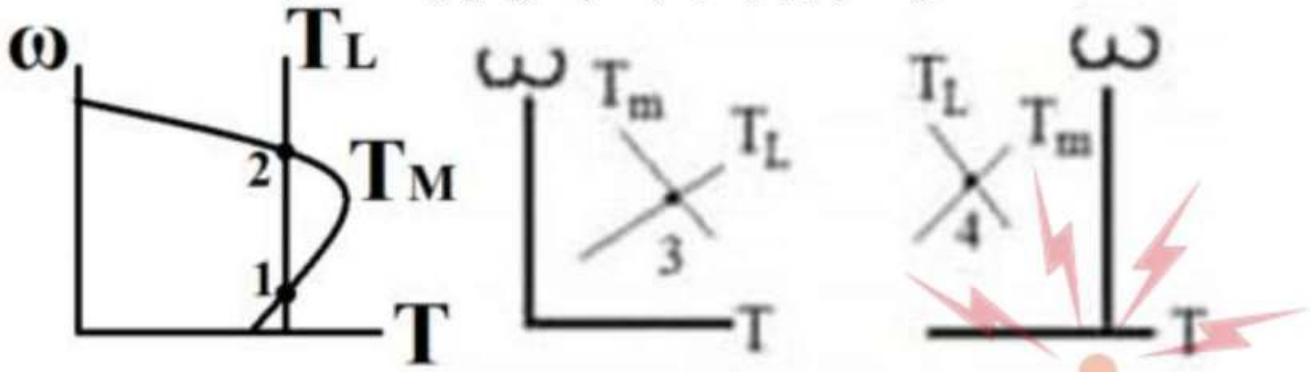


a. 1 only
b. 2 only
c. 3 only
d. 2 and 4

Correct answer: 2 and 4

12. Unstable points of equilibrium are:

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a. 1 and 2
b. 1 and 3
c. 1 and 4
d. 2 and 3

Correct answer: 1 and 4

13. Loading diagram is load torque as function of:

a. Moment of inertia
b. Gear ratio
c. Time
d. Speed

Correct answer: Time

14. The condition for the steady-state operation of the electric drive system is:

a. Load torque < Motor torque
b. Load torque = Motor torque
c. Load torque <<<< Motor torque
d. Load torque > Motor torque

Correct answer: Load torque = Motor torque

15. If the torque developed by the motor is $T = T_L \pm J\frac{d\omega}{dt}$ and all of the equation components have values, the motor runs at:

a. Generating mode
b. Transient condition
c. No-load mode
d. Steady-state condition

Correct answer: Transient condition

Torque developed by the motor: $T = T_L + J\frac{d\omega}{dt}$

According to the above expression the steady state operation takes place when the motor torque equals to the load torque

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Answer:- Transient Condition.

$T = T_L + J\, d\omega/dt$

T = Motor Torque.

T_L = Load Torque.

J = Moment of Inertia.

$\frac{d\omega}{dt}$ = Rate of change of angular velocity.

→ T_L exist in equation means there is Load on the motor.

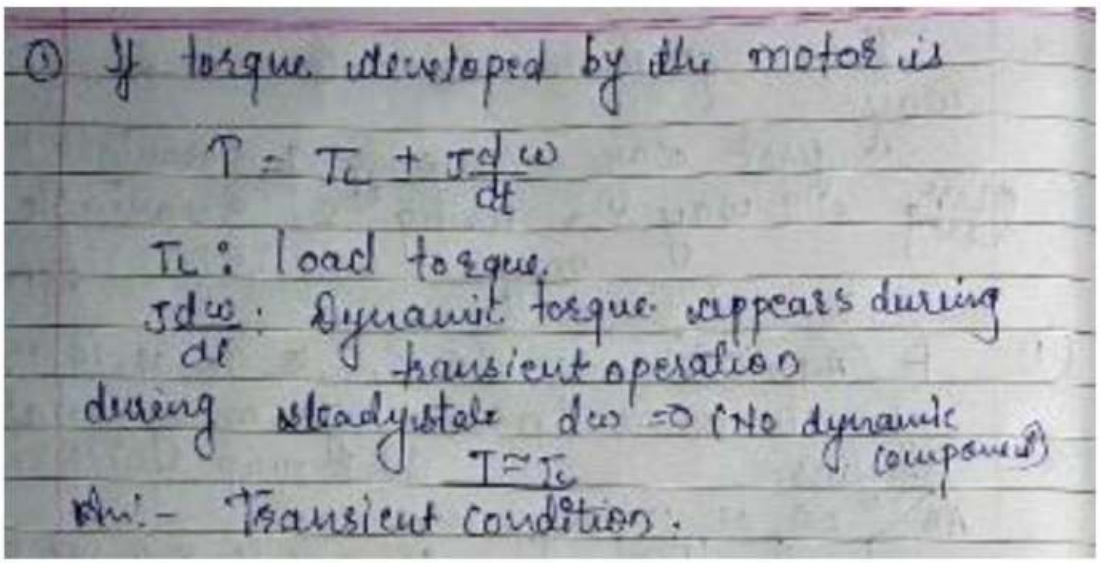
→ $\frac{d\omega}{dt}$ is Zero for steady state Condition.

→ The above equation Not for generating mode.

Answer:- Transient Condition.

ALL THE BEST. Rate the Answer.

Or



16. If the torque developed by the motor is $T = T_L + J\frac{d\omega}{dt}$, the drive is:

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ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

a. decelerated
b. accelerated
c. working in S.S.P (Steady-State Point)
d. None of these

Correct answer: accelerated

17. If the torque developed by the motor is $T = T_L - J\frac{d\omega}{dt}$, the drive is:

a. decelerated
b. accelerated
c. working in S.S.P (Steady-State Point)
d. None of these

Correct answer: decelerated

18. If the torque developed by the motor is $T = T_L$, the drive is:

a. decelerated
b. accelerated
c. working in S.S.P (Steady-State Point)
d. None of these

Correct answer: working in S.S.P (Steady-State Point)

19. The electric drive system with following motor torque and load torque expressions is stable:

a. $T = 25 + 5\omega,\ T_L = 2 + 2\omega^2$
b. $T = 25 - 3\omega,\ T_L = 10$
c. $T = 15 + 4\omega,\ T_L = 15 + \omega$
d. $T = 10 + 2\omega,\ T_L = 10$

Correct answer: $T = 25 - 3\omega,\ T_L = 10$

20. The drive system linear (translational) motion load power is calculated by multiplying:

a. Voltage and current
b. Angular speed and torque
c. Linear velocity and force
d. Rotational speed and torque

Correct answer: Linear velocity and force

21. The drive system rotational motion output power is calculated by multiplying:

a. Voltage and current
b. Rotational speed and torque
c. Angular speed and efficiency
d. Angular speed and torque

Correct answer: Angular speed and torque

22. What is the relationship between torque and speed in constant type loads?

a. Torque linearly increases with an increase in speed

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b. Torque non-linearly decreases with an increase in speed
c. Torque non-linearly increases with an increase in speed
d. Torque is independent of speed (Torque is not depending on the speed)

Correct answer: Torque is independent of speed (Torque is not depending on the speed)

23. The equivalent moment of inertia referred to motor shaft calculation is based on:
a. Mechanical power conservation law
b. Heating (electric) power conservation law
c. Electro-magnetic energy conservation law
d. Kinetic energy conservation law

Correct answer: Kinetic energy conservation law

24. The equivalent load torque referred to motor shaft T_{eq} calculation is based on:
Or
The equivalent load torque referred to motor shaft calculated depending on:
a. Electro-magnetic energy conservation law
b. Mechanical power conservation law
c. Kinetic energy conservation law
d. Heating (electric) power conservation law

Correct answer: Mechanical power conservation law

25. In the figure below, diameters of pulley 1 and pulley 2 are 10cm and 50cm respectively, if the belt speed is 1.257m/sec the motor rotational speed is:



a. 12.56 rad/sec
b. 25 rad/sec
c. 240 rpm
d. 120 rpm

Correct answer: 240 rpm

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(A) Belt Speed = 1.257 m/se

$$\text{Belt Speed (rpm)} = \frac{60}{\pi D} V_{m/s}$$

$$= \frac{60}{\pi (0.5\ mtr)} (1.257)$$

$$\therefore \text{Belt speed (rpm)} = 48$$

$$\rightarrow \frac{RPM1}{RPM2} = \frac{Dia2}{Dia1}$$

$$\Rightarrow \frac{\text{RPM of motor}}{\text{RPM of Belt}} = \frac{50cm}{10cm}$$

$$\therefore \text{RPM of motor} = 5\ (\text{Belt RPM})$$

$$= 5\ (48)$$

$$\therefore \text{Motor Speed} = 240\ rpm$$

option 3

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26. Electric motor has moment of inertia of 1 $kg.m^2$ connected through gearbox of 5 gear ratio to drum with moment of inertia 4 $kg.m^2$. The drive system total (equivalent) moment of inertia equals:

a. 21 $kg.m^2$
b. 6 $kg.m^2$
c. 1.16 $kg.m^2$
d. None of these

Correct answer: 1.16 $kg.m^2$

$$J_\Sigma = J_m + \frac{J_D}{i^2} = 1 + \frac{4}{5^2} = 1 + 0.16 = 1.16\ kg.m^2$$

27. Electric motor has moment of inertia of 1 $kg.m^2$ directly connected to motor shaft drum with moment of inertia 4 $kg.m^2$. The drive system total (equivalent) moment of inertia equals:

a. 3 $kg.m^2$

بسواعدنا نبنيها

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b. $5\ kg.m^2$
c. $4\ kg.m^2$
d. $1\ kg.m^2$

Correct answer: $5\ kg.m^2$

(1) Given

Electric motor has moment of inertia (J_1) $= 1\ kg-m^2$

Motor shaft drum moment of inertia (J_2) $= 4\ kg\ m^2$

Then equivalent Moment of inertia $= J_{eq} = J_1 + J_2$

$J_{eq} = (1+4)\ kg.m^2$

$J_{eq} = 5\ kg.m^2$

28. The active load torque industrial machinery is:
a. Crane
b. Ventilator
c. Pump
d. Turning machine

Correct answer: Crane

29. Turning machine is an active load.
a. True
b. False

Correct answer: False

Production machineries, which are raw materials (wood, plastic and metal) manufacturing machines, including drilling machines, cutting and milling, turning, sawing, and others, have Reactive Torque as it is friction torque.

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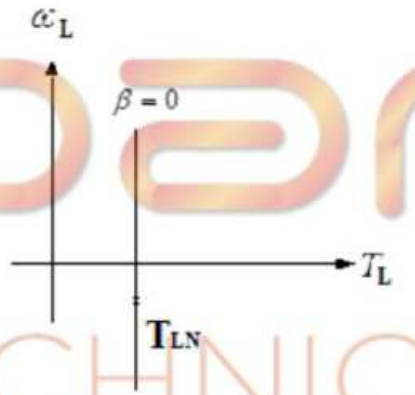
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30. Production machineries are reactive loads.
a. True
b. False

Correct answer: True
Production machineries, which are raw materials (wood, plastic and metal) manufacturing machines, including drilling machines, cutting and milling, turning, sawing, and others, have Reactive Torque as it is friction torque.

31. Transporting Machineries is:
a. active loads
b. reactive loads

Correct answer: active loads
For the Transporting Machineries, the load torque has a fixed value and does not depend on the value and direction of the speed as shown in figure and this torque is called Active Torque and the machinery is called constant load torque machinery.



32. Which of the following Industrial machinery types has an Active Torque?
Or
Which of the following Industrial machinery types hasn't a Reactive Torque?
a. Transporting machineries
b. Production machineries
c. Fan-load characteristic machineries
d. Machineries with linear incremental characteristics

Correct answer: Transporting machineries

33. Industrial machineries as mechanical load have only torque-speed (mechanical) characteristics
a. True
b. False

Correct answer: True

34. The mechanical characteristics of industrial machineries are the relationship of angular or rotational speed with load torque
a. True
b. False

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Correct answer: True

35. The torque-speed characteristics of different types of industrial machineries can be derived using an experimental relationship in _____.
a. Transient state (Unstable mode)
b. Steady state (Stable mode)
c. Steady state and Transient state (Stable and Unstable modes)
d. It depends on the industrial machinery type

Correct answer: Steady state (Stable mode)

36. The torque-speed characteristics of different types of industrial machineries can be derived using an experimental relationship in steady state (stable mode), this experimental relationship is:

$$T_L = T_O + (T_L - T_O)\left(\frac{\omega_L}{\omega_{Ln}}\right)^X$$

a. True
b. False

Correct answer: True

The torque-speed characteristics of different types of industrial machineries can be derived using an experimental relationship in steady state (stable mode), this experimental relationship is:

$$T_L = T_O + (T_{Ln} - T_O)\left(\frac{\omega_L}{\omega_{Ln}}\right)^X$$

Where:

T_L: Load Torque
T_O: Initial friction torque
T_{Ln}: Nominal load torque
ω_L: Load angular speed
ω_{Ln}: Load nominal angular speed
X: A parameter that determines the relationship between torque change and velocity change according to the type of machinery.

37. Rotating mechanical load of 10 N.m connected to the motor through gearbox, which has gear ratio of 5. If the system efficiency equals 0.9, the referred load torque to motor shaft is:
Or
Crane motor lowers load of 10 N.m connected to the motor through gearbox, which has gear ratio of 5. If the system efficiency equals 0.9, the referred load torque to motor shaft is:
a. 50 N.m
b. 2.22 N.m
c. 1.8 N.m
d. 2 N.m

Correct answer: 2.22 N.m

$$\frac{10\ \text{N.m}}{5 * 0.9}$$

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38. Rotating mechanical load of 10 N.m connected to the motor through gearbox, which has gear ratio of 5. If the system efficiency equals 0.8, the referred load torque to motor shaft is:

a. 2.5 N.m
b. 1.6 N.m
c. 2 N.m
d. 40 N.m

Correct answer: 2.5 N.m

$$\frac{10\ \text{N.m}}{5 * 0.8}$$

Here, the given load torque = $T_2 = 10$ N-m

gear Ratio = 5

∴ Referred load torque to motor, $T_1 = T_2 \times \left(\frac{1}{5}\right)$

$= \frac{10}{5} = 2$ N-m

Now;

Efficiency = 0.8

So, The Required Referred torque = $\frac{2}{0.8} = 2.5$ N-m

Answer is A. 2.5 N-m

Or

$T_{developed} = \frac{T_{out}}{\eta}$

$T_{developed} = \frac{10}{0.8} = 12.5$ N-m

$T_{Referred} = \frac{12.5}{5} = 2.5$ N-m option (A).

Or

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η = Efficiency = 0.8

a = Gear ratio = 5

T_{l_1} = load torque connected to motor through gear box = 10 Nm

T_1 = load torque refered to motor shaft

$$T_1 = \frac{a T_{l_1}}{\eta} = \frac{5 \times 10}{0.8} = 62.5 \text{ Nm}$$

The correct answer is 62.5 Nm. and the options given are wrong.

However if the gear is a reduction gear, then

$a = \frac{1}{5}$, in that case

$$T_1 = \frac{\left(\frac{1}{5}\right) \times 10}{0.8} = 2.5 \text{ Nm}$$ which matches option i)

(i) 2.5 Nm

39. The stability condition of electric drive system operation is:

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a. Hardness of load torque-speed characteristic is less than hardness of motor torque-speed characteristic
b. Motor speed is more than load speed
c. Hardness of motor torque-speed characteristic is less than Hardness of load torque-speed characteristic
d. Motor electro-magnetic torque is less than load torque

Correct answer: Hardness of motor torque-speed characteristic is less than Hardness of load torque-speed characteristic

40. Transporting machineries load torque is:

a. Inversely proportional with speed
b. Directly proportional with speed square
c. Directly proportional with speed
d. Independent on speed

Correct answer: Independent on speed

41. Fan load torque is:

a. Directly proportional with speed
b. Directly proportional with speed square
c. Inversely proportional with speed
d. Inversely proportional with speed square

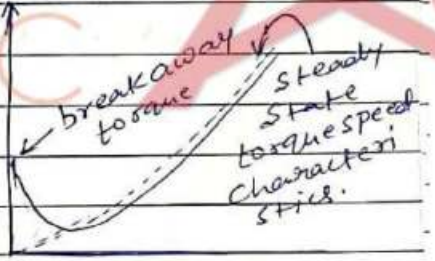
Correct answer: Directly proportional with speed square

Fan load torque

the load torque is proportional to the square of the speed of fan. this type of loads have variable load (torque) torque characteristics.

It requires very low torque at low speed than at high speeds.

So load torque is low when starting it.



$$T = T_{rated} \times \left(\frac{n}{n_{rated}}\right)^2$$

where T_{rated} → rated torque & n_{rated} → speed of a fan load

So $T \propto n^2$ option 'c' is correct.

Page No.

42. Compressor Machinery Torque-Speed Characteristic is:

a. Non-Linear Second Order Incremental Characteristic
b. Non-Linear Decaying Characteristic
c. Non-Linear First Order Incremental Characteristic
d. Linear Incremental Characteristic

Correct answer: Non-Linear Second Order Incremental Characteristic

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43. Transporting Machinery Torque-Speed Characteristic is:
a. Non-Linear Second Order Incremental Characteristic
b. Linear Incremental Characteristic
c. Non-Linear Decaying Characteristic
d. Speed Independent Characteristic

Correct answer: Speed Independent Characteristic

44. Drilling machine speed is:
a. Inversely proportional with torque square
b. Independent on torque
c. Directly proportional with torque
d. Inversely proportional with torque

Correct answer: Inversely proportional with torque

45. In paper rewinding machine:
a. Torque is independent of the speed
b. Torque is proportional to speed square
c. Torque is directly proportional to the speed
d. Torque is inversely proportional to speed

Correct answer: Torque is inversely proportional to speed

46. Production machineries torque-speed characteristic is:
a. Increasing non-linear characteristic
b. Decreasing non-linear characteristic
c. Decreasing linear characteristic
d. Independent on speed

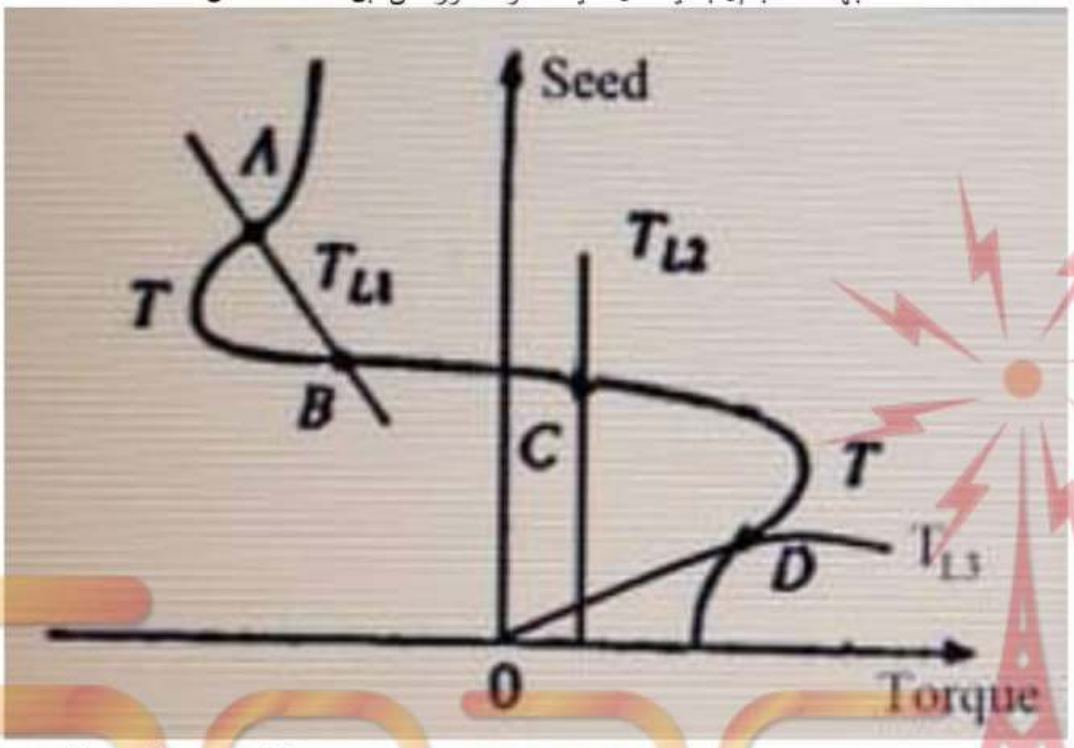
Correct answer: Decreasing non-linear characteristic

47. The hardness of torque-speed characteristic is the relationship between the changes of:
a. Speed to torque
b. Torque to speed
c. Speed to current
d. Current to torque

Correct answer: Torque to speed

48. Operating points A, B, C and D are:

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a. unstable, stable, stable, unstable
b. unstable, unstable, stable, unstable
c. unstable, stable, stable, stable
d. unstable, unstable, stable, unstable

Correct answer: unstable, stable, stable, unstable

49. Speed profile is a speed as function of:
a. Moment of inertia
b. Load torque
c. Time
d. Motor torque

Correct answer: Time

50. To select the motor rating accurately, the most suitable method is:
a. equivalent current
b. equivalent torque
c. equivalent power
d. equivalent losses

Correct answer: equivalent power

51. Absolute hard speed-torque characteristics can be obtained in dc motor drive in closed-loop systems
a. True
b. False

Correct answer: True

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52. Squirrel-Cage motors are the most widely used in industry.
a. True
b. False

Correct answer: True

The most widely used machine in industry is the three-phase induction motor. Its characteristic features are:

1. Simple and rugged construction
2. Low cost and minimum maintenance
3. High reliability and sufficiently high efficiency
4. Needs no extra starting motor and need not be synchronized

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Unit three: DC Motors static characteristics

53. One of the following dc motors does not operate in no-load mode:
a. Parallel field dc motor
b. Series field dc motor
c. Compound field dc motor
d. Separately excited dc motor

Correct answer: Series field dc motor

54. One of the following dc motors does not operate in regenerating mode:
a. Compound field dc motor
b. Series field dc motor
c. Separately excited dc motor
d. Parallel field dc motor

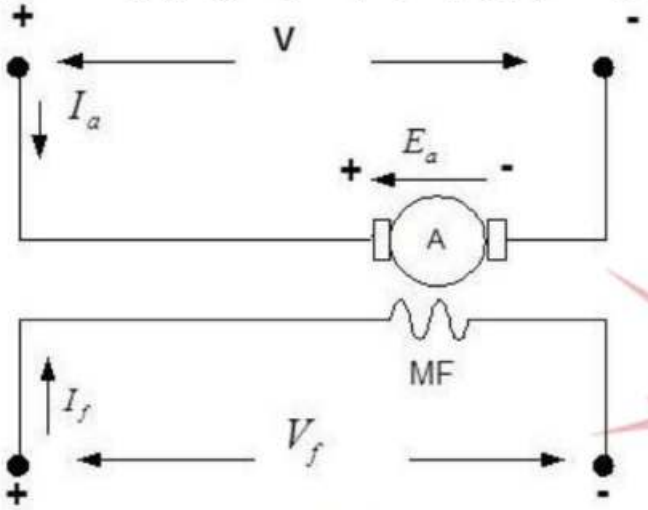
Correct answer: Series field dc motor

55. When varying load torque, separately excited dc motor has constant:
a. Torque
b. Back emf
c. Armature current
d. Flux
e. Speed

Correct answer: Flux

Figure below shows the electrical circuit of separately excited dc motor in which the Field Windings (symbolized by MF) is separated from Armature, and it is fed from an independent source and the motor flux is independent on motor load variation, so the motor flux is constant.

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The electrical circuit of separately excited dc motor

56. Plugging mode is very dangerous mode due to overheating, but it used under nominal load in:
    Or
    Plugging mode in the 4th quadrant is very dangerous mode due to overheating, but it used under nominal load in:
    a. Escalator
    b. Pump machinery
    c. Drilling machine
    d. Crane

Correct answer: Crane
The plugging mode can be used practically in cranes under nominal load torque by adding resistances in series with the armature operating on regulated characteristic.

57. Series field dc motor has non-linear static characteristics, because:
    a. Motor flux is variable
    b. Load torque is variable
    c. Armature current is variable
    d. Motor speed is variable

Correct answer: Motor flux is variable

58. Adding resistance in series with armature of dc motor will cause:
    a. Increasing of torque-speed characteristic hardness
    b. Increasing of short-circuit current
    c. Decreasing of torque-speed characteristic hardness
    d. Increasing of starting torque

Correct answer: Decreasing of torque-speed characteristic hardness

59. Adding resistance in series with armature of dc motor will cause:
    a. Additional heat in the armature circuit
    b. Increasing of motor torque

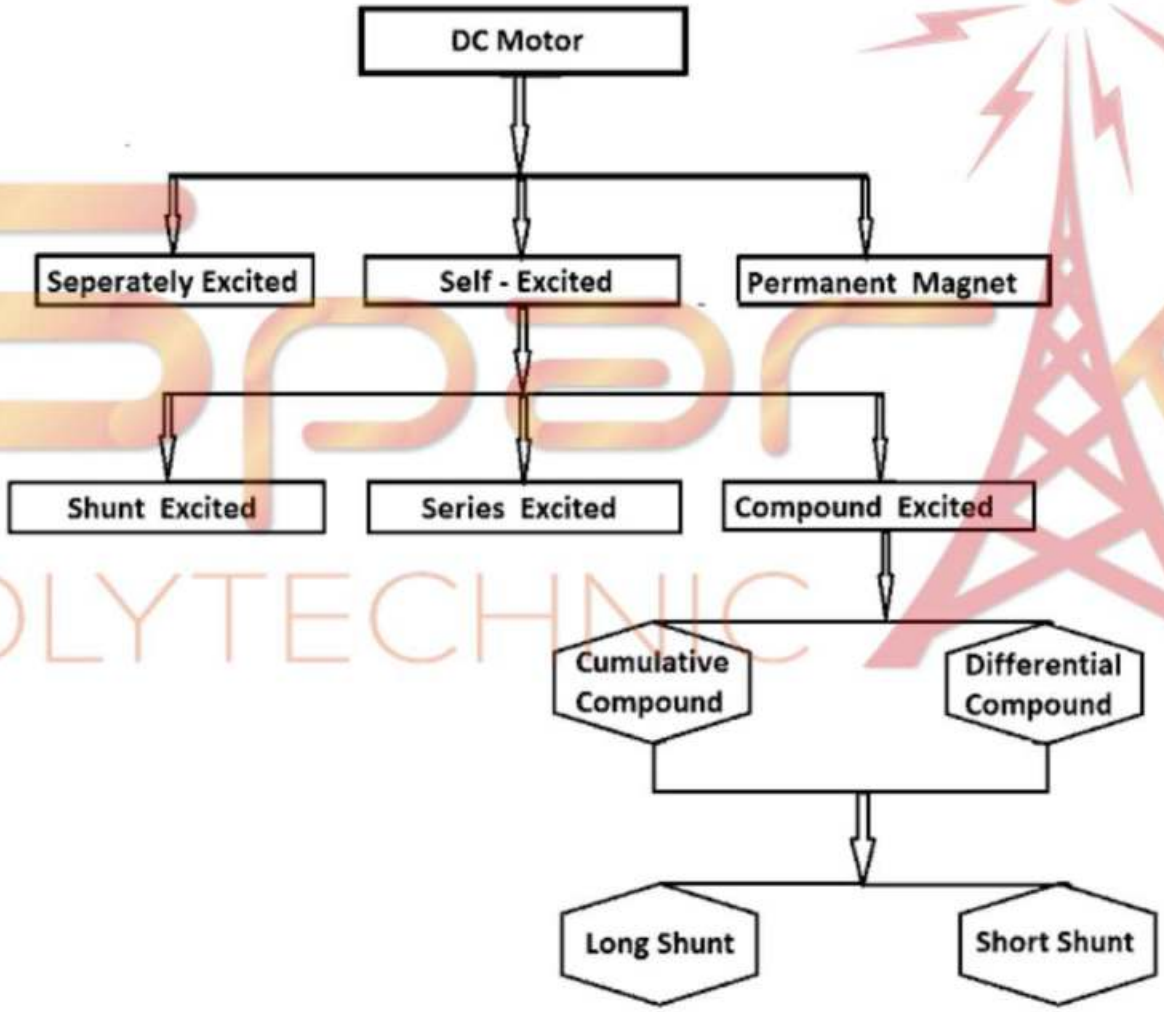
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c. Increasing of short-circuit current
d. Increasing of starting torque

Correct answer: Additional heat in the armature circuit

60. All dc motors are self-excited.
a. True
b. False

Correct answer: False



61. Brushed dc motors are not used in modern electric vehicles.
a. True
b. False

Correct answer: True

DC motors are robust and allow simple control. They can be made as brushed and brushless DC motors. Brushed DC motors are a mature technology that provides low cost, high torque at low speed, and easy speed control. These features are very important for traction motors. However, brushed DC motors are not widely used in EVs because of

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their disadvantages, which include large size, low efficiency, and requirement for frequent maintenance due to the brush and collector structure. Brushless DC motors have a much higher efficiency. These motors use an electronic commutator/inverter instead of the brushes.

62. The most dangerous loading mode of motor operating modes is:

a. Plugging (motor is series with the line circuit)
b. Regenerative (motor is parallel to the line circuit)
c. Forward direction motoring
d. Backward direction motoring

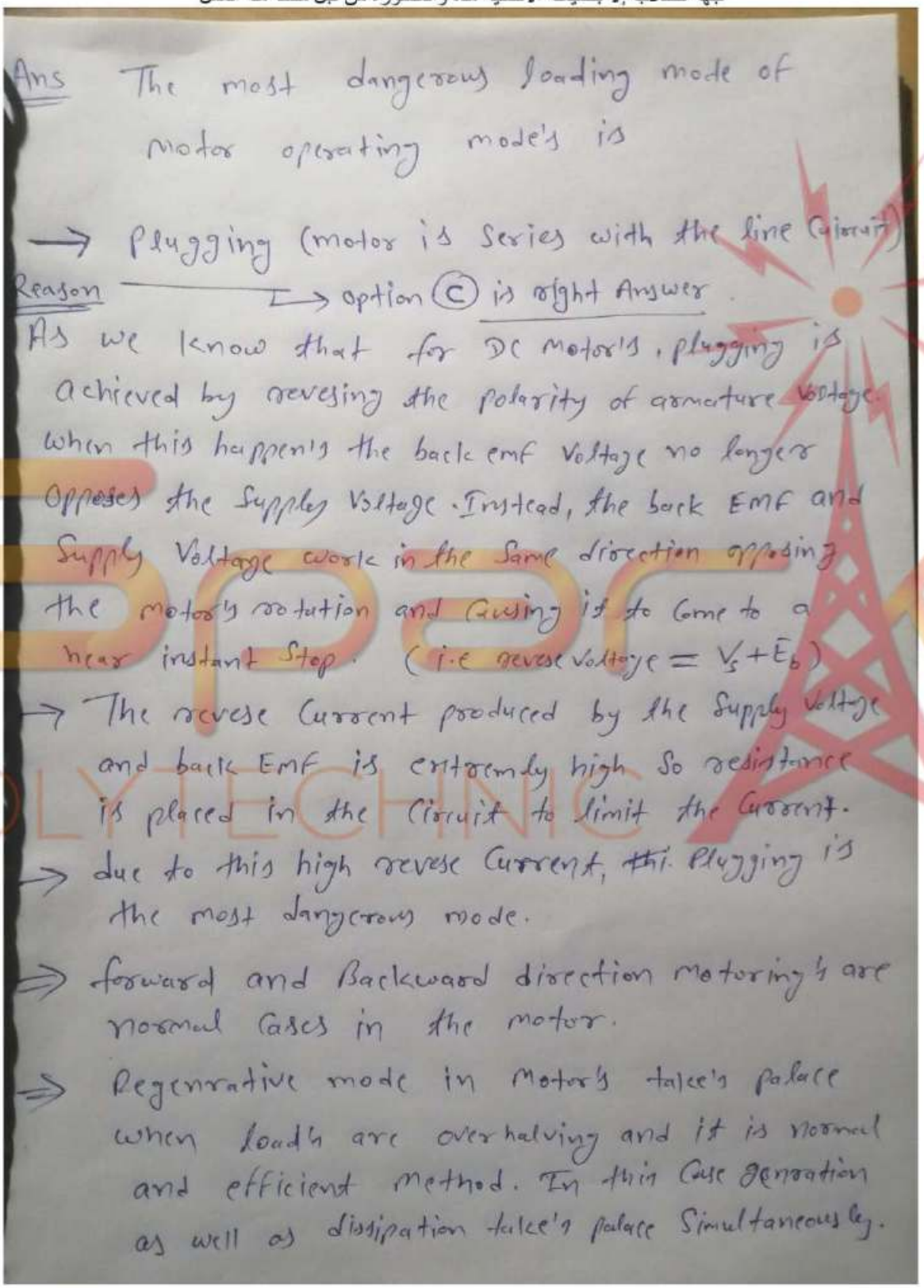
Correct answer: Plugging (motor is series with the line circuit)

Note that: Plugging is a highly inefficient method of braking because, in addition to the power supplied by the load, the power supplied by the source is wasted in resistances

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Ans The most dangerous loading mode of motor operating mode's is

→ Plugging (motor is series with the line Circuit)

Reason ——— I→ option (C) is right Answer.

As we know that for DC motor's, plugging is achieved by revesing the polarity of armature voltage. When this happen's the back emf voltage no longer opposes the supply voltage. Instead, the back EMF and supply voltage work in the same direction opposing the motor's rotation and Causing it to come to a near instant stop. (i.e revese voltage = $V_s + E_b$)

→ The revese Current produced by the supply voltage and back EMF is extremly high So resistance is placed in the Circuit to limit the Current.

→ due to this high revese Current, ~~thi~~ Plugging is the most dangerous mode.

⇒ forward and Backward direction motoring's are normal Cases in the motor.

⇒ Regenrative mode in motor's take's palace when loads are overhalving and it is normal and efficient method. In this Case genration as well as dissipation take's palace Simultaneously.

63. The DC motor that should never be started on no-load is:

a. series

b. permanent magnet

c. compound

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d. shunt

Correct answer: series

64. For loads which require high starting torque, the suitable motor is:

a. Permanent magnet dc motors
b. Separately excited dc motors
c. Series field dc motors
d. Compound field dc motors

Correct answer: Series field dc motors

65. A 200 V, 10 KW, 2000 rpm, separately excited dc motor with a 200Ω field resistance consumes 1.1A from the power supply. Determine the nominal torque:

a. $477.5\ N.m$
b. $4.775\ N.m$
c. $47.75\ KN.m$
d. $47.75\ N.m$

Correct answer: $47.75\ N.m$

$$\frac{P}{\omega} = \frac{10000}{\frac{2000}{9.55}} = 47.75\ N.m$$

66. Separately excited dc motor with following ratings:

$$\mathrm{P_{2N}} = 10\mathrm{kW}, \mathrm{V_a} = \mathrm{V_f} = 220\ \mathrm{V}, \omega_\mathrm{s} = 100\frac{\mathrm{rad}}{\mathrm{s}}, \mathrm{R_a} = 0.4\Omega, \mathrm{I_{an}} = 50\ \mathrm{A}, \mathrm{k}\emptyset_\mathrm{N} = 2\mathrm{V.s}$$

Motor starting current is:

a. 150 A
b. 550 A
c. 50 A
d. 100 A

Correct answer: 550 A

$$I_{st} = \frac{V_a}{R_a} = \frac{220\ V}{0.4\ \Omega} = 550\ A$$

67. Separately excited dc motor with following ratings:

$$\mathrm{P_{2N}} = 10\mathrm{kW}, \mathrm{V_a} = \mathrm{V_f} = 220\ \mathrm{V}, \omega_\mathrm{s} = 100\frac{\mathrm{rad}}{\mathrm{s}}, \mathrm{R_a} = 0.4\Omega, \mathrm{I_{an}} = 50\ \mathrm{A}, \mathrm{k}\emptyset_\mathrm{N} = 2\mathrm{V.s}$$

Nominal angular speed drop is:

a. 20 rad/s
b. 80 rad/s
c. 90 rad/s
d. 10 rad/s
e. 100 rad/s

Correct answer: 10 rad/s

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$$\Delta\omega = \frac{I_{AR} * R_a}{K\Phi_N} = \frac{50\ A * 0.4\ \Omega}{2\ V.s} = 10\ \text{rad/s}$$

68. Separately excited dc motor with following ratings:

$$P_{2N} = 10\text{kW}, V_a = V_f = 220\ \text{V}, \omega_s = 100\frac{\text{rad}}{\text{s}}, R_a = 0.4\Omega, I_{an} = 50\ \text{A}, k\emptyset_N = 2\text{V.s}$$

No-load angular speed is:

a. 110 rad/s
b. 200 rad/s
c. 220 rad/s
d. 100 rad/s

Correct answer: 110 rad/s

$$\omega_o = \frac{V_t}{K\Phi_N} = \frac{220\ V}{2\ V.s} = 110\ \text{rad/s}$$

69. Separately excited dc motor with following ratings:

$$P_{2N} = 10\text{kW}, V_a = V_f = 220\ \text{V}, \omega_s = 100\frac{\text{rad}}{\text{s}}, R_a = 0.4\Omega, I_{an} = 50\ \text{A}, k\emptyset_N = 2\text{V.s}$$

Nominal rotational speed is:

a. 955 rpm
b. 1000 rpm
c. 981 rpm
d. 100 rpm

Correct answer: 955 rpm

$$n_N = \frac{60}{2\pi}\omega_N = \frac{60}{2\pi} * 100\ rpm = 954.92965856\ \text{rpm} \approx 955\ \text{rpm}$$

70. Hardness (β) of PM dc-motor mechanical characteristics is:

a. β>0
b. β<0
c. β=0

Correct answer: β<0

71. Compound field dc motor total flux is cumulative to:

a. Increase motor torque
b. Increase motor speed
c. Decrease motor torque
d. Increase back emf

Correct answer: Increase motor torque

72. A 220 V shunt motor has armature and field resistance of 0.2 Ω and 220 Ω respectively. The motor is driving a constant load torque and running at 1000 rpm drawing 10 A current from the supply. Find the output power of the motor:

a. 236 W
b. 2.2 kW

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c. 1900 W
d. 1.964 kW

Correct answer: 2.2 kW

$$P_{2n} = V * I = 220 * 10 = 2.2\,kW$$

73. When decreasing separately excited dc motor armature voltage by 50%, the no-load speed will be:
a. same value
b. 50% less
c. 25% less
d. 50% more

Correct answer: 50% less

$$V_a \propto \omega$$
$$V_f \propto \frac{1}{\omega}$$

74. When decreasing separately excited dc motor field voltage by 50%, the no-load speed will be:
a. 50% less
b. 50% more
c. 25% less
d. same value

Correct answer: 50% more

$$V_a \propto \omega$$
$$V_f \propto \frac{1}{\omega}$$

75. The hardness of series field dc motors torque-speed characteristic is:
a. Negative
b. Zero
c. Infinity
d. Positive

Correct answer: Negative

76. When decreasing separately excited dc field voltage by 50%, the starting current will be:
a. 25% less
b. 50% less
c. Doubled
d. Same value

Correct answer: 50% less

77. When decreasing separately excited dc field voltage by 50%, the starting torque will be:
a. 25% less
b. 50% less
c. Same value
d. Doubled

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Correct answer: 50% less

78. When decreasing separately excited dc field voltage by 50%, the no-load speed will be:

a. 25% less
b. 50% less
c. Same value
d. Doubled

Correct answer: Doubled

79. Separately excited dc motor with following ratings:

$$P_{2N} = 10\text{kW}, V_a = V_f = 220\text{ V}, \omega_s = 100\frac{\text{rad}}{\text{s}}, R_a = 0.4\Omega, I_{an} = 50\text{ A}, k\emptyset_N = 2\text{V.s}$$

Nominal back emf is:

a. $220\ V$
b. $240\ V$
c. $200\ V$
d. $120\ V$

Correct answer: $200\ V$

$$\omega_n = \frac{V}{k\emptyset_N} - \frac{I_a R_a}{k\emptyset_N} = 100 rad/sec$$

$$E_n = k\emptyset_N \omega_n = 2 * 100 = 200V$$

80. Separately excited dc motor with following ratings:

$$P_{2N} = 10\text{kW}, V_a = V_f = 220\text{ V}, \omega_s = 100\frac{\text{rad}}{\text{s}}, R_a = 0.4\Omega, I_{an} = 50\text{ A}, k\emptyset_N = 2\text{V.s}$$

Nominal motor torque is:

a. $100\ N.m$
b. $55\ N.m$
c. $50\ N.m$
d. $25\ N.m$
e. $12.5\ N.m$
f. $27.5\ N.m$

Correct answer: $100\ N.m$

$$T_n = k\emptyset_N I_{an} = 2 * 50 = 100\text{N.m}$$

81. Separately excited dc motor with following ratings:

$$P_{2N} = 10\text{kW}, V_a = V_f = 220\text{ V}, \omega_s = 100\frac{\text{rad}}{\text{s}}, R_a = 0.4\Omega, I_{an} = 50\text{ A}, k\emptyset_N = 2\text{V.s}$$

Nominal input power is:

a. 9000 W
b. 5000 W
c. 11000 W
d. 10000 W

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Correct answer: 11000 W

$$V_a * I_{an} = 220V * 50A = 11000\ W$$

82. Separately excited dc motor has linear static characteristics, because:
a. Load torque is variable
b. Motor flux is constant
c. Motor speed is variable
d. Armature current is variable

Correct answer: Motor flux is constant

83. The dc motor loading mode in third quadrant is called:
a. Short circuit mode
b. Motoring mode
c. Plugging mode
d. Regenerating mode

Correct answer: Motoring mode

84. The dc motor loading mode in second quadrant is called:
a. Plugging mode
b. Short circuit mode
c. Regenerating mode
d. Motoring mode

Correct answer: Regenerating mode

85. The dc motor loading mode in fourth quadrant is called:
a. Regenerating mode
b. Motoring mode
c. Plugging mode
d. Short circuit mode

Correct answer: Plugging mode

86. When adding resistance in series with armature of dc motor, the following parameter will not change its value:
a. No-load motor speed
b. Short-circuit current
c. Speed drop
d. Starting torque

Correct answer: No-load motor speed

87. Torque overload coefficient is equal to:
a. maximum torque divided by actual no-load torque
b. maximum torque divided by starting torque
c. maximum torque divided by rated torque
d. starting torque divided by rated torque

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Correct answer: maximum torque divided by rated torque

88. The dc motors allowed starting current value is:

a. (5-7) times of nominal current
b. Nominal current
c. Less than nominal current
d. (2-3) times of nominal current

Correct answer: (2-3) times of nominal current

89. For D.C. series motor which of the following is incorrect:

a. suitable for heavy duty starting
b. torque varies as armature current
c. has non-linear characteristics
d. has hard characteristics

Correct answer: has hard characteristics

90. If the motor works in the 4th quadrant. This mode of operation is a:

a. reverse braking mode
b. forward regenerative mode
c. motoring mode
d. reverse motoring mode

Correct answer: reverse braking mode

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Unit Four: AC Motors static characteristics

91. Reducing stator voltage of three-phase ac motor by 20% will cause rotor current reducing by:

a. 80%
b. 64%
c. 40%
d. 20%

Correct answer: 20%

Note that: We know that: Torque $\propto V^2$

And also, Torque $\propto {I_2}^2 ; I_2 = rotor\ current$

So $V \propto I_2$

So, reducing stator voltage of three phase ac motor by 20 percent will cause rotor current reducing by 20 percent

Note that: The rotor current will be reduced by 20% because current and voltage are directly proportional

$$\frac{I_2}{I_1} = \frac{E_2}{E_1}$$
$$E_2 = E_1 - 20\%E_1$$
$$E_2 = 80\%E_1$$
$$\frac{I_2}{I_1} = \frac{80\%E_1}{E_1}$$
$$I_2 = 80\%I_1$$

So, decreases by 20%

92. Reducing stator voltage of three-phase ac motor by 30% will cause rotor current reducing by:

a. 15%
b. 30%
c. 60%
d. 80%

Correct answer: 30%

Note that: We know that: Torque $\propto V^2$

And also, Torque $\propto {I_2}^2 ; I_2 = rotor\ current$

So $V \propto {I_2}^2$

So, reducing stator voltage of three phase ac motor by 30 percent will cause rotor current reducing by 30 percent

Note that: The rotor current will be reduced by 20% because current and voltage are directly proportional

$$\frac{I_2}{I_1} = \frac{E_2}{E_1}$$
$$E_2 = E_1 - 30\%E_1$$
$$E_2 = 70\%E_1$$
$$\frac{I_2}{I_1} = \frac{70\%E_1}{E_1}$$

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$$I_2 = 80\%I_1$$

So, decreases by 30%

93. Slip losses are the:

a. Magnetic circuit losses
b. Air gap losses
c. Stator circuit losses
d. Rotor circuit losses

Correct answer: Rotor circuit losses

94. The three-phase synchronous motor has:

a. Three loading modes
b. Four loading modes
c. Five loading modes
d. Two loading modes

Correct answer: Three loading modes

95. The three-phase squirrel cage asynchronous motor has:

a. Five loading modes
b. Four loading modes
c. Two loading modes
d. Three loading modes

Correct answer: Four loading modes

Because three-phase squirrel cage asynchronous motor hasn't plugging mode

96. The three-phase wound rotor asynchronous motor has:

OR

The three-phase slip ring asynchronous motor has:

a. Five loading modes
b. Three loading modes
c. Two loading modes
d. Four loading modes

Correct answer: Five loading modes

97. When the ac wound rotor motor slip value more than one, the ac motor operates in: (When the slip value more than one, the ac motor operates in:)

a. Third quadrant
b. First quadrant
c. Fourth quadrant
d. Second quadrant

Correct answer: Fourth quadrant (Plugging Mode)

Note that: Slip greater than one then wound rotor motor operates in fourth quadrant in braking mode

98. The ac slip ring motor of slip value more than one operates in:

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a. Motoring mode
b. Short circuit mode
c. Regenerating mode
d. Plugging mode

Correct answer: Fourth quadrant (Plugging Mode)

99. When the ac wound rotor motor slip value is negative, the ac motor operates in:

a. Third quadrant
b. First quadrant
c. Second quadrant
d. Fourth quadrant

Correct answer: Second quadrant (Regenerative Mode)

100. When the ac wound rotor motor slip value equal zero, the ac motor operates in:

a. Motoring Mode
b. Plugging Mode
c. Regenerative Mode
d. Short Circuit Mode
e. No Load Mode

Correct answer: No Load Mode

101. When the ac wound rotor motor slip value between zero and one, the ac motor operates in:

a. Plugging Mode
b. Motoring Mode
c. No Load Mode
d. Short Circuit Mode
e. Regenerative Mode

Correct answer: Motoring Mode

102. When the ac wound rotor motor slip value equal one, the ac motor operates in:

a. Plugging Mode
b. Motoring Mode
c. No Load Mode
d. Short Circuit Mode
e. Regenerative Mode

Correct answer: Short Circuit Mode

103. If the synchronous motor during operation over loaded more than maximum torque, the motor will:

a. Be out of synchronization and operate as synchronous generator
b. Stay operate as synchronous motor
c. Be out of synchronization and operate as asynchronous motor
d. Run in short circuit mode

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Correct answer: Be out of synchronization and operate as asynchronous motor

104. The ac motor critical (maximum) torque independent on:

a. Rotor resistance
b. Stator resistance
c. Source frequency
d. Source voltage

Correct answer: Rotor resistance

$$T_{e\,max} = \frac{3}{2\omega_{sm}} \frac{V_{th}^2}{\left\{R_{th} + \sqrt{R_{th}^2 + (X_{th} + X_2')^2}\right\}} \rightarrow (1)$$

if stator impedance is neglected, then

$$T_{e\,max} = \frac{3}{2\omega_{sm}} \times \frac{V_1^2}{X_2'}$$

V_1 = source voltage

ω_s = synchronous speed

from (1), T_{emax} depends on source voltage
stator resistance
source frequency

and is independent of rotor resistance

(d) rotor resistance

105. The ac motor critical (maximum) torque does not depend on the: (The ac motor critical (maximum) torque is not depend on the:/ The ac motor critical (maximum) torque is not depending on the:)

a. Rotor resistance

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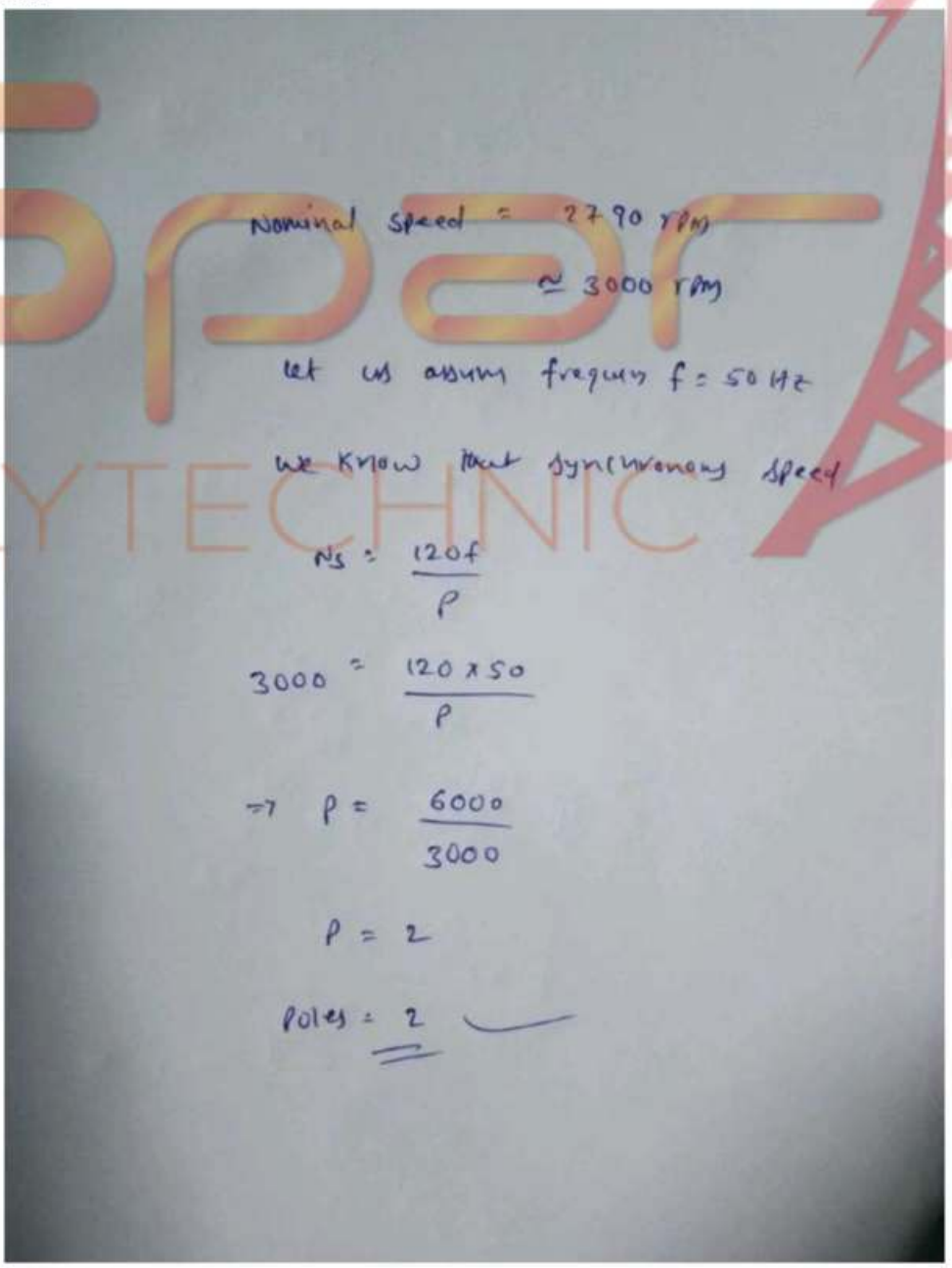
b. Source frequency
c. Stator resistance
d. Source voltage

Correct answer: Rotor resistance

106. If the nominal speed of squirrel- cage motor is 2790 rpm, the motor has:

a. 2 poles
b. 6 poles
c. 4 poles
d. 8 poles

Correct answer: 2 poles



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107. If the nominal speed of squirrel-cage motor is 1470 rpm, the motor has:

a. 6 poles
b. 8 poles
c. 4 poles
d. 2 poles

Correct answer: 4 poles

108. Three-phase induction motor, its poles number = 4, $f_1 = 50\text{Hz}$, $S_n = 0.02$, the nominal rotational speed is:

a. $2940\ rpm$
b. $1470\ rpm$
c. $1530\ rpm$
d. $3060\ rpm$

Correct answer: $1470\ rpm$

If the motor rotational speed n_r equals 1470 rpm, then the nearest synchronous speed n_S for it is 1500 rpm.

$$n_{rN} = (1 - S_n)n_S = (1 - 0.02) * 1500\ \text{rpm} = 0.98 * 1500\ \text{rpm} \approx 1470\ \text{rpm}$$

109. The highest hardness of torque-speed characteristic belongs to:

a. Series-field DC motor
b. Synchronous ac motor
c. Shunt field DC motor
d. Squirrel-cage ac motor

Correct answer: Synchronous ac motor

110. Induction motor magnetic losses for 50 Hz frequency vary:

a. Directly proportional to stator phase voltage
b. Directly proportional to load torque
c. Directly proportional to stator phase voltage square
d. Inversely proportional to stator phase voltage

Correct answer: Directly proportional to stator phase voltage square

$$\Delta P_m = \Delta P_{mn}(\frac{V_1}{V_{1N}})^2(\frac{f_1}{f_{1N}})^{1.2}$$

111. The stable mode of synchronous motor occurs (happens) when the angular characteristic angle value is:

a. $\frac{\pi}{2} < \theta < \pi$
b. $\theta > \pi$
c. $\theta < 0$
d. $0 < \theta < \frac{\pi}{2}$

Correct answer: $0 < \theta < \frac{\pi}{2}$

112. The unstable (out of synchronization) mode of synchronous motor occurs (happens) when the angular characteristic angle value is:

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a. $\frac{\pi}{2} < \theta < \pi$
b. $\theta > \pi$
c. $\theta < 0$
d. $0 < \theta < \frac{\pi}{2}$

Correct answer: $\frac{\pi}{2} < \theta < \pi$

113. Three-phase ac motor has nominal torque of 75 N.m and maximum torque of 225 N.m. The motor loading capacity (coefficient) equals:

a. 0.5
b. 3
c. 300 N.m
d. 150 N.m

Correct answer: 3

114. If the synchronous motor load torque value is more than synchronizing torque during starting, the motor will:

a. Be synchronized and operate as synchronous generator
b. Not be synchronized and operate as asynchronous motor
c. Be synchronized and operate as synchronous motor
d. Run in short circuit mode

Correct answer: Not be synchronized and operate as asynchronous motor

115. If the synchronous motor load torque value is less than synchronizing torque during starting, the motor will:

a. Run in short circuit mode
b. Not be synchronized and operate as asynchronous motor
c. Be synchronized and operate as synchronous generator
d. Be synchronized and operate as synchronous motor

Correct answer: Be synchronized and operate as synchronous motor

Note that: The motor will be synchronized and operate at as synchronous motor. It will be synchronized till pull out torque which is approx. 150% of full load torque

Or

Reason, synchronous motor rotor having a high inertia due to high inertia rotor cannot catch synchronous speed of the stator, for these reason

116. The slip value at starting point of ac motor equals:

a. Zero
b. One
c. Infinity
d. Minus two

Correct answer: One

117. If motor rotational speed equals 950 rpm, the slip is:

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Or

If ac motor rotational speed equals 950 rpm, the slip is:

a. 0.005
b. 0.05
c. 0.1
d. 0.95

Correct answer: 0.05

$$S = \frac{n_o - n}{n_o}$$

Where:

S: Slip value

n_o: Rotational speed of stator field (synchronous speed)

n: Rotational speed of rotor

If the motor rotational speed equals 950 rpm, then the nearest synchronous speed for it is 1000 rpm

$$S = \frac{1000 - 950}{1000} = \frac{50}{1000} = 0.05$$

118. If motor rotational speed equals 950 rpm, the no-load angular speed is:

Or

If ac motor rotational speed equals 950 rpm, the no-load angular speed is:

a. 104.72 rad/s
b. 209.44 rad/s
c. 1000 rpm
d. None of these

Correct answer: 104.72 rad/s

If the motor rotational speed equals 950 rpm, then the nearest synchronous speed for it is 1000 rpm which is approximately equals the no-load speed, so no-load rotational speed is 1500 rpm:

$$\omega_o = \frac{2\pi}{60} n = \frac{2\pi}{60} * 1000\ rpm = 104.71975512\ \text{rad/s} \approx 104.72\ \text{rad/s}$$

119. If motor rotational speed equals 950 rpm, the number of poles is:

Or

If ac motor rotational speed equals 950 rpm, the number of poles is:

a. 2
b. 4
c. 6
d. 8

Correct answer: 6

If the motor rotational speed equals 950 rpm, then the nearest synchronous speed for it is 1000 rpm, and according to Synchronous Speed Formula:

$$n = \frac{120 * f}{P}$$

Where:

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n: Synchronous Speed

f: frequency

P: Number of Poles

If the frequency does not mention, then it's considered to be the same as the power grid frequency of your country (in Jordan the power grid frequency is 50 Hz)

$$P = \frac{120 * f}{n} = \frac{120 * 50Hz}{1000\ rpm} = 6\ poles$$

120. If motor rotational speed equals 1450 rpm, the slip is:

Or

If ac motor rotational speed equals 1450 rpm, the slip is:

a. 0.0333
b. 0.9666
c. 0.0033
d. 0.0666

Correct answer: 0.0033

$$S = \frac{n_o - n}{n_o}$$

Where:

S: Slip value

n_o: Rotational speed of stator field (synchronous speed)

n: Rotational speed of rotor

If the motor rotational speed equals 1450 rpm, then the nearest synchronous speed for it is 1500 rpm

$$S = \frac{1500 - 1450}{1500} = \frac{50}{1500} = 0.033$$

121. If motor rotational speed equals 1450 rpm, the no-load angular speed is:

Or

If ac motor rotational speed equals 1450 rpm, the no-load angular speed is:

a. 314 rad/s
b. 1500 rpm
c. 209.42 rad/s
d. 157.07 rad/s

Correct answer: 157.07 rad/s

If the motor rotational speed equals 1450 rpm, then the nearest synchronous speed for it is 1500 rpm which is approximately equals the no-load speed, so no-load rotational speed is 1500 rpm:

$$\omega_o = \frac{2\pi}{60} n = \frac{2\pi}{60} * 1500\ rpm = 157.079632679\ \text{rad/s} \approx 157.07\ \text{rad/s}$$

122. If motor rotational speed equals 1450 rpm, the number of poles is:

Or

If ac motor rotational speed equals 1450 rpm, the number of poles is:

a. 1
b. 2
c. 3

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d. 4

Correct answer: 4

If the motor rotational speed equals 1450 rpm, then the nearest synchronous speed for it is 1500 rpm, and according to Synchronous Speed Formula:

$$n = \frac{120 * f}{P}$$

Where:

n: Synchronous Speed

f: frequency

P: Number of Poles

If the frequency does not mention, then it's considered to be the same as the power grid frequency of your country (in Jordan the power grid frequency is 50 Hz)

$$P = \frac{120 * f}{n} = \frac{120 * 50Hz}{1500\ rpm} = 4\ poles$$

123. When the slip value is negative, the ac motor operates in:

a. Third quadrant
b. Second quadrant
c. First quadrant
d. Fourth quadrant

Correct answer: Second quadrant

124. The electric motor is selected to operate for long period of time without overheating on:

a. Natural characteristic at light load
b. Natural characteristic at overload
c. Regulated characteristic at nominal load
d. Natural characteristic at nominal load

Correct answer: Natural characteristic at nominal load

125. The synchronization conditions of synchronous motor are:

a. $motor\ speed < synchronization\ speed$
b. $load\ torque < synchronization\ torque\ and\ Motor\ speed > synchronization\ speed$
c. $load\ torque > synchronization\ torque$
d. $load\ torque > synchronization\ torque\ and\ Motor\ speed > synchronization\ speed$

Correct answer: $load\ torque < synchronization\ torque\ and\ Motor\ speed > synchronization\ speed$

126. T-form per phase equivalent open circuit describes:

a. Regenerating mode
b. Motoring mode
c. Short circuit mode
d. No-load mode

Correct answer: No-load mode

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127. T-form per-phase equivalent short circuit describes:

a. Motoring mode
b. Regenerating mode
c. No-load mode
d. Short circuit mode

Correct answer: Short circuit mode

128. The following expression $\frac{3V_1{}^2R_2{}'}{\omega_o((R_1+R_2{}')^2+(X_{S.C})^2)}$ determines:

a. Motoring mode torque
b. Critical (Maximum) Torque
c. Starting torque
d. No-load motor torque

Correct answer: Starting torque (Short Circuit Torque)

129. The following expression $\frac{3V_1{}^2}{2\omega_o[R_1\pm\sqrt{R_1{}^2+(X_{S.C})^2}]}$ determines:

a. Motoring mode torque
b. Starting torque
c. Short circuit mode
d. No-load motor torque
e. Critical (Maximum) Torque

Correct answer: Critical (Maximum) Torque

130. No-load mode is in-efficient, but it is efficiently used in:

a. Drilling machine
b. Elevator
c. Pump machinery
d. Escalator

Correct answer: Elevator

131. The slip ring three-phase ac motor allowed starting current is:

a. Nominal current
b. (5-7) times of nominal current
c. (2-3) times of nominal current
d. Less than nominal current

Correct answer: (2-3) times of nominal current

132. The squirrel cage three-phase ac motor allowed starting current is:

a. Nominal current
b. (2-3) times of nominal current

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

c. Less than nominal current
d. (5-7) times of nominal current

Correct answer: (5-7) times of nominal current



ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Unit Five: Heating and Cooling of Electric Motors

133. The insulating (material/materials) class A temperature is:

Or

The insulating (material/materials) class A temperature rise is:

Or

The insulating (material/materials) class A temperature limit is:

a. $120°C$
b. $140°C$
c. $105°C$
d. $130°C$

Correct answer: $105°C$

134. The insulating (material/materials) class B temperature is:

Or

The insulating (material/materials) class B temperature rise is:

Or

The insulating (material/materials) class B temperature limit is:

a. $90°C$
b. $120°C$
c. $105°C$
d. $130°C$
e. $110°C$
f. $140°C$

Correct answer: $130°C$

135. The insulating (material/materials) class C temperature is:

Or

The insulating (material/materials) class C temperature rise is:

Or

The insulating (material/materials) class C temperature limit is:

a. 130 °C
b. 180 °C
c. >180°C
d. 150 °C

Correct answer: >180°C

136. The insulating (material/materials) class E temperature is:

Or

The insulating (material/materials) class E temperature rise is:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Or

The insulating (material/materials) class E temperature limit is:

a. $140°C$
b. $120°C$
c. $105°C$
d. $130°C$

Correct answer: $120°C$

137. The insulating (material/materials) class F temperature is:

Or

The insulating (material/materials) class F temperature rise is:

Or

The insulating (material/materials) class F temperature limit is:

a. $135°C$
b. $145°C$
c. $155°C$
d. $125°C$

Correct answer: $155°C$

138. The insulating (material/materials) class H temperature is:

Or

The insulating (material/materials) class H temperature rise is:

Or

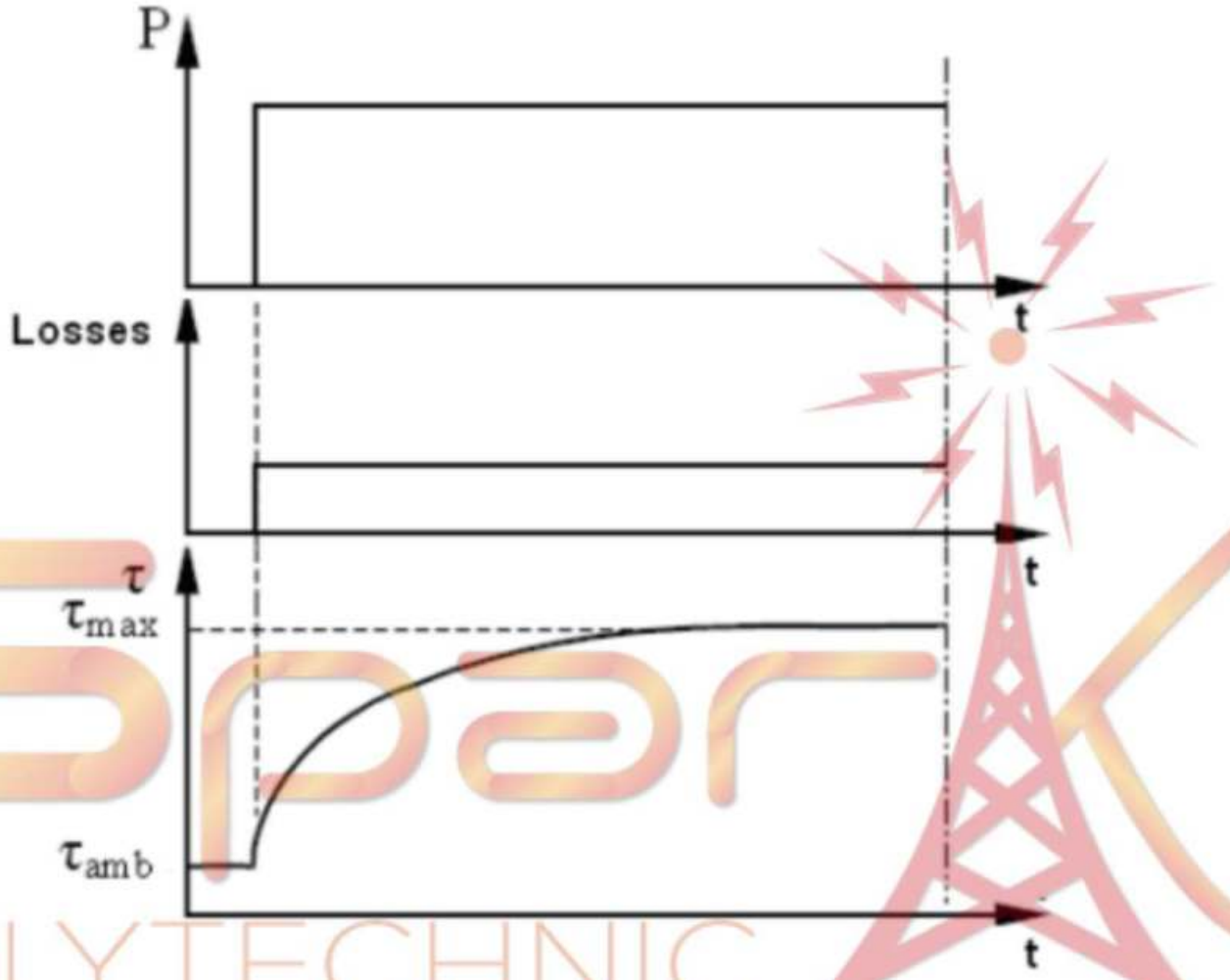
The insulating (material/materials) class H temperature limit is:

a. $150°C$
b. $> 180°C$
c. $130°C$
d. $180°C$

Correct answer: $180°C$

139. The following motor duty type is:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

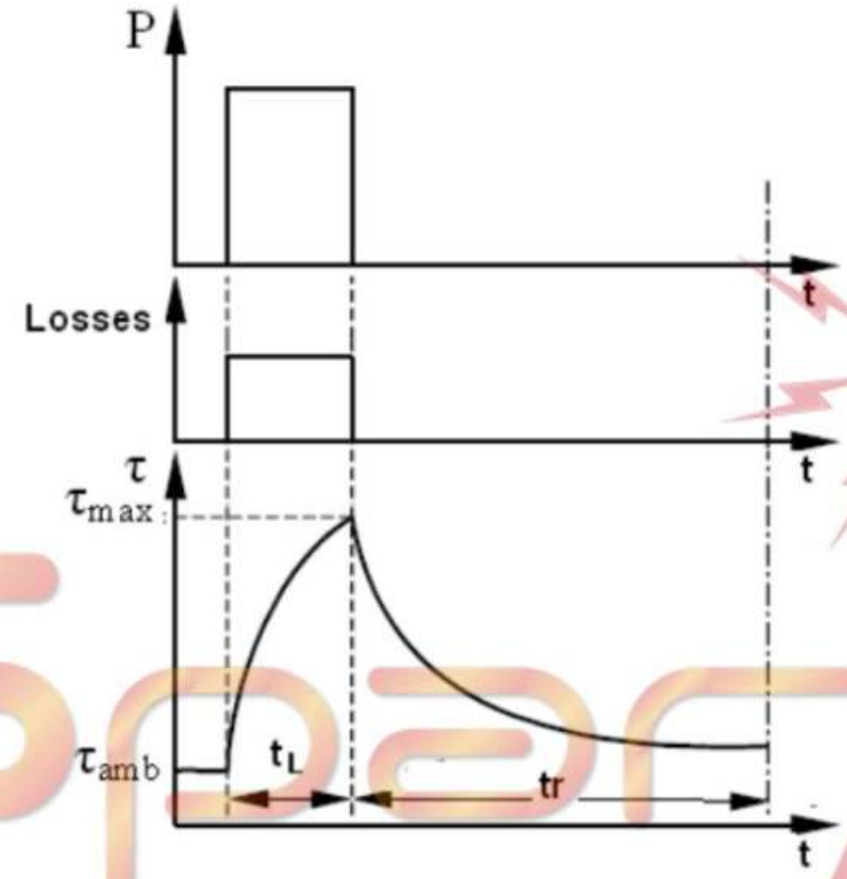


a. Continuous time duty (duty type S1)
b. Short time duty (duty type S2)
c. Intermittent periodic duty (duty type S3)
d. Intermittent periodic duty with starting (duty type S4)
e. Intermittent periodic duty with starting and braking (duty type S5)
f. Continuous operation with intermittent load (duty type S6)
g. Continuous operation with periodic starting and braking (duty type S7)
h. Continuous operation with periodic changes in load and speed (duty type S8)

Correct answer: Continuous time duty (duty type S1)

140. The following motor duty type is:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

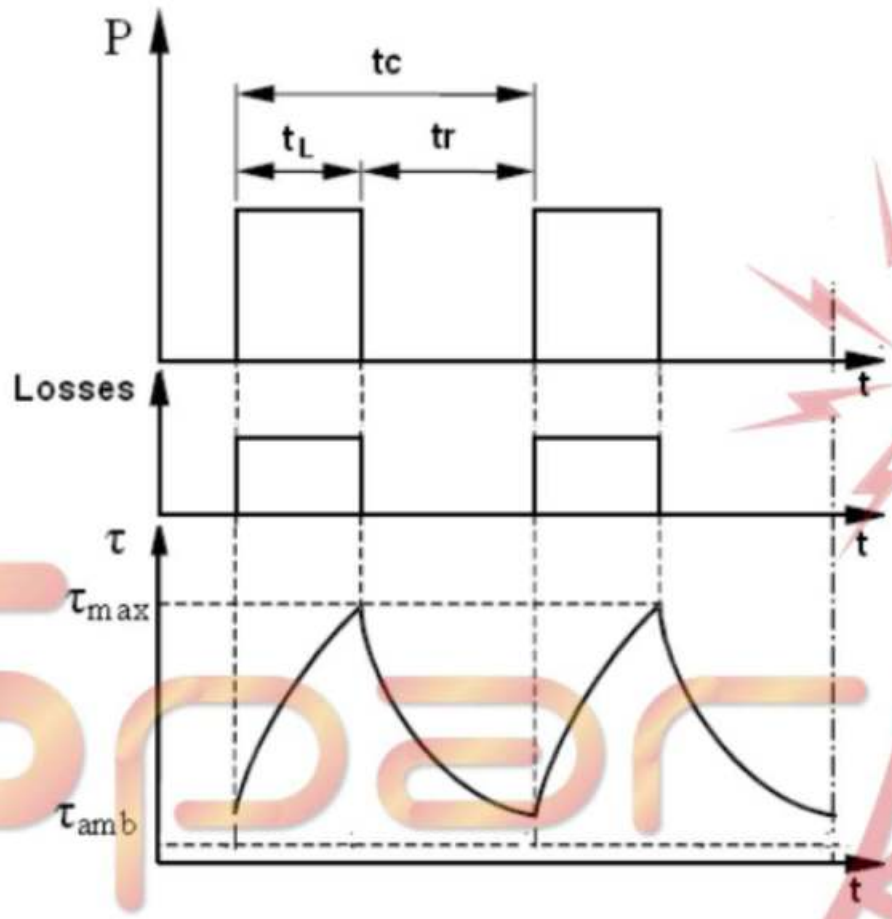


a. Continuous time duty (duty type S1)
b. Short time duty (duty type S2)
c. Intermittent periodic duty (duty type S3)
d. Intermittent periodic duty with starting (duty type S4)
e. Intermittent periodic duty with starting and braking (duty type S5)
f. Continuous operation with intermittent load (duty type S6)
g. Continuous operation with periodic starting and braking (duty type S7)
h. Continuous operation with periodic changes in load and speed (duty type S8)

Correct answer: Short time duty (duty type S2)

141. The following motor duty type is:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

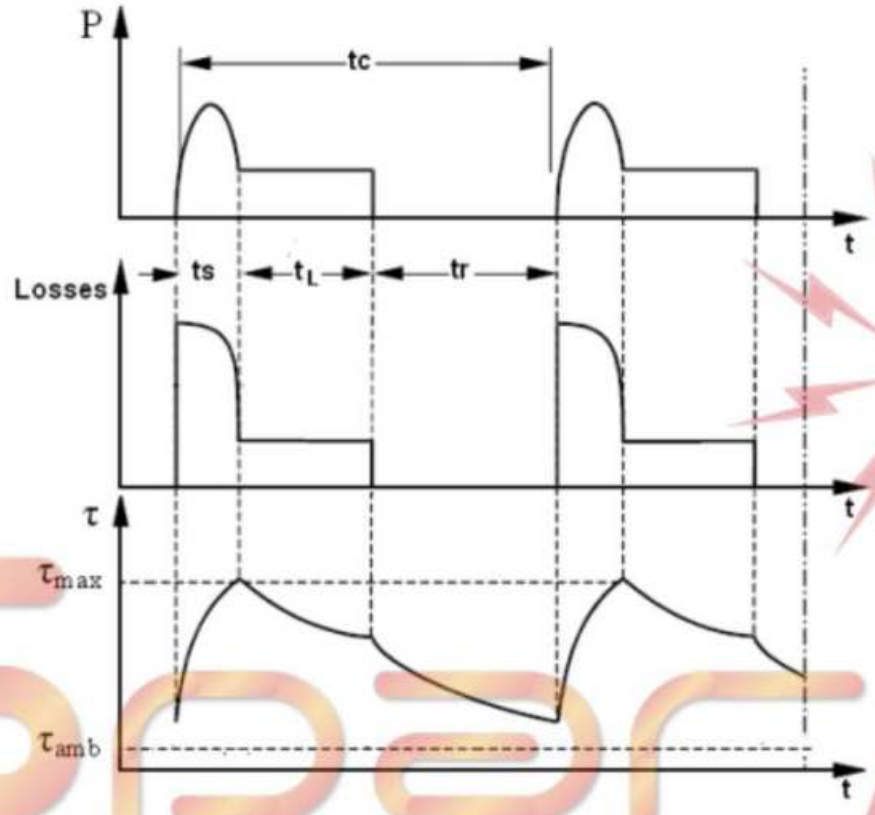


a. Continuous time duty (duty type S1)
b. Short time duty (duty type S2)
c. Intermittent periodic duty (duty type S3)
d. Intermittent periodic duty with starting (duty type S4)
e. Intermittent periodic duty with starting and braking (duty type S5)
f. Continuous operation with intermittent load (duty type S6)
g. Continuous operation with periodic starting and braking (duty type S7)
h. Continuous operation with periodic changes in load and speed (duty type S8)

Correct answer: Intermittent periodic duty (duty type S3)

142. The following motor duty type is:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

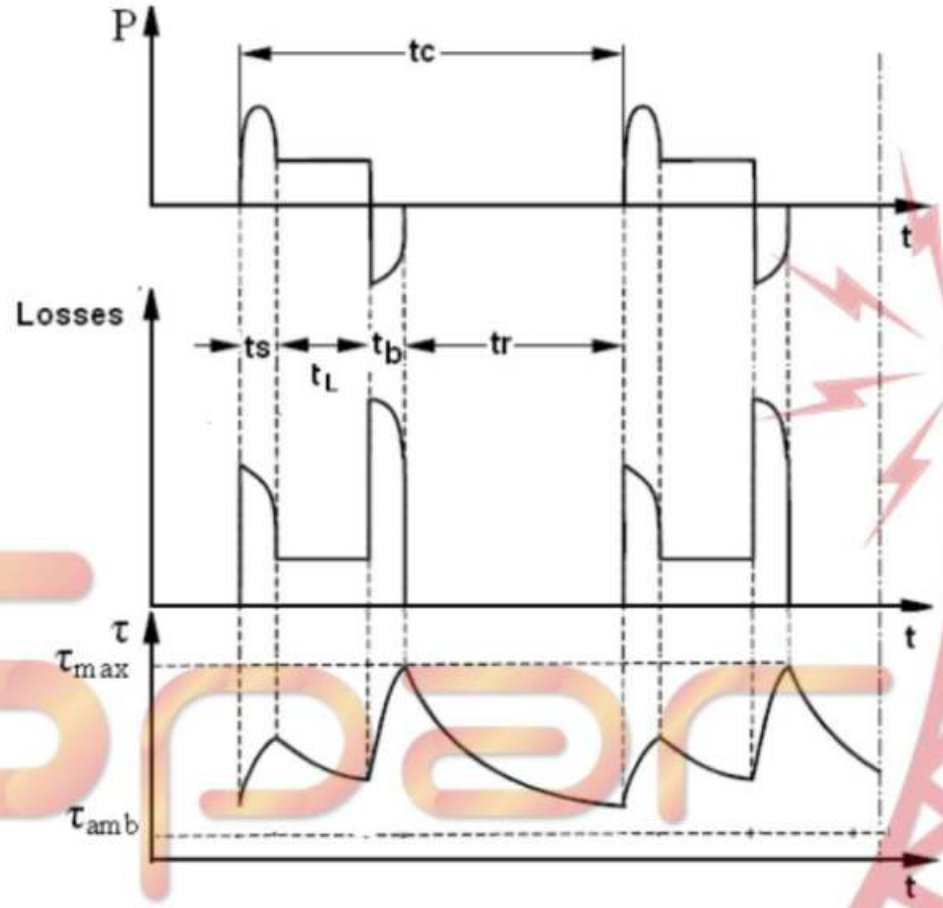


a. Continuous time duty (duty type S1)
b. Short time duty (duty type S2)
c. Intermittent periodic duty (duty type S3)
d. Intermittent periodic duty with starting (duty type S4)
e. Intermittent periodic duty with starting and braking (duty type S5)
f. Continuous operation with intermittent load (duty type S6)
g. Continuous operation with periodic starting and braking (duty type S7)
h. Continuous operation with periodic changes in load and speed (duty type S8)

Correct answer: Intermittent periodic duty with starting (duty type S4)

143. The following motor duty type is:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

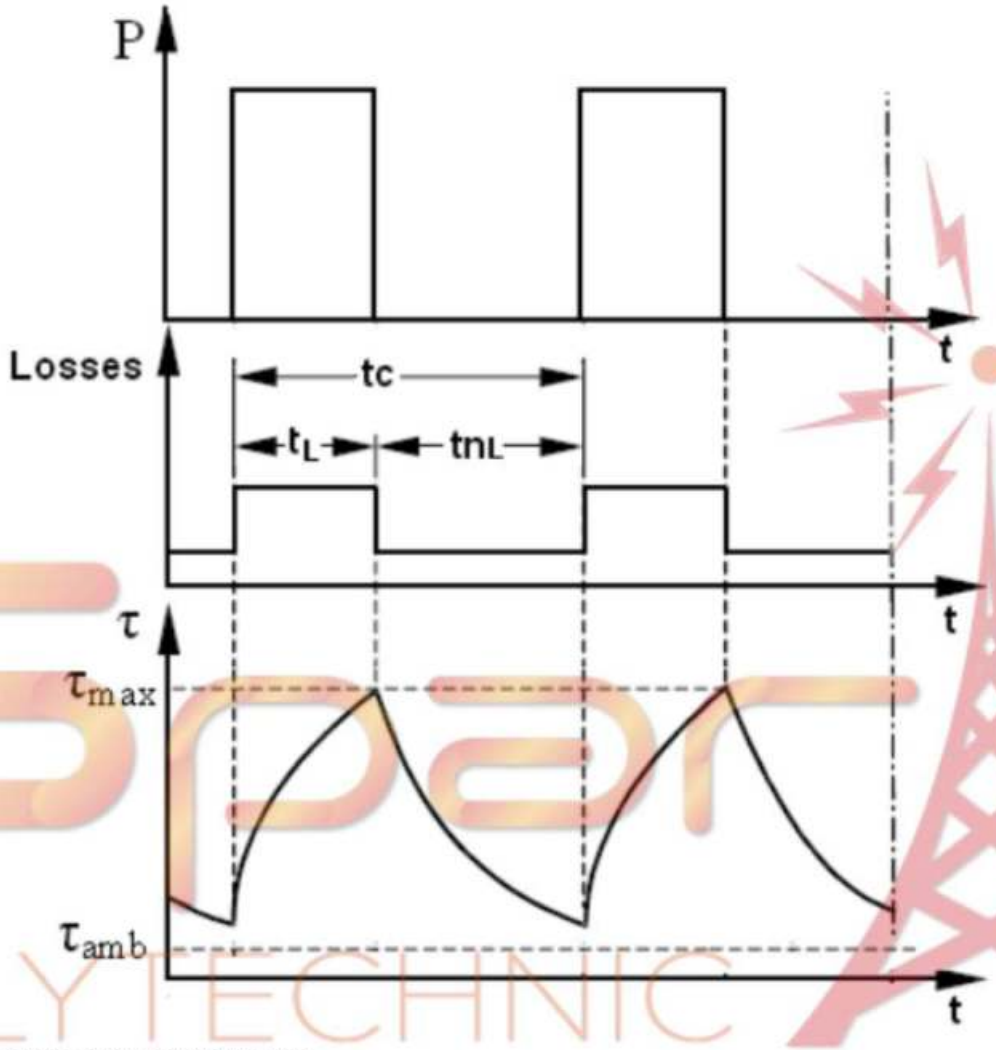


a. Continuous time duty (duty type S1)
b. Short time duty (duty type S2)
c. Intermittent periodic duty (duty type S3)
d. Intermittent periodic duty with starting (duty type S4)
e. Intermittent periodic duty with starting and braking (duty type S5)
f. Continuous operation with intermittent load (duty type S6)
g. Continuous operation with periodic starting and braking (duty type S7)
h. Continuous operation with periodic changes in load and speed (duty type S8)

Correct answer: Intermittent periodic duty with starting and braking (duty type S5)

144. The following motor duty type is:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

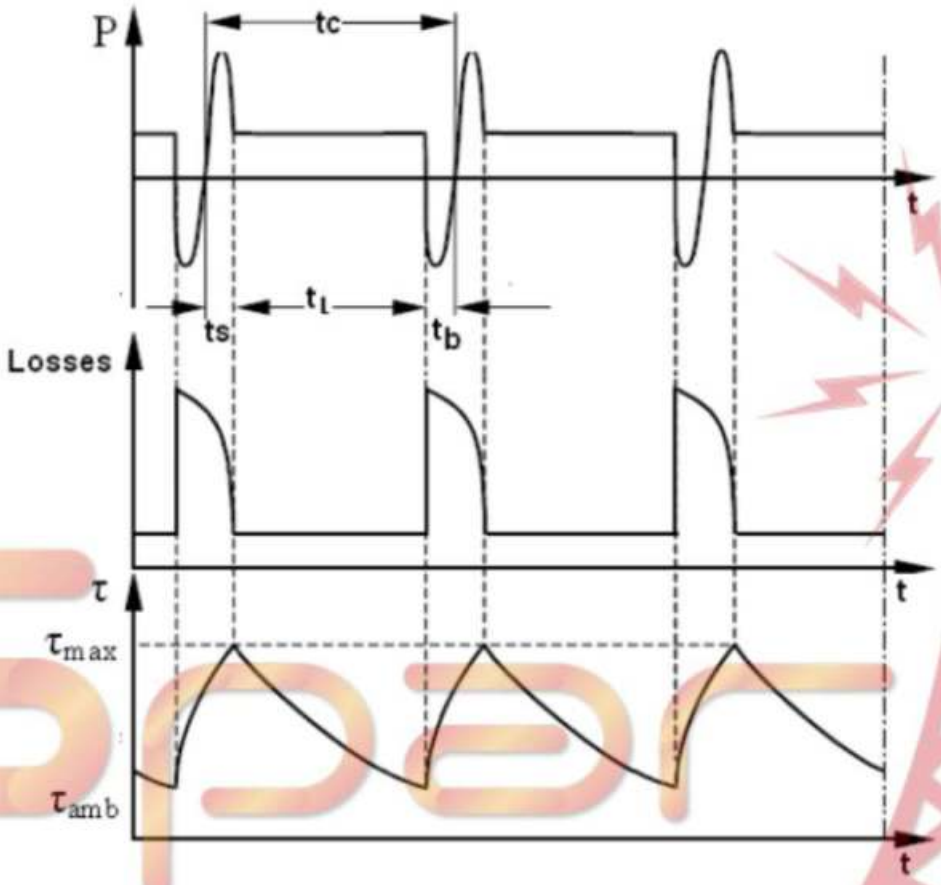


a. Continuous time duty (duty type S1)
b. Short time duty (duty type S2)
c. Intermittent periodic duty (duty type S3)
d. Intermittent periodic duty with starting (duty type S4)
e. Intermittent periodic duty with starting and braking (duty type S5)
f. Continuous operation with intermittent load (duty type S6)
g. Continuous operation with periodic starting and braking (duty type S7)
h. Continuous operation with periodic changes in load and speed (duty type S8)

Correct answer: Continuous operation with intermittent load (duty type S6)

145. The following motor duty type is:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

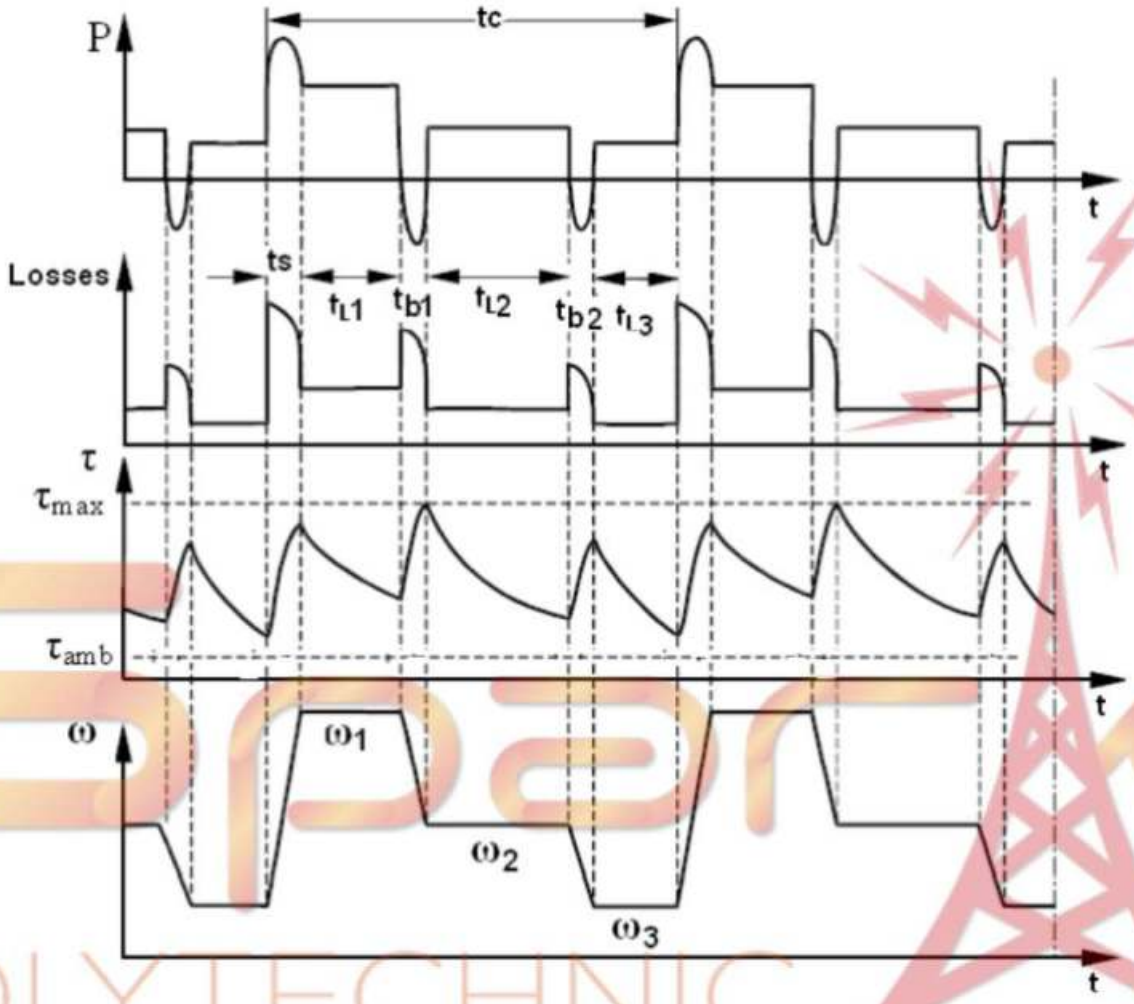


a. Continuous time duty (duty type S1)
b. Short time duty (duty type S2)
c. Intermittent periodic duty (duty type S3)
d. Intermittent periodic duty with starting (duty type S4)
e. Intermittent periodic duty with starting and braking (duty type S5)
f. Continuous operation with intermittent load (duty type S6)
g. Continuous operation with periodic starting and braking (duty type S7)
h. Continuous operation with periodic changes in load and speed (duty type S8)

Correct answer: Continuous operation with periodic starting and braking (duty type S7)

146. The following motor duty type is:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



POLYTECHNIC

a. Continuous time duty (duty type S1)
b. Short time duty (duty type S2)
c. Intermittent periodic duty (duty type S3)
d. Intermittent periodic duty with starting (duty type S4)
e. Intermittent periodic duty with starting and braking (duty type S5)
f. Continuous operation with intermittent load (duty type S6)
g. Continuous operation with periodic starting and braking (duty type S7)
h. Continuous operation with periodic changes in load and speed (duty type S8)

Correct answer: Continuous operation with periodic changes in load and speed (duty type S8)

147. Temperature equilibrium is never reached because these periods being too short to attain thermal equilibrium during one duty cycle. This is satisfied for:

a. Continuous time duty with fluctuating load
b. Short-time duty
c. Continuous-time duly with constant load
d. Intermittent-periodic duty

Correct answer: Intermittent-periodic duty

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

148. In intermittent periodic duty (duty type S 3):

a. Temperature equilibrium (steady-state) is reached during many duty cycles
b. Temperature equilibrium (steady-state) is never reached
c. Temperature equilibrium (steady-state) is reached during one duty cycle
d. Ambient temperature at rest (off) period is reached

Correct answer: Temperature equilibrium (steady-state) is never reached

149. The relationship $K = \sqrt{\frac{1-e^{-[t_{on}/\tau_h + t_{off}/\tau_c]}}{1-e^{-t_{on}/\tau_h}}}$ used to calculate overloading factor of:

a. Intermittent periodic duty cycle
b. Continuous duty cycle
c. Continuous duty cycle with fluctuating periodic loading
d. Short time duty cycle

Correct answer: Intermittent periodic duty cycle

150. Which duty cycle has on-time and off-time period?

a. Short time duty cycle
b. Continuous duty cycle with constant load
c. Continuous duty cycle with variable load
d. Intermittent periodic duty cycle

Correct answer: Intermittent periodic duty cycle

Note that: Speed hoist is a perfect example of constant type loads in which torque variation is independent of speed. The speed-torque characteristics of this type of load are given by T=K where K is a constant

151. To determine motor rating for continuous duty cycle with fluctuating loads, it is better to use:

a. Equivalent torque method
b. Equivalent power method
c. Equivalent current method
d. Maximum load only for choosing the motor rating

Correct answer: Equivalent power method

152. Heating time constant of electric motors is:

a. Equal to the mechanical time constant
b. More than the electromagnetic and mechanical time constants
c. Less than the electrical and mechanical time constants
d. Equal to the electro-magnetic time constant

Correct answer: More than the electromagnetic and mechanical time constants

$$\text{Heating time constant} > \text{Mechanical time constant} >> \text{Electromagnetic time constant}$$

What does a motor's electrical time constant represent? - Quora

Understanding Servo Motor Time Constants - Electromechanical Knowledge Base - Electromechanical Group - Parker Community

بسواعدنا نبنيها

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For more information about Electric Motor Cooling and Heating:

Electric Motor Cooling and Heating | Heating time constant | Cooling Curve (eeeguide.com)

Soln:-

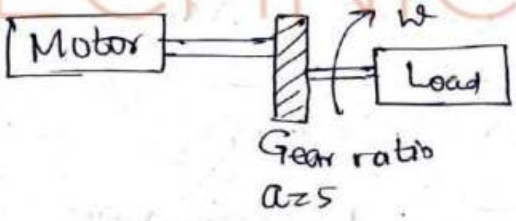
⇒ Heating time constant of electric motor is

① More than electromagnetic and mechanical time constant.

Explanation:- Electromagnetic and mechanical system are faster in dynamics compared to thermal system. It takes much larger time for a ~~thermal~~ motor to reach a particular temperature than the time taken to reach a particular torque (mechanical part) and/or the current to reach a particular value (electrical part).

⇒ Therefore the heating time constant of electric motor is much larger than electrical, mechanical and electromagnetic time constant

⇒



$$T_1 = \frac{a\,T_{l1}}{\eta}$$

153. Coefficient of heat transfer is:

a. the amount of heat dissipated into the ambient medium from 1 m^2 of motor area when the motor temperature rises 1°C
b. the amount of heat dissipated into the ambient medium from 1 kg of motor mass per second when the motor temperature rises 1°C
c. the amount of heat dissipated into the ambient medium from 1 m^2 of motor area per second when the motor temperature rises 1°C
d. the amount of heat dissipated into the ambient medium from 1 m^2 of motor area per second

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Correct answer: the amount of heat dissipated into the ambient medium from 1 m^2 of motor area per second when the motor temperature rises 1°C

154. Continuous duty cycle operates under constant load with following parameters:

a. Temperature rise during operation under load reaches steady-state value
b. Temperature rise during operation under load does not reach steady-state value
c. Starting time is not neglected
d. Starting losses are not neglected

Correct answer: Temperature rise during operation under load reaches steady-state value

155. Short time duty cycle operates under constant load with following parameters:

a. Temperature rise during operation under load does not reach steady-state value
b. Starting and braking losses are not neglected
c. Temperature rise during operation under load reaches steady-state value
d. Starting time and braking time is not neglected

Correct answer: Temperature rise during operation under load does not reach steady-state value

156. The relationship $P_{eq} = \sqrt{\frac{1}{Period}\int_0^{Period} {P_L}^2\, dt}$ used to calculate equivalent load power of:

a. Continuous duty cycle with constant load
b. Short time duty cycle
c. Continuous duty cycle with intermittent periodic loading
d. Continuous duty cycle with fluctuating periodic loading

Correct answer: Continuous duty cycle with fluctuating periodic loading

157. If the motor has a short-time duty, then the relation between heating time constant and cooling time constant is:

a. the sum of both constants is equal to the mechanical constant
b. the heating time constant is more than the cooling time constant
c. the heating time constant is equal to the cooling time constant
d. the heating time constant is less than the cooling time constant

Correct answer: the heating time constant is less than the cooling time constant

158. In the short time duty cycle, the motor temperature can:

a. Reach ambient temperature at rest interval
b. Not reach ambient temperature at rest interval
c. Reach steady-state temperature at run interval
d. Reach maximum temperature at run interval

Correct answer: Reach ambient temperature at rest interval

159. Motor duty is considered continuous duty (S 1), if the motor operates with the following conditions:

a. Temperature rise during operation under load reaches steady-state value
b. Temperature rise during operation under load does not reach steady-state value

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

c. Starting losses are not neglected
d. Starting time is not neglected

Correct answer: Temperature rise during operation under load reaches steady-state value

160. The electric motor windings insulating material temperature rise is determined by assuming ambient temperature value equals: (The temperature rise is related to ambient temperature which value is equal to:)

a. Ten degrees centigrade (10°C)
b. Fourty degrees centigrade (40°C)
c. Fifty degrees centigrade (50°C)
d. Twenty degrees centigrade (20°C)
e. Thirty degrees centigrade (30°C)

Correct answer: Fourty degrees centigrade (40°C)

161. The relationship used $K = \sqrt{\frac{\alpha+1}{1-e^{-t_{on}/\tau_h}} - \alpha}$ to calculate over loading factor of:

a. Short time duty cycle
b. Continuous duty cycle with intermittent periodic loading
c. Intermittent periodic duty cycle
d. Continuous duty cycle

Correct answer: Short time duty cycle

162. The electric motor heat exchange expression is:

a. $C\frac{d\Theta}{dt} = -D\Theta - P_1$
b. $C\frac{d\Theta}{dt} = P_1 - D\Theta$
c. $C\frac{d\Theta}{dt} = P_1 + D\Theta$
d. $D\Theta - C\frac{d\Theta}{dt} = P_1$

Correct answer: $C\frac{d\Theta}{dt} = P_1 - D\Theta$

163. The electric motor thermal model transfer function represents:

a. Integral element
b. Proportional element
c. Second order element
d. First order lag element

Correct answer: First order lag element

164. Frequency of operation (switching) parameter is used to specify the following duty:

a. Intermittent periodic cycle

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

b. Continuous duty cycle with intermittent periodic loading
c. Intermittent periodic duty cycle with starting and braking
d. Continuous duty cycle with starting and braking

Correct answer: Intermittent periodic duty cycle with starting and braking

165. Frequency of operation (switching) parameter measuring unit is:

a. Starting (cycle) per minute
b. Starting (cycle) per second
c. Starting (cycle) per hour
d. Starting (cycle) per shift

Correct answer: Starting (cycle) per hour

166. The relationship $P_{eq} = \sqrt{\frac{\sum_{i=1}^{n} {P_i}^2 t_i}{\sum_{i=1}^{n} t}}$ used to calculate equivalent load power of:

a. Continuous duty cycle with intermittent periodic loading
b. Continuous duty cycle with constant load
c. Short time duty cycle
d. Continuous duty cycle with fluctuating periodic loading

Correct answer: Continuous duty cycle with intermittent periodic loading

167. The thermal time constant of electric motor is:

a. The ratio of the temperature rise and the dissipation constant
b. The ratio of the dissipation constant and the thermal capacity
c. The ratio of the temperature rise and the thermal capacity
d. The ratio of the thermal capacity and the dissipation constant

Correct answer: The ratio of the thermal capacity and the dissipation constant

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Unit Six: DC Motors Starting, Braking and Reversing Methods

168. Indirect starting of electric motor is used when:

a. The motor is medium and high power and the source is limited power supply
b. The number of starting during operation shift is small
c. The number of starting during operation shift is large
d. The motor is fractional power and the source is unlimited power supply
e. The motor is medium and high power and the source is unlimited power supply

Correct answer: The motor is medium and high power and the source is limited power supply

169. The three-phase ac motor plugging braking achieved by:

OR

The three-phase ac motor Reversing mode achieved by:

a. Switching from high speed windings to low speed windings
b. Reducing stator voltage with fixed frequency
c. Switching off stator windings ac source and adding dc source with stator windings
d. Reversing the motor speed direction

Correct answer: Reversing the motor speed direction

Note that: For ac motor:

The regenerative braking is accomplished by switching from high speed windings to low speed windings

The dynamic braking is accomplished by disconnecting the stator windings from the source and connecting the direct current source with two phases (Switching off stator windings ac source and adding dc source with stator windings)

This method is called speed control method with constant load power, and it is used to control the speed of production machinery.

170. Braking of dc motor, where motor speed equals zero but motor torque is not equal zero is called:

a. Dynamic braking
b. Regenerative braking
c. Mechanical braking
d. Plugging (reversing) braking

Correct answer: Plugging (reversing) braking

171. Electric motors indirect starting method used to:

a. Reduce load torque
b. Reduce starting current
c. Increase motor speed
d. Keep electrical losses constant

Correct answer: Reduce starting current

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172. If mechanical time-constant related to electro-magnetic time-constant of drive system value is more than four, then drive system time response process is:

a. Under-damped
b. Over-damped
c. Critically damped
d. Oscillatory process

Correct answer: Over-damped

173. If mechanical time-constant related to electro-magnetic time-constant of drive system value is less than four, then drive system time response process is:

a. Under-damped
b. Over-damped
c. Critically damped
d. Oscillatory process

Correct answer: Under-damped

174. If mechanical time-constant related to electro-magnetic time-constant of drive system value equals four, then drive system time response process is:

a. Under-damped
b. Over-damped
c. Critically damped
d. Oscillatory process

Correct answer: Critically damped

175. Electric motor torque during mechanical braking is:

a. Negative
b. Variable
c. Zero
d. Positive

Correct answer: Zero

176. If mechanical time-constant related to electro-magnetic time-constant of drive system value is greater than four, then drive system characteristic equation has:

a. Two equal complex roots
b. Two different real roots
c. Two equal real roots
d. Two different complex roots

Correct answer: Two different real roots

177. If mechanical time-constant related to electro-magnetic time-constant of drive system value is equal four, then drive system characteristic equation has:

a. Two equal complex roots
b. Two different real roots

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c. Two equal real roots
d. Two different complex roots

Correct answer: Two equal real roots

178. If mechanical time-constant related to electro-magnetic time-constant of drive system value is less than four, then drive system characteristic equation has:

a. Two equal complex roots
b. Two different real roots
c. Two equal real roots
d. Two different complex roots

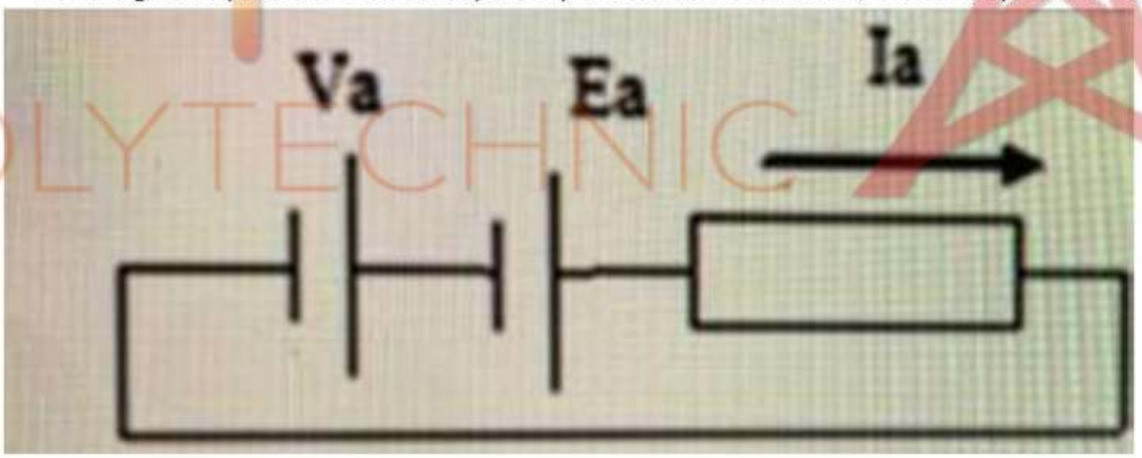
Correct answer: Two different complex roots

179. If mechanical time-constant is much larger than electro-magnetic time-constant, then drive system characteristic equation has:

a. Two equal complex roots
b. Two different real roots
c. Two equal real roots
d. Two different complex roots

Correct answer: Two equal complex roots

180. For the given equivalent circuit of separately excited dc motor below, the drive system is working in the:



a. Third quadrant
b. Second quadrant
c. First quadrant
d. Fourth quadrant

Correct answer: Second quadrant (Regenerative braking)

181. Electrical braking in comparison with mechanical braking leads to:

a. Decrease braking time
b. improve braking torque - Speed characteristic
c. Increase the hardness of braking torque-speed characteristic
d. Increase braking time

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Correct answer: Increase braking time

182. The electric motor torque value is zero, when using:

a. Dynamic braking
b. Regenerating braking
c. Mechanical braking
d. Plugging braking

Correct answer: Mechanical braking

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Unit Seven: AC Motors Starting, Braking and Reversing Methods

183. Series field dc motor can operate at no-load mode by inserting resistance in:

a. Parallel with the field circuit
b. Series with the armature circuit
c. Series with the field circuit
d. Parallel with armature circuit

Correct answer: Parallel with armature circuit

184. Regenerative braking occurs when:

a. the motor speed control equals the no-load speed
b. the motor speed exceeds the nominal value
c. the motor speed exceeds the synchronous speed
d. the motor speed changes its direction

Correct answer: the motor speed exceeds the nominal value

185. The ac motor regenerating braking achieved by:

a. Reducing stator voltage with fixed frequency
b. Reversing the motor speed direction
c. Switching from high speed windings to low speed windings
d. Switching stator windings ac source and adding dc source with stator windings

Correct answer: Switching from high speed windings to low speed windings

186. The slip ring ac motor dynamic braking achieved by:

a. Reducing stator voltage with fixed frequency
b. Switching off stator windings ac source and adding dc source with stator windings
c. Switching off stator windings ac source and adding dc source with rotor windings
d. Reversing the motor speed direction

Correct answer: Switching off stator windings ac source and adding dc source with stator windings

187. The synchronous motor dynamic braking achieved by:

a. Switching off stator windings ac source and adding dc source with stator windings
b. Reversing the motor speed direction
c. Switching off stator windings ac source and adding dc source with rotor windings
d. Switching off stator windings ac source then adding star connected rheostat with stator windings and keeping dc source with field windings switched on

Correct answer: Switching off stator windings ac source then adding star connected rheostat with stator windings and keeping dc source with field windings switched on

188. The most efficient braking method is:

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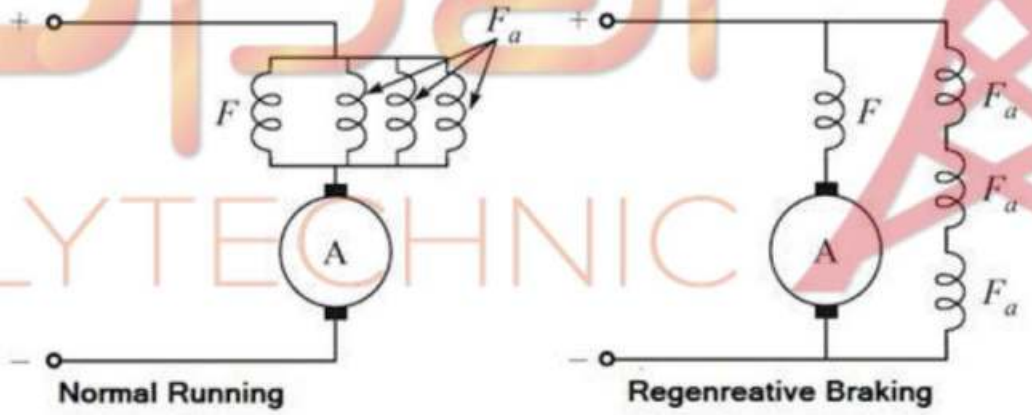
a. Dynamic braking
b. Reversing (plugging) braking
c. Electromechanical braking
d. Regenerative braking

Correct answer: Regenerative braking

Note that: Regenerative braking

In regenerative braking, the motor is not disconnected from the supply but is made to run as a generator by utilizing the Kinetic energy of the moving train. Electrical energy is feedback to the supply. The magnetic drag produced on account of generator action offers the braking torque. It is the most efficient method of braking.

- In regenerative braking, the motor, instead of being disconnected from the supply, remains connected to the supply and returns the braking energy to the supply line. So, the wastage of kinetic energy in the plugging and rheostatic braking methods is prevented in the regenerative braking. This method the drives cannot be brought to the standstill. It reduces the speed to a minimum permissible value.
- In case of a d.c. series motor, increase in excitation results decrease in speed. As such it is not possible to get e.m.f. more than voltage. It is not possible to make field current more than the armature current. Hence regeneration braking with series motors is not possible. But can be used with traction motors with some special arrangements.
- In one method of regenerative braking with a single series motor, the motor is provided with a main series field winding and a few auxiliary field windings.



- For normal running, all the field windings are connected in parallel as shown in above Fig.
- During braking period the auxiliary windings are connected in series with each other and this series connected windings are connected in parallel with the series motor, and then, connected across the supply.
- The machine acts as a compound generator, slightly differential. Such an arrangement is quite stable. A change in the line voltage causes a corresponding change in excitation which produces a change in the induced emf. Thus, the change in line voltage is compensated.

Advantages:-

Following are the advantages of regenerative braking:

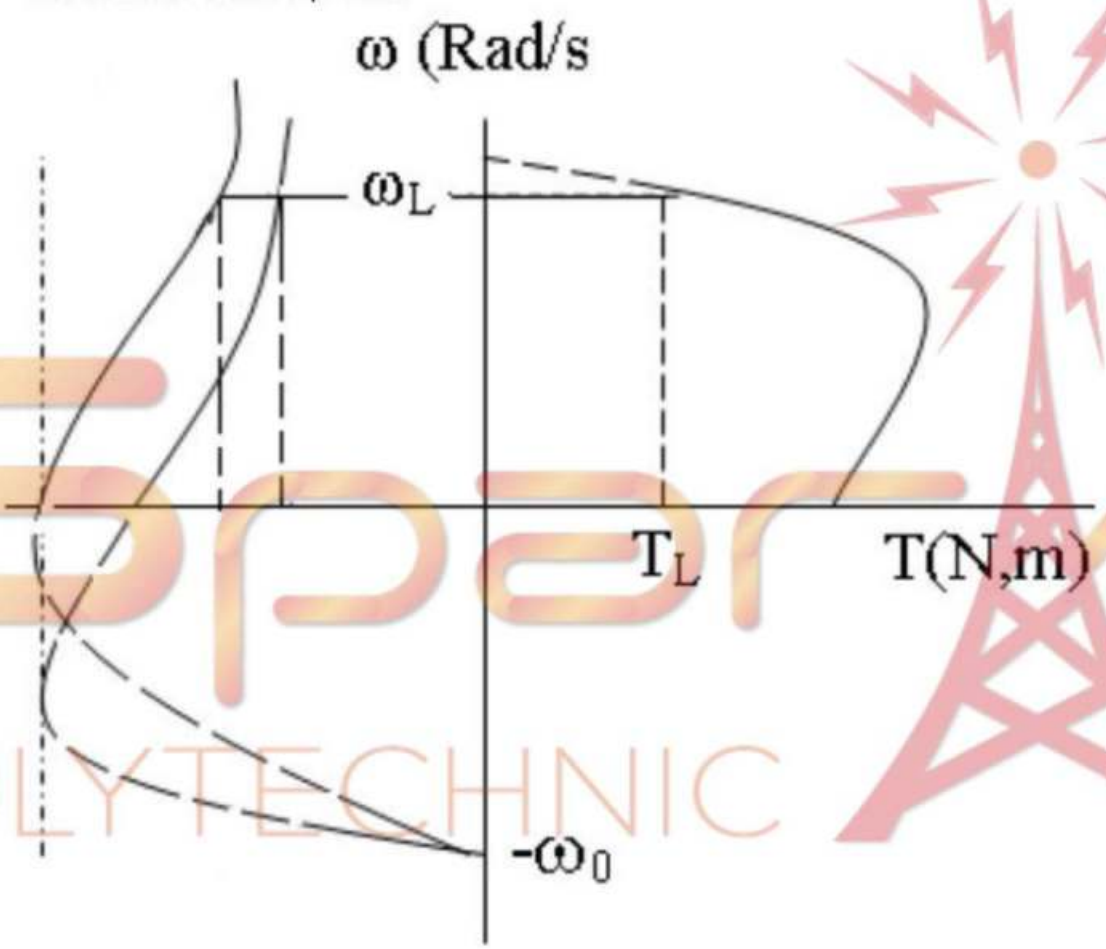
- Reduced energy consumption.
- Less wear and tear of brake blacks, wheels and, track, consequently less maintenance required by these items.
- A relatively small amount of brake block dust formed which increases the life of bearings.
- Ease and safety with which heavy loads can be hauled over steep gradients.
- The higher value of braking retardation is obtained 60 that the vehicle can be brought to rest quickly and running time is considerably reduced.
- Higher speeds are possible while going down the gradients.
- It is the most efficient method of braking.

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Disadvantages:

- Increased capital cost as motors of heavier size is employed.
- Additional cost on control equipment with consequent complications in methods of operation and control.
- Mechanical braking is required at a speed of 6.5 Km/h to bring the locomotive to rest.
- Due to recuperated energy, the operation of the sub-stations become complicated and difficult.

189. Characteristics below represent:



a. dynamic braking of induction motor for variable resistance in the dc circuit
b. counter current braking of induction motor for variable stator voltage
c. counter-current braking of induction motor for variable rotor resistance
d. counter-current braking of induction motor for variable stator resistance

Correct answer: counter-current braking of induction motor for variable rotor resistance

190. Auto transformer starting method of three-phase ac motors leads to:

a. Vary stator voltage and vary stator frequency
b. Vary stator voltage with fixed nominal frequency
c. Vary stator frequency with fixed stator voltage
d. Fixed stator voltage and fixed stator frequency

Correct answer: Vary stator voltage with fixed nominal frequency

191. A drive system has the following parameters:

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$$J = 0.25kg.m^2, \qquad T_L = 3N.m$$

Initially the drive system is working in steady-state. The steady-state speed is 1374 rpm. Now the drive is disconnected from the main supply. Calculate the time taken by the drive system to stop (consider that the stopping process is finished at 0.05 of the final value).

a. $23.4\ s$
b. $5.85\ s$
c. $11.7\ s$
d. $11.4\ s$

Correct answer: $11.4\ s$

Braking time without brake:

$$t_{stopping} = J_\Sigma \frac{\omega_L}{T_L} = 0.25 * \frac{\frac{1374}{9.55}}{3} * (1 - 0.05(\text{of the final value}))$$

192. A drive system has the following parameters:

$$J = 0.5kg.m^2, \qquad T_L = 5N.m$$

Initially the drive system is working in steady-state. The steady-state speed is 955 rpm. Now the drive is disconnected from the main supply. Calculate time taken by the drive to stop (consider that the stopping process is finished at 0.05 of the final value).

a. $9.5\ s$
b. $19\ s$
c. $90.725\ s$
d. $28.5\ s$

Correct answer: $9.5\ s$

Braking time without brake:

$$t_{stopping} = J_\Sigma \frac{\omega_L}{T_L} = 0.5 * \frac{\frac{955}{9.55}}{5} * (1 - 0.05(\text{of the final value}))$$

193. In case of star/delta starting method, starting current of three-phase squirrel cage motor delta connected stator windings is 173 A. For star connected stator windings, the starting current is:

a. 519 A
b. 260 A
c. 100 A
d. 57.67 A

Correct answer: 100 A

194. In case of star/delta starting method, critical torque of three-phase squirrel cage motor delta connected stator windings is 150 N.m. For star connected stator windings, the critical torque is:

a. $50\ N.m$
b. $86.7\ N.m$
c. $450\ N.m$
d. $259.5\ N.m$

Correct answer: $50\ N.m$

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$$\frac{150}{3} = 50$$

195. Plugging braking of induction motors is obtained by:

a. interchanging connections of any two phases of stator with respect to supply terminals
b. dissipating the stored energy of rotating mass in an electrical resistance
c. reducing the stator voltage to a value less than the nominal value
d. connecting of two phases of stator with dc voltage supply

Correct answer: interchanging connections of any two phases of stator with respect to supply terminals

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Unit Eight: DC Motors Automatic Starting, Braking and Reversing Principles

196. The suitable principle for automatic starting, braking and reversing of elevators is:

a. Displacement (position) principle
b. Speed principle
c. Time principle
d. Current principle

Correct answer: Displacement (position) principle

197. Centrifugal relay is used for automatic starting and braking of electric motors according to:

a. Displacement principle
b. Time principle
c. Speed principle
d. Current Principle

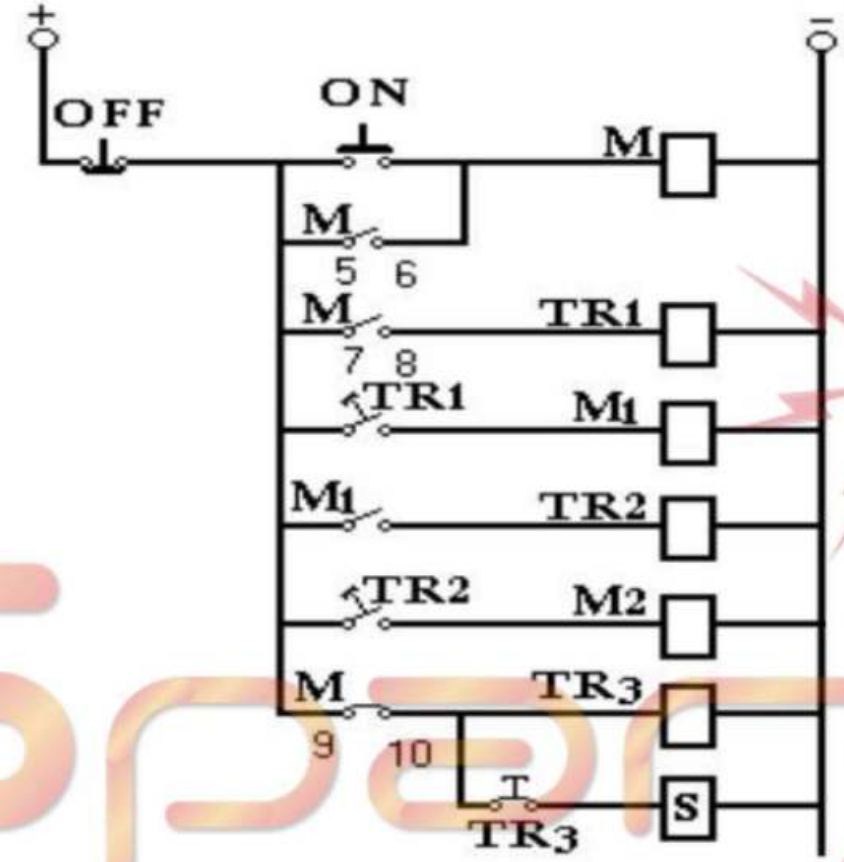
Correct answer: Speed principle

198. Normally closed contact M(9-10) in control circuit used to prevent .....of drive system:

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a. Indirect starting in two steps
b. Indirect starting in one step
c. Dynamic braking during starting
d. Direct starting

Correct answer: Dynamic braking during starting

199. Push button ON is:

a. Normally closed contact used to deactivate the circuit
b. Normally closed contact used to activate the circuit
c. Normally open contact used to deactivate the circuit
d. Normally open contact used to activate the circuit

Correct answer: Normally open contact used to activate the circuit

200. Push button OFF is:

a. Normally closed contact used to activate the circuit
b. Normally open contact used to activate the circuit
c. Normally closed contact used to deactivate the circuit
d. Normally open contact used to deactivate the circuit

Correct answer: Normally closed contact used to deactivate the circuit

201. Tacho-Generator is used for automatic starting and braking of electric motors according to:

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OR

Speed Generator is used for automatic starting and braking of electric motors according to:

a. Time principle
b. Current principle
c. Displacement principle
d. Speed principle

Correct answer: Speed principle

202. One of the most superlative advantages of dc drives compared to ac drives is:

a. Presence of commutator in dc motors
b. Cheap control for high-performance drives
c. Ability to have high softness without expensive control
d. Ability to be controlled and coupled with low-speed loads without transmission device

Correct answer: Ability to have high softness without expensive control

Starting torque and speed range – these are both much higher for DC – an inherent characteristic. AC motors require more sophisticated and expensive controls to be equivalent, possible even encoder feedback from the motor

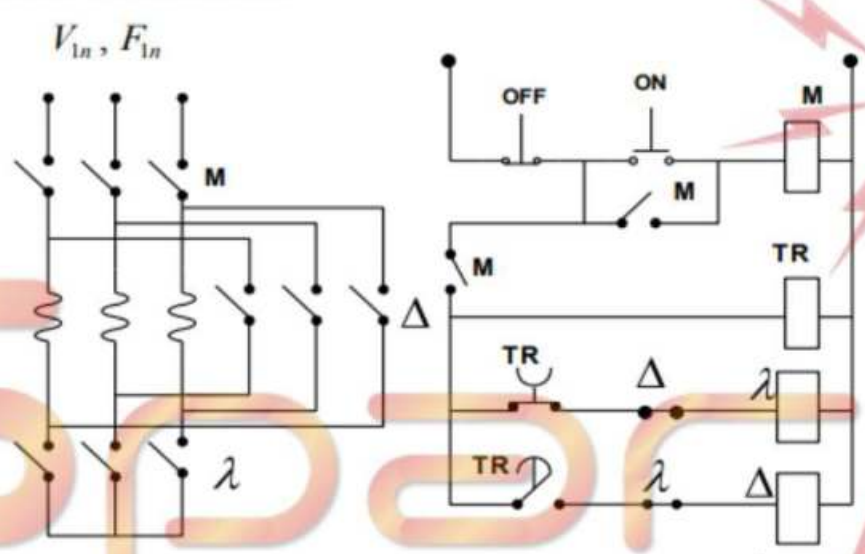
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Unit Nine: AC Motors Automatic Starting, Braking and Reversing

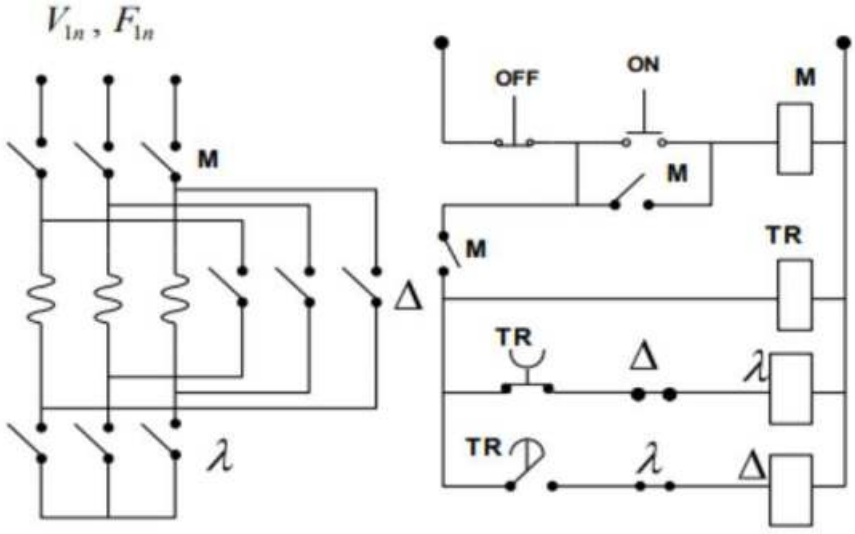
203. In shown automatic starting electrical circuit of three-phase squirrel cage ac motor, the time delay on timer normally closed contact used for:



a. Delta/ star starting sequence
b. Star /delta starting sequence
c. Squirrel cage stator windings short circuiting protection
d. Power supply short circuiting protection

Correct answer: Power supply short circuiting protection

204. In shown Star/Delta automatic starting electrical circuit of three-phase squirrel cage ac motor, the normally closed contacts of star and delta contactors are used to prevent:



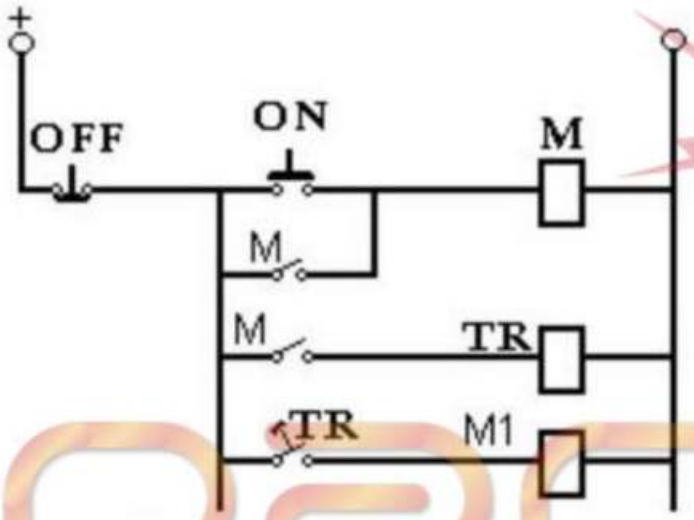
a. Control Circuit short circuiting

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b. Squirrel cage ac motor overloading
c. Squirrel cage stator windings short circuiting
d. Power circuit power supply short circuiting

Correct answer: Power circuit power supply short circuiting

205. The given control circuit performs ..... for the drive system:



a. Indirect starting in one step and mechanical braking
b. Direct starting and dynamic braking
c. Indirect starting in two steps and mechanical braking
d. Direct starting and mechanical braking

Correct answer: Indirect starting in one step and mechanical braking

206. When controlling the squirrel-cage motor speed by using a solid-state voltage controller, the permissible (allowed) load torque is:

OR

When controlling the squirrel-cage motor speed by using a impulse regulator (chopper), the permissible (allowed) load torque is:

a. $T_{per} = T_n S_n / S$
b. $T_{per} = T_n S / S_n$
c. $T_{per} = T_n S_{sc} / S$
d. $T_{per} = T_{cr} S_{cr} / S$

Correct answer: $T_{per} = T_n S_n / S$

207. When controlling the ac motor speed by using autotransformer, the permissible (allowed) load torque is:

a. $T_{per} = T_n S / S_n$
b. $T_{per} = T_n S_n / S$
c. $T_{per} = T_{cr} S_{cr} / S$
d. $T_{per} = T_n S_{sc} / S$

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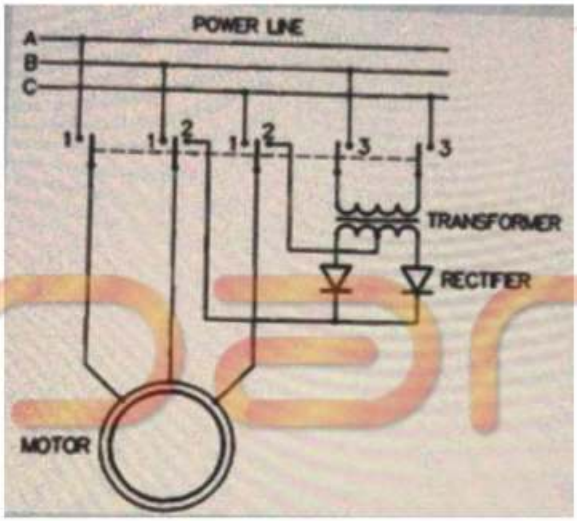
Correct answer: $T_{per} = T_n S_n / S$

208. Three-phase ac motor reversing is accomplished by:

a. Switching off two phases
b. Changing the polarity of load torque
c. Interchanging between two phases only
d. Interchanging between three phases

Correct answer: Interchanging between two phases only

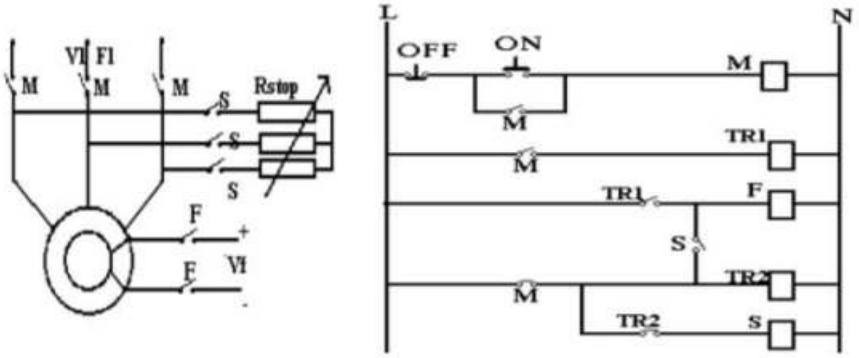
209. The following circuit is used for:



a. Reversing of induction motor
b. Automatic starting by inserting reactance into the stator circuit
c. Dynamic braking
d. Plugging braking

Correct answer: Automatic starting by inserting reactance into the stator circuit

210. The normally open contact (S) used in shown control circuit to:



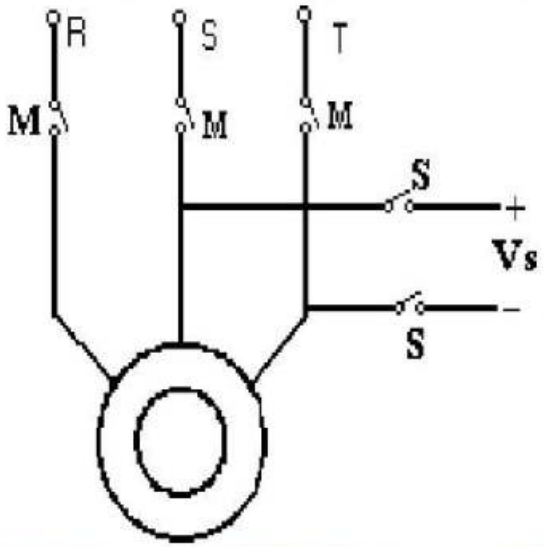
a. Keep synchronization during dynamic braking
b. Keep synchronization during motoring mode
c. Prevent dynamic braking during starting process
d. Keep DC source connected during dynamic braking

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Correct answer: Keep DC source connected during dynamic braking

211. The given power circuit of three-phase squirrel cage ac motor represents:



a. Direct starting and dynamic braking
b. Indirect starting and dynamic braking
c. Direct starting and mechanical braking
d. Indirect starting and mechanical braking

Correct answer: Direct starting and dynamic braking

212. Star-delta stator windings connection starting method used when the ac motor runs at natural characteristic of:

a. Star connection with 400 V, line voltage
b. Delta connection with 400 V, line voltage
c. Star connection with 230 V, phase voltage
d. Delta connection with 230 V, line voltage

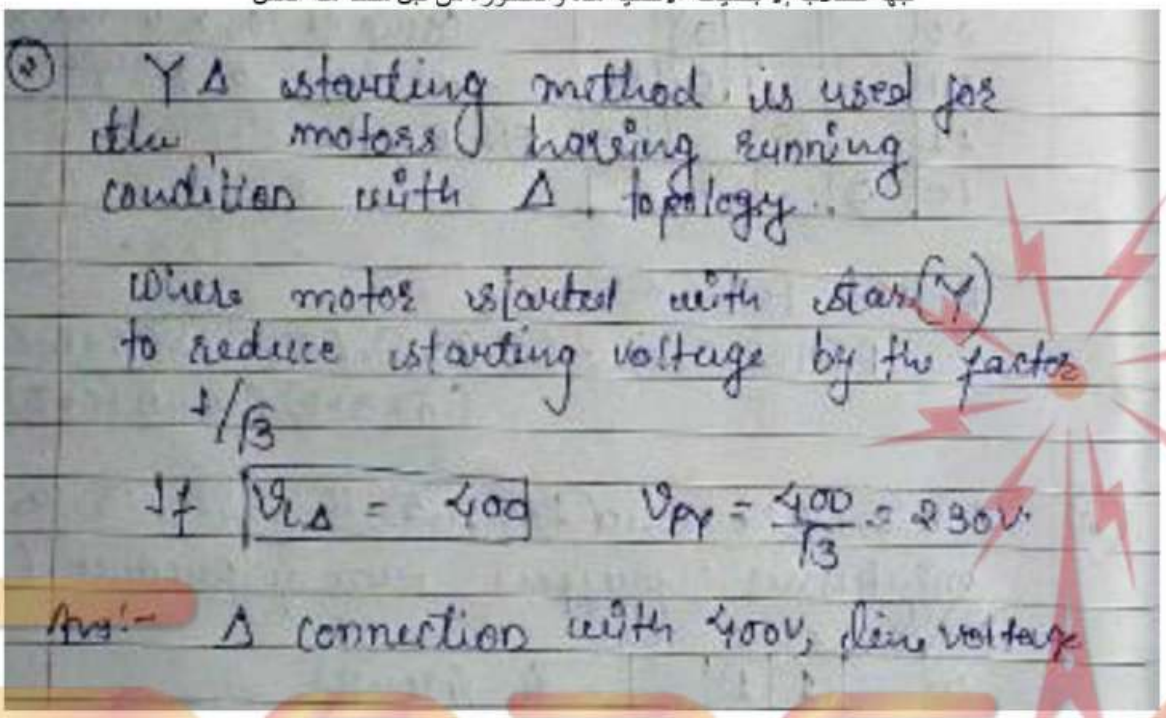
Correct answer: Delta Connection with 400V, Line Voltage

Note that: Star-Delta Starting Method:

This method is used in the case of motors which are built to run normally with a delta-connected stator winding. It consists of a two-way switch which connects the motor in star for starting and then in delta for normal running. Under the running condition the starter switch will be thrown to run position (Delta Connected starter) hence the stator winding should also be of Delta connected in order to withstand . we know that in 3-Phase motors irrespective of type of winding the line voltage will be around 400V, the stator if the 3-phase induction motor in this case is Delta connected and hence Line Voltage = Phase Voltage

A star delta starter will start a motor with a star connected stator winding. When motor reaches about 80 percent of its full load speed, it will begin to run in a delta connected stator winding.

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



② Y Δ starting method is used for the motors having running condition with Δ topology.

Where motor is started with star (Y) to reduce starting voltage by the factor $1/\sqrt{3}$

If $V_{L\Delta} = 400$ $V_{PY} = \frac{400}{\sqrt{3}} = 230V$.

Ans:- Δ connection with 400V, line voltage

or

→ As the winding is star connected, the voltage across each phase of winding is $V_L/\sqrt{3}$.

$$\frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} \simeq 230V.$$

→ During running condition (delta connection) the voltage across each phase of the winding is

$$V_L = 400V$$

✓a) Delta connection with 400V, line voltage.

or

بسواعدنا نبنيها

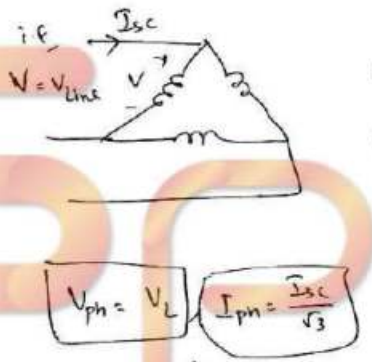
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Given,

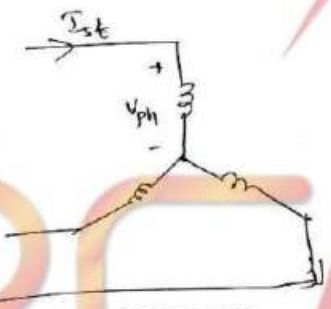
"Star - delta starter".

In this we connect motor winding in star(Y) at the time of starting.

i.e, If motor is designed to operate in Δ, to limit the starting current, we connect it in Y at starting.

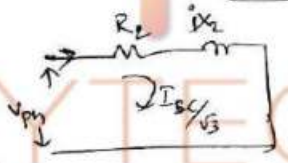


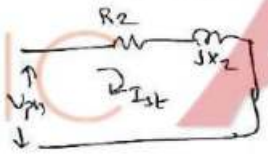
at the time of starting



$V_{ph} = V_L$, $I_{ph} = \frac{I_{sc}}{\sqrt{3}}$

$V_{ph} = \frac{V_L}{\sqrt{3}}$, $I_{ph} = I_{st}$





$$V_{ph} = \frac{I_{sc}}{\sqrt{3}} (Z_2)$$

$$V_{ph} = I_{st} \cdot (Z_2)$$

$$Z_2 = \frac{\sqrt{3}\, V_L}{I_{sc}}$$

$$\frac{V_L}{\sqrt{3}} = I_{st} \left[\frac{\sqrt{3}\, V_L}{I_{sc}} \right]$$

$$\frac{1}{\sqrt{3}} = \sqrt{3} \left(\frac{I_{st}}{I_{sc}} \right)$$

$$\therefore I_{st} = \frac{1}{3} I_{sc} \quad -(1)$$

from (1),

$\therefore$ I_{st} is reduced by 3-times.

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→ It is only used for motors with delta connection only.

∴ Delta connection with 400 V, line voltage.

Ans: option: a //

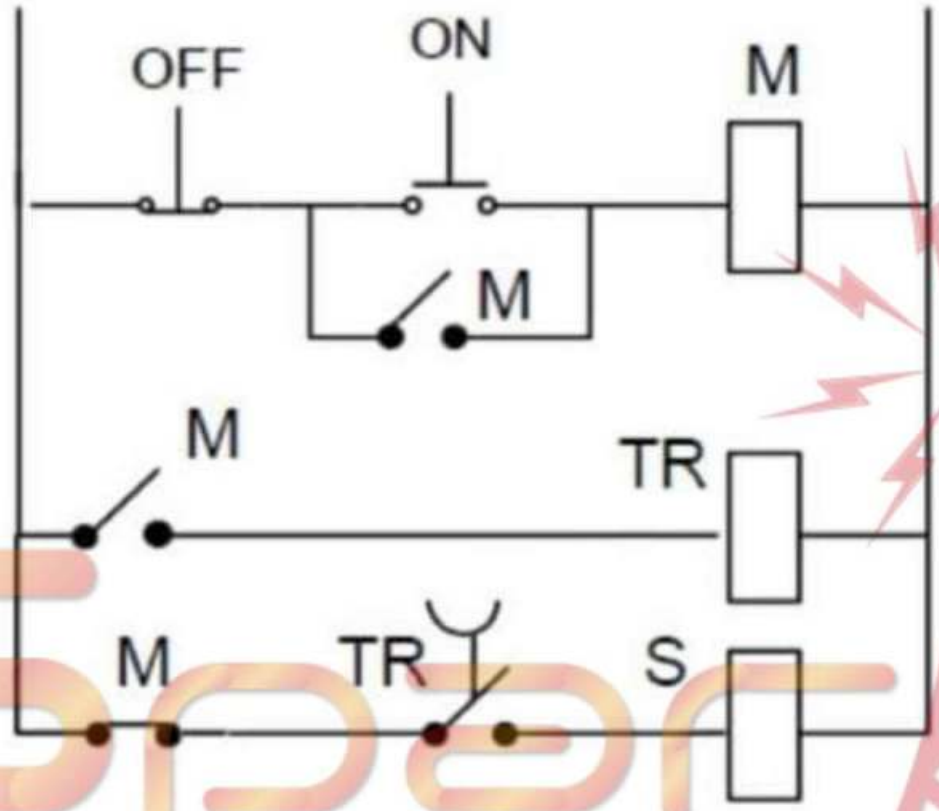
213. Time principle starting in one step and mechanical braking control circuit uses: (Time principle starting in one step and mechanical braking uses:)

a. Two time delay on timers
b. One time delay off timer
c. Two time delay off timers
d. One time delay on timer

Correct answer: One time delay on timer

214. The shown control circuit performs:

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل



a. Indirect starting in one step using time delay on timer and mechanical braking
b. Direct starting and dynamic braking using time delay off timer
c. Indirect starting in one step using time delay off timer and mechanical braking
d. Direct starting and dynamic braking using time delay on timer

Correct answer: Direct starting and dynamic braking using time delay off timer

215. The functions of the shunt contactor (Sh) in the diagrams below are:

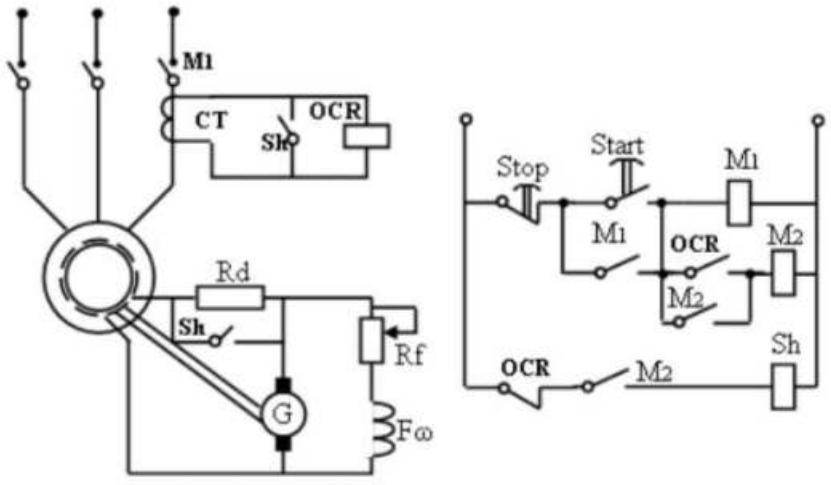


Figure: Starting of synchronous motor with an over current relay and exciter

a. to prevent the OCR from energizing when the motor is getting into synchronization, in addition, to pass the whole generator voltage to the field winding

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

b. to prevent the in-rush starting current from acting on the motor windings
c. to prevent the motor from getting out the synchronization at load fluctuation
d. to protect the motor from phase-failure

Correct answer: to prevent the OCR from energizing when the motor is getting into synchronization, in addition, to pass the whole generator voltage to the field winding

Note that: As shunt contactor is in parallel to OCR it prevents OCR from energized when motor getting into synchronization

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Unit Ten: DC Motors Open- Loop Speed Control

216. The natural and regulated characteristics of separately excited dc motor have same hardness, when using:

a. Adding resistance in series with field
b. Motor flux variation
c. Adding resistance in series with armature
d. Armature voltage variation

Correct answer: Armature voltage variation

217. The natural and regulated characteristics of separately excited dc motor have same speed drop, when using:

a. Adding resistance in series with armature
b. Adding resistance in series with field
c. Motor flux variation
d. Armature voltage variation

Correct answer: Armature voltage variation

218. Speed control method by varying the flux of dc motors is used to regulate the speed of:

a. Constant load power machineries
b. Constant load torque machineries
c. Variable losses drive
d. Constant losses drive

Correct answer: Constant load power machineries
This method is used to control the speed of production machinery.

219. Controlling the separately excited dc motor speed above natural characteristic achieved by:

a. Increasing armature voltage more than nominal value
b. Decreasing armature voltage less than nominal value
c. Decreasing motor flux less than nominal value
d. Increasing motor flux more than nominal value

Correct answer: Decreasing motor flux less than nominal value

220. Speed control of series field dc motor by adding resistance in series with armature circuit is suitable for:

a. Drilling machine
b. Pump
c. Crane
d. Compressor

Correct answer: Crane

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

221. Series field dc motor speed control by adding resistance in parallel with armature circuit will lead to increase:

a. Motor flux
b. Armature current
c. Motor speed
d. No-load speed

Correct answer: Armature current

222. Separately excited dc motor speed control by adding resistance in series with armature circuit keeps the same value of:

a. speed drop
b. No-load speed
c. Torque-speed characteristic hardness
d. Short circuit current

Correct answer: No-load speed

223. Speed regulation of separately excited dc motor by motor flux variation, the motor flux should not exceed nominal value to protect the motor from:

a. Under loading
b. Over heating
c. Over voltage
d. Over loading

Correct answer: Over heating

224. Speed regulation of separately excited dc motor by Armature voltage variation, the armature voltage should not exceed nominal value to protect the motor from:

a. Over heating
b. Under loading
c. Over loading
d. Over voltage

Correct answer: Over heating

225. Speed regulation of series field dc motor by Armature voltage variation, the armature voltage should not exceed nominal value to protect the motor from:

a. Under loading
b. Over voltage
c. Over heating
d. Over loading

Correct answer: Over heating

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Unit Eleven: AC Motors Open-Loop Speed Control

226. Star- double star stator windings connection torque-speed characteristics have:

a. $\omega_{o\lambda\lambda} < \omega_{o\lambda}\ and\ T_{\max\lambda\lambda} = T_{\max\lambda}$
b. $\omega_{o\lambda\lambda} > \omega_{o\lambda}\ and\ T_{\max\lambda\lambda} < T_{\max\lambda}$
c. $\omega_{o\lambda\lambda} > \omega_{o\lambda}\ and\ T_{\max\lambda\lambda} = T_{\max\lambda}$
d. $\omega_{o\lambda\lambda} > \omega_{o\lambda}\ and\ T_{\max\lambda\lambda} > T_{\max\lambda}$

Correct answer: $\omega_{o\lambda\lambda} > \omega_{o\lambda}\ and\ T_{\max\lambda\lambda} = T_{\max\lambda}$

227. Delta- double star stator windings connection torque-speed characteristics have:

a. $\omega_{o\lambda\lambda} > \omega_{o\Delta}\ and\ T_{\max\lambda\lambda} = T_{\max\Delta}$
b. $\omega_{o\lambda\lambda} < \omega_{o\Delta}\ and\ T_{\max\lambda\lambda} = T_{\max\Delta}$
c. $\omega_{o\lambda\lambda} > \omega_{o\Delta}\ and\ T_{\max\lambda\lambda} > T_{\max\Delta}$
d. $\omega_{o\lambda\lambda} > \omega_{o\Delta}\ and\ T_{\max\lambda\lambda} < T_{\max\Delta}$

Correct answer: $\omega_{o\lambda\lambda} > \omega_{o\Delta}\ and\ T_{\max\lambda\lambda} < T_{\max\Delta}$

228. The ac motor $V_1/{f_1}^2 =$ constant speed control principle is suitable for:

a. Turning machines
b. Milling machines
c. Elevators
d. Compressors

Correct answer: Compressors

229. The $v/(f)^2 =$ constant speed control is suitable for:

a. Turning machines
b. Milling machines
c. Fans
d. Conveyor belts

Correct answer: Fans

230. Delta-double star stator windings connection suitable for speed control of:

a. Pumping machineries
b. Production machineries
c. Transporting machineries
d. Ventilating machineries

Correct answer: Production machineries

231. AC motor speed control by varying stator voltage with fixed frequency leads to have constant:

a. Motor critical torque
b. Motor starting torque

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

c. Motor loading capacity
d. Load torque

Correct answer: Motor loading capacity

232. AC motor speed control using auto-transformer leads to have constant:
a. Starting torque
b. Motor maximum torque
c. Short circuit torque
d. Critical slip
e. Motor electromagnetic torque

Correct answer: Critical slip

The direction of peed control in this way is below the natural characteristic, with constant critical slip value.

233. Speed regulation of three-phase ac motor operating under nominal load, variation of stator voltage and stator frequency is implemented as following:
a. $V_1 > V_{1N}\ and\ f_1 < f_{1N}$
b. $V_1 > V_{1N}\ and\ f_1 > f_{1N}$
c. $V_1 < V_{1N}\ and\ f_1 < f_{1N}$
d. $V_1 < V_{1N}\ and\ f_1 > f_{1N}$

Correct answer: $V_1 < V_{1N}\ and\ f_1 < f_{1N}$

234. AC motor speed control by varying stator voltage with fixed frequency is suitable for: (AC motor speed control by varying stator voltage with fixed frequency is suitable for)
a. Conveyor belts
b. Milling machines
c. Compressors
d. Drilling machines
e. Ventilating machineries
f. Transporting machineries
g. Turning machines (Turning machineries)

Correct answer: Turning machines

Note that: Stator voltage is suitable where load torques decrease with speed.

Conveyor and compressor are constant torque machine.

Milling machine torque is inversely proportional to speed.

So, in Turning machines stator voltage control with fixed frequency is used

235. AC motor speed control by varying stator voltage with fixed frequency is suitable for:
a. Cutting machine
b. Escalator
c. Compressor
d. Milling machines

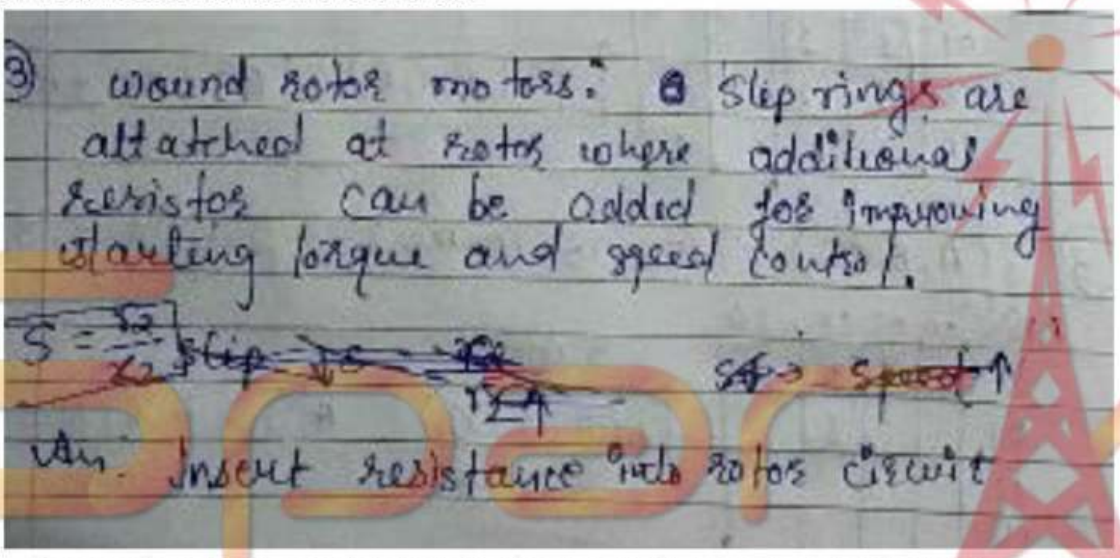
Correct answer: Milling machines

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

236. For wound rotor motors, in order to improve the speed control range when you use the stator voltage method, you have to:

a. insert reactance into the stator circuit
b. insert resistance into the stator circuit
c. decrease the load down the nominal value
d. insert resistance into the rotor circuit

Correct answer: insert resistance into the rotor circuit

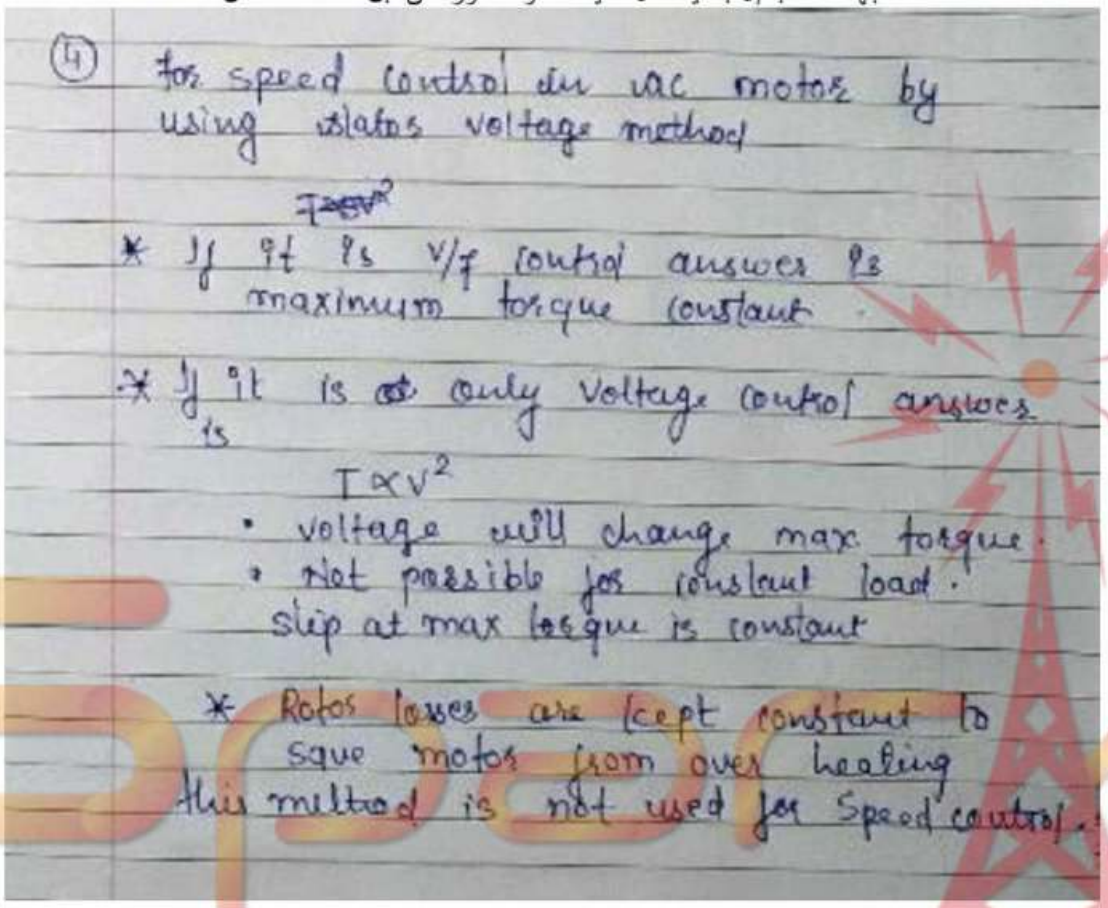


237. For speed control in ac motors by using the stator voltage method, you have to keep:

a. the critical slip constant
b. the rotor losses constant
c. the maximum torque constant
d. the nominal load constant

Correct answer: the critical slip constant

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238. The ac motor $V_1/f_1 =$ constant speed control principle is suitable for: *(The ac motor v/f =* constant speed control principle is suitable for:)

a. Milling machines
b. Compressors
c. Turning machines
d. Conveyor belts

Correct answer: Conveyor belts

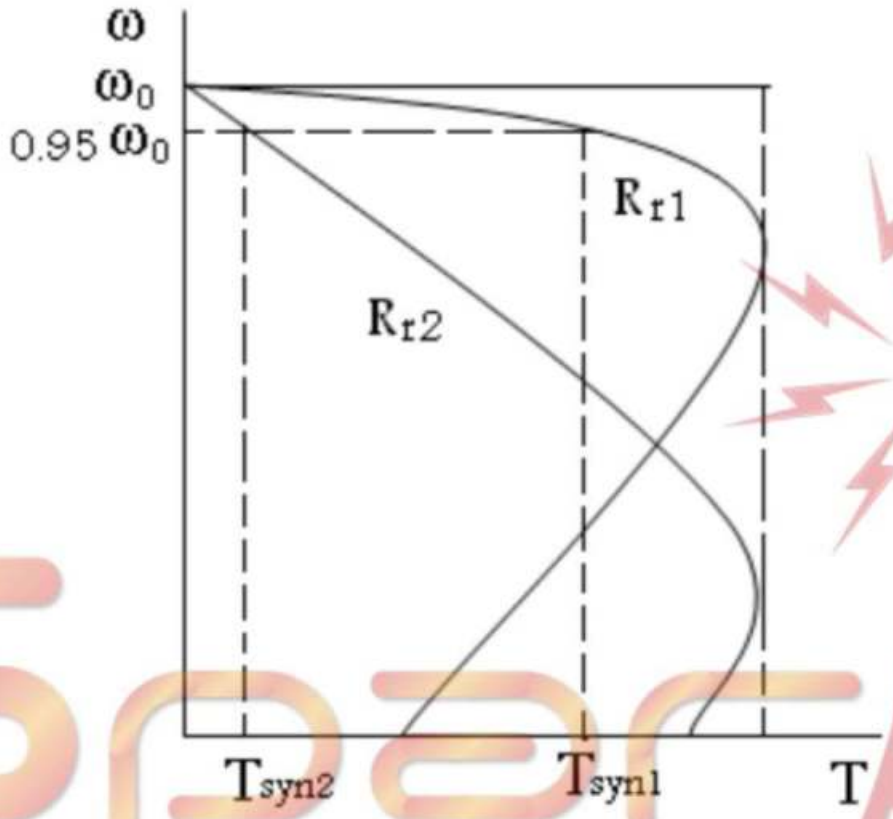
239. Allowed load torque of ac motor speed control method depends on the variation of:

a. Speed direction
b. Motor speed
c. Motor loading capacity (λ)
d. Motor flux
e. Armature voltage

Correct answer: Motor loading capacity (λ)

240. The following characteristics represent:

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a. mechanical characteristics of synchronous motor with different values of rotor circuit resistance
b. characteristics of wound rotor induction motor for different values of rotor circuit resistance
c. characteristics of induction motor for constant load when V/F control is used
d. automatic starting characteristics of synchronous motor

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Correct answer: characteristics of wound rotor induction motor for different values of rotor circuit resistance

241. A 50Hz, 22 kW, 2970 rpm induction motor has a speed of 2700 rpm when the rotor resistance control is used. Find the output power:

a. 22 kW
b. 20 kW
c. 21.77 kW
d. 18 kW

Correct answer: 20 kW

$$22kW * \frac{2700}{2970} = 20kW$$

242. Elevator torque-speed characteristic is:

a. Non-Linearly rising torque characteristic
b. Linearly rising torque characteristics
c. Non-Linearly decreasing torque characteristic
d. Constant torque characteristic

Correct answer: Constant torque characteristic

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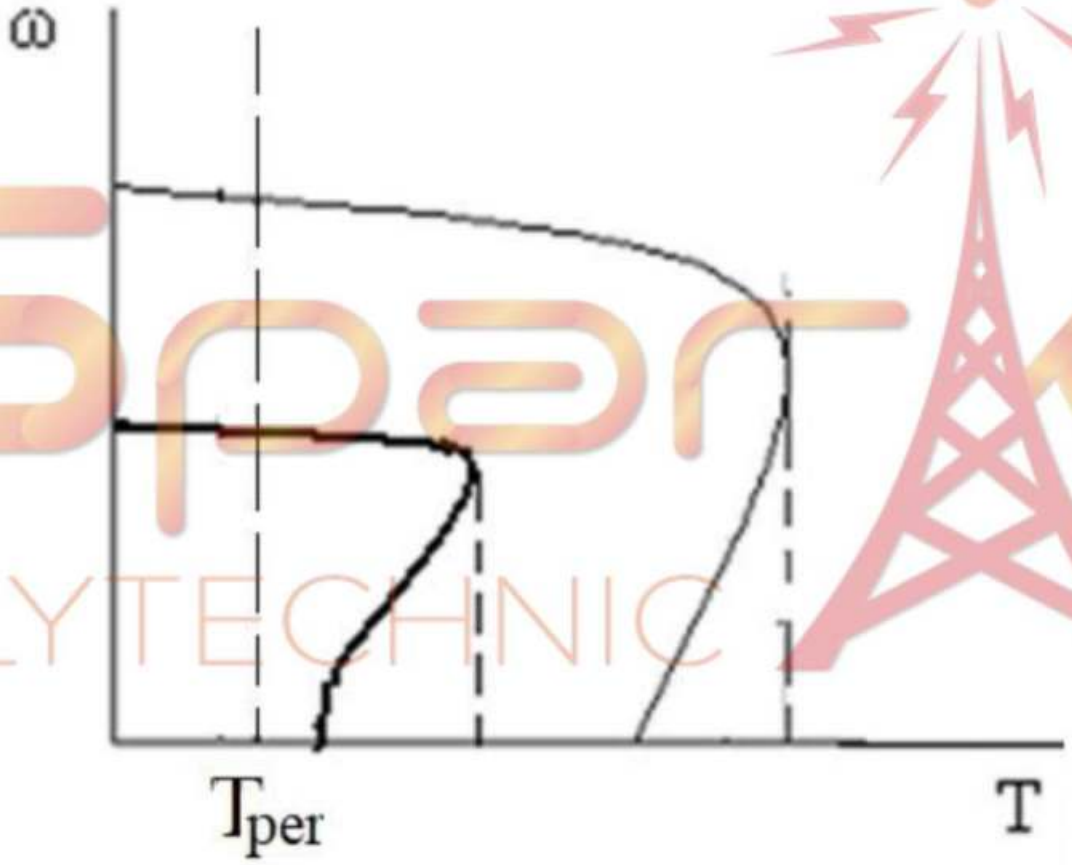
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243. Higher starting torque in induction motors can be obtained by:

a. V/f control or inserting resistance into the rotor circuit
b. pole changing control
c. stator voltage control by using an autotransformer or chopper
d. inserting resistance into the stator circuit

Correct answer: V/f control or inserting resistance into the rotor circuit

244. Characteristics below are:



a. regulated characteristics of induction motor when V/F method is used
b. regulated characteristics of Induction motor when pole changing method is used
c. regulated characteristics of induction motor when y/A method is used
d. regulated characteristics of induction motor when pole changing or V/F methods are used

Correct answer: regulated characteristics of induction motor when V/F method is used

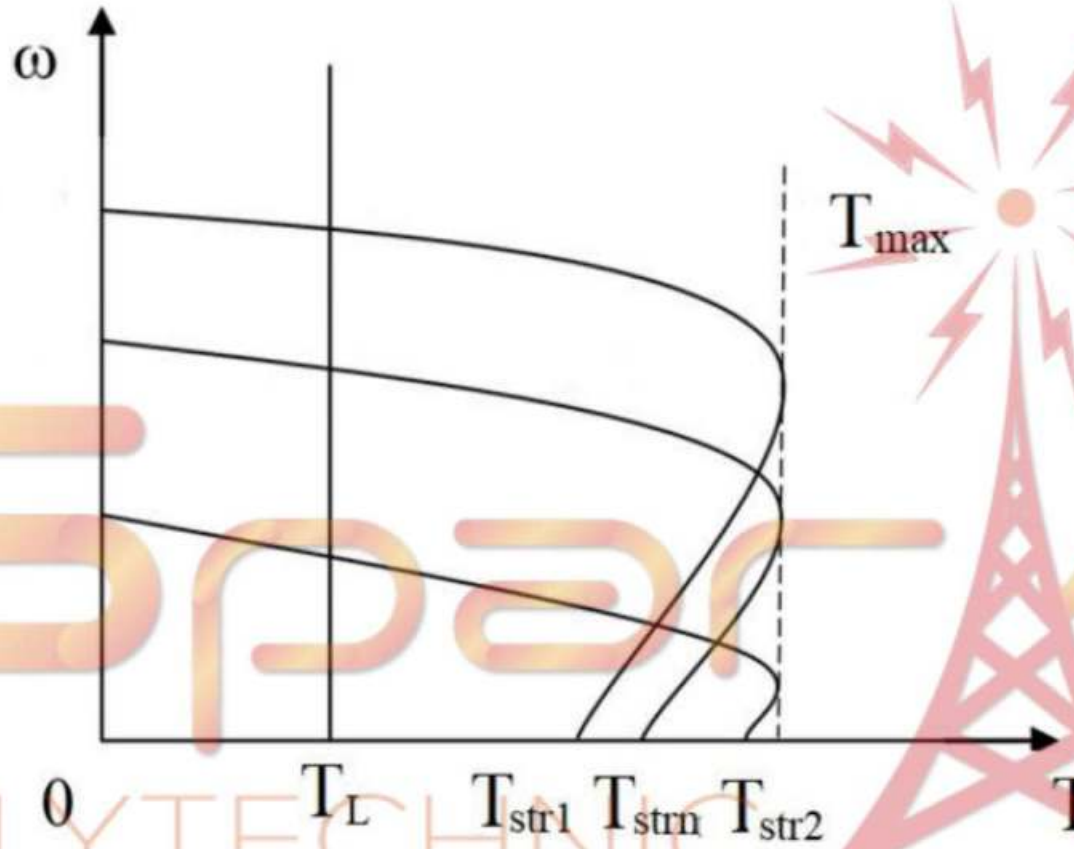
245. Speed regulation of three-phase ac motors by stator voltage variation with nominal frequency, the stator voltage should not exceed nominal value to protect the motor from:

a. Over heating
b. Under loading
c. Over loading
d. Over voltage

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Correct answer: Over heating

246. Characteristics below are:



a. regulated characteristics of induction motor for V/F control
b. regulated characteristics of induction motor when pole changing method is used
c. regulated characteristics of induction motor for NIF or rotor resistance control
d. regulated characteristics of induction motor for rotor resistance control

Correct answer: regulated characteristics of induction motor for V/F control

247. A 50Hz, 22 kw, 2970 rpm induction motor has speed of 2700 rpm when it is controlled by stator voltage controller. Find the permissible load torque:

a. $64.3N.m$
b. $0.7074N.m$
c. $70.74N.m$
d. $7.074N.m$

Correct answer: $70.74N.m$

$$\frac{22000}{\frac{2970}{9.55}} = 70.74N.m$$

248. A 50Hz, 220V, 2970 rpm induction motor has a speed of 2700 rpm when it is controlled by V/F controller. Find the stator frequency if the load is constant (crane):

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a. $45.45\ Hz$
b. $40\ Hz$
c. $35\ Hz$
d. $45.5\ Hz$

Correct answer: $45.45\ Hz$

$$\frac{2700}{2970} * 50 = 45.45\ Hz$$

249. A 50Hz, 220V, 2970 rpm induction motor has a speed of 2700 rpm when it is controlled by V/F controller. Find the stator voltage if the load is constant (crane):

a. $190\ V$
b. $199\ V$
c. $200.2\ V$
d. $210\ V$

Correct answer: $200.2\ V$

$$\frac{2700}{2970} * 220 \cong 200.2\ V$$

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Previous Exams Forms

نماذج اختبارات سابقة

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ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

Fundamentals of Electric Drive Mid Term Exam

Number of Questions: 3 Number of pages: 5

Faculty of Engineering Technology

Mechatronics Engineering Department

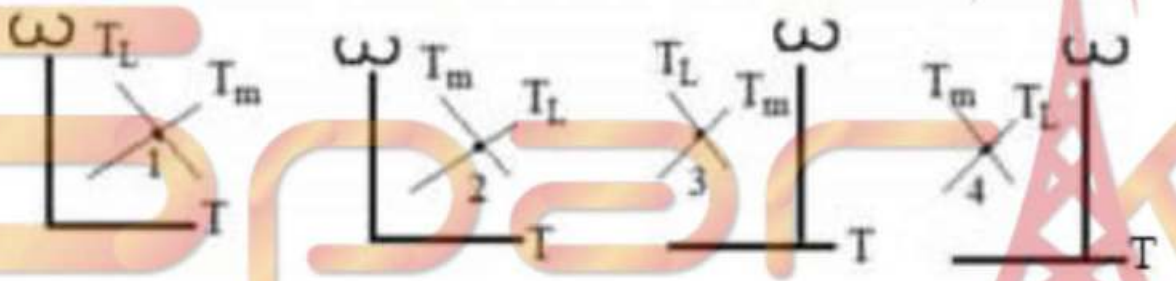
50 Minutes **30 Marks**

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Q1: (20 marks)

Answer the following questions by circling the right answer in the table at the end of this question

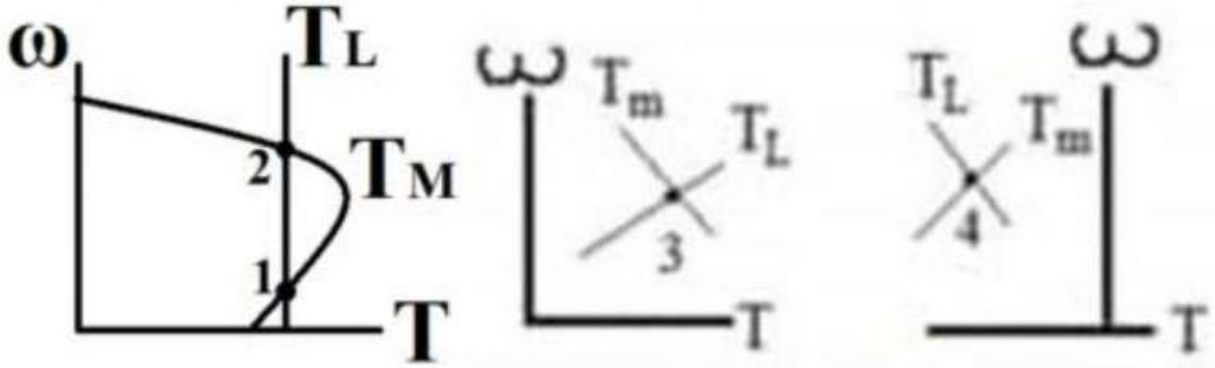
1. Stable points of equilibrium are:



a. 1 only
b. 2 only
c. 3 only
d. 2 and 4

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2. Unstable points of equilibrium are:



a. 1 and 2
b. 1 and 3
c. 1 and 4
d. 2 and 3

3. If the torque developed by the motor is $T = T_L + J\frac{d\omega}{dt}$, the drive is:

a. decelerated

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Fundamentals of Electric Drive Mid Term Exam

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b. accelerated
c. working in S.S.P (Steady-State Point)
d. None of these

4. In electric drive system, the main function of the controller is:
a. Matching the motor and power converter to meet the load requirements
b. Adapting to the changes of the environment only
c. Sensing the feedback variables only
d. Controlling the power converter only

5. Loads which require coupling with the motor shaft through transmission mechanical system (TMS) are:
a. Loads which need low speeds and high torques
b. Loads which need high speeds and low torques
c. Loads which need high speeds and high torques
d. Loads which need low speeds and low torques

6. The electric drive system with following motor torque and load torque expressions is stable:
a. $T = 10 + 2\omega,\ T_L = 10$
b. $T = 25 - 3\omega,\ T_L = 10$
c. $T = 15 + 4\omega,\ T_L = 15 + \omega$
d. $T = 25 + 5\omega,\ T_L = 2 + 2\omega^2$

7. The drive system output power is:
a. Thermal power
b. Electrical power
c. Mechanical power
d. Electromagnetic power

8. Component determines the motor starting time is:
a. motor inertia
b. motor load
c. speed change rate
d. all of these

9. The DC motor that should never be started on no-load is:
a. series
b. compound
c. shunt
d. permanent magnet

10. The active load torque industrial machinery is:

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a. Pump
b. Crane
c. Ventilator
d. Turning machine

11. Production machineries torque-speed characteristic is:
a. Decreasing linear characteristic
b. Increasing non-linear characteristic
c. Decreasing non-linear characteristic
d. Independent on speed

12. For D.C. series motor which of the following is incorrect:
a. suitable for heavy duty starting
b. torque varies as armature current
c. has non-linear characteristics
d. has hard characteristics

13. Crane motor lowers load of 10 N.m connected to the motor through gearbox, which has gear ratio of 5. If the system efficiency equals 0.9, the referred load torque to motor shaft is:
a. 2.22 N.m
b. 50 N.m
c. 2 N.m
d. 1.8 N.m

14. Electric motor has moment of inertia of 1 $kg.m^2$ connected through gearbox of 5 gear ratio to drum with moment of inertia 4 $kg.m^2$. The drive system total (equivalent) moment of inertia equals:
a. 21 $kg.m^2$
b. 6 $kg.m^2$
c. 1.16 $kg.m^2$
d. None of these

15. When varying load torque, separately excited dc motor has constant:
a. Speed
b. Flux
c. Back emf
d. Armature current

16. If the motor works in the 4th quadrant. This mode of operation is a:
a. reverse braking mode
b. forward regenerative mode
c. motoring mode

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d. reverse motoring mode

17. Separately excited dc motor with following ratings:

$$P_{2N} = 10\text{kW}, V_a = V_f = 220\text{ V}, \omega_s = 100\frac{\text{rad}}{\text{s}}, R_a = 0.4\Omega, I_{an} = 50\text{ A}, k\emptyset_N = 2\text{V.s}$$

Nominal motor torque is:

a. $50\ N.m$
b. $100\ N.m$
c. $12.5\ N.m$
d. $27.5\ N.m$

18. Separately excited dc motor with following ratings:

$$P_{2N} = 10\text{kW}, V_a = V_f = 220\text{ V}, \omega_s = 100\frac{\text{rad}}{\text{s}}, R_a = 0.4\Omega, I_{an} = 50\text{ A}, k\emptyset_N = 2\text{V.s}$$

Nominal angular speed drop is:

a. 100 rad/s
b. 20 rad/s
c. 10 rad/s
d. 80 rad/s

19. Plugging mode in the 4th quadrant is very dangerous mode due to overheating, but it used under nominal load in:

a. Pump machinery
b. Drilling machine
c. Escalator
d. Crane

20. Adding resistance in series with armature of dc motor will cause:

a. Additional heat in the armature circuit
b. Increasing of motor torque
c. Increasing of short-circuit current
d. Increasing of starting torque

Ques	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans																				

Q2) Answer with yes or no (5 marks)

1. All dc motors are self-excited ( )
2. Brushed dc motors are not used in modern electric vehicles ( )
3. Turning machine is an active load ( )

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Fundamentals of Electric Drive Mid Term Exam

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4. Absolute hard speed-torque characteristics can be obtained in dc motor drive in closed-loop systems ( )
5. Squirrel-Cage motors are the most widely used in industry ( )

Q3) (5 marks)

Calculate the starting time of the drive system with the following data:

$$J_\Sigma = 1.0\ kg.m^2, \quad T_m = 15 + 0.5\omega, \quad T_L = 5 + 0.6\omega$$

Consider final speed $\omega_2 = 0.95\ \omega_{ss}$

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Fundamentals of Electric Drive Mid Term Exam

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Correct answers:

Q1)

Question Number	Correct answer symbol	Correct answer
1	d	2 and 4
2	c	1 and 4
3	b	accelerated
4	a	Matching the motor and power converter to meet the load requirements
5	b	Loads which need high speeds and low torques
6	b	$T = 25 - 3\omega, \quad T_L = 10$
7	c	Mechanical power
8	d	all of these (motor inertia, motor load, speed change rate) The motor torque equation is given by: $T = T_L + T_{dyn}$ Where: T: Motor Torque T_L: Load Torque T_{dyn}: Dynamic Acceleration, which is given by: $T_{dyn} = J_\Sigma \frac{d\omega}{dt}$ $T = T_L + J_\Sigma \frac{d\omega}{dt}$ $T - T_L = J_\Sigma \frac{d\omega}{dt}$ Rearrange the last equation and let dt on one side alone $dt = \frac{J_\Sigma}{T - T_L} d\omega$ And now integrate both two sides $\int_{t_o}^{t} dt = \int_{w_o}^{w} \frac{J_\Sigma}{T - T_L} d\omega$ $t_{st} = \int_{w_o}^{w} \frac{J_\Sigma}{T - T_L} d\omega$

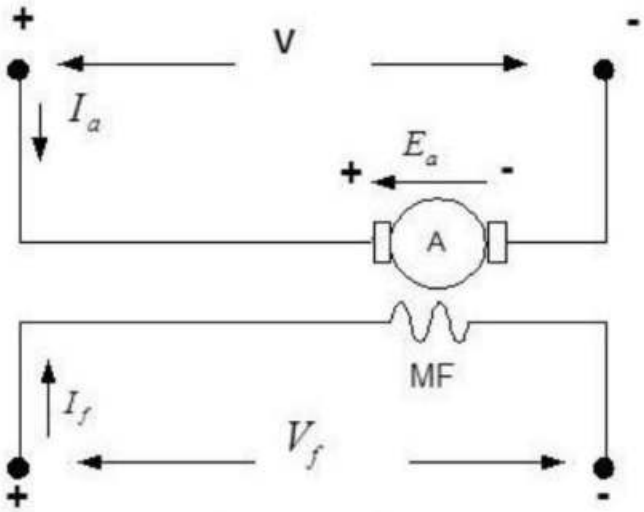
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Fundamentals of Electric Drive Mid Term Exam

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		According to the last equation, the starting time depends on the following: a. motor moment of Inertia (motor inertia) J_m b. Motor Torque T c. Load Torque T_L d. Motor angular speed rate of change $d\omega$ e. Initial operating motor angular speed ω_o, usually motor rotates from rest $\omega_o = 0\frac{rad}{s}$ f. Steady-state operating motor angular speed ω g. Motor efficiency η_m, $\eta_m = J_m\frac{\omega^2}{2}$
9	a	series
10	b	Crane
11	c	Decreasing non-linear characteristic
12	d	has hard characteristics
13	a	2.22 N.m $\frac{10\text{ N. m}}{5 * 0.9}$
14	c	$1.16\ kg.m^2$ $J_\Sigma = J_m + \frac{J_D}{i^2} = 1 + \frac{4}{5^2} = 1 + 0.16 = 1.16\ kg.m^2$
15	b	Flux Figure below shows the electrical circuit of separately excited dc motor in which the Field Windings (symbolized by MF) is separated from Armature, and it is fed from an independent source and the motor flux is independent on motor load variation, so the motor flux is constant.  **The electrical circuit of separately excited dc motor**

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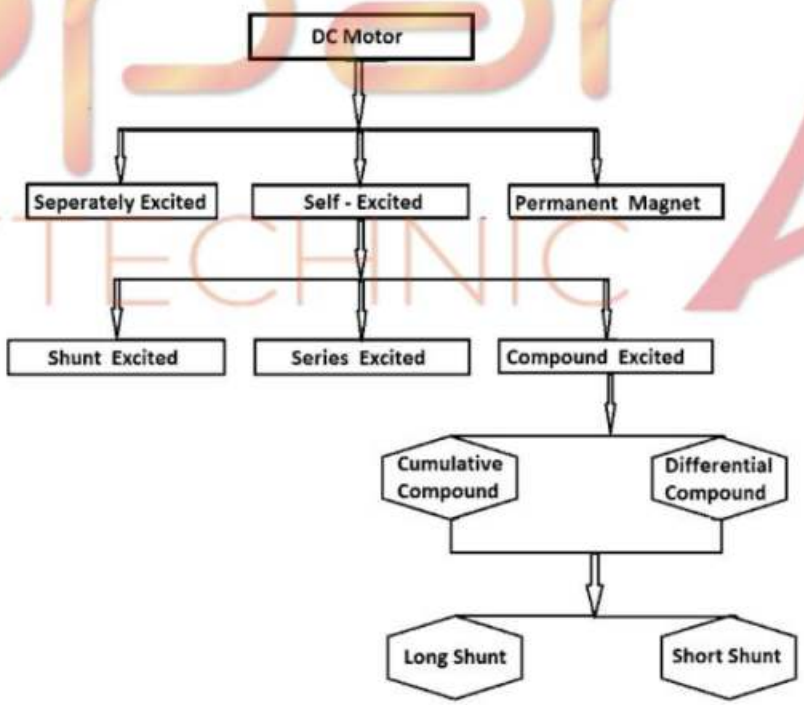
Fundamentals of Electric Drive Mid Term Exam

Number of Questions: 3 Number of pages: 5

16	a	reverse braking mode
17	b	$100\ N.m$ $T_n = k\emptyset_N I_{an} = 2 * 50 = 100\text{N.m}$
18	c	10 rad/s $\Delta\omega = \frac{I_{AR} * R_a}{K\Phi_N} = \frac{50\ A * 0.4\ \Omega}{2\ V.s} = 10\ \text{rad/s}$
19	d	Crane The plugging mode can be used practically in cranes under nominal load torque by adding resistances in series with the armature operating on regulated characteristic.
20	a	Additional heat in the armature circuit

Q2)

1. NO



2. YES

DC motors are robust and allow simple control. They can be made as brushed and brushless DC motors. Brushed DC motors are a mature technology that provides low cost, high torque at low speed, and easy speed control. These features are very important for traction motors. However, brushed DC motors are not widely used in EVs because of their disadvantages, which include large size, low efficiency, and requirement for frequent maintenance due to the brush and collector structure. Brushless DC motors have a much higher efficiency. These motors use an electronic commutator/inverter instead of the brushes.

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3. NO

Production machineries, which are raw materials (wood, plastic and metal) manufacturing machines, including drilling machines, cutting and milling, turning, sawing, and others, have Reactive Torque as it is friction torque.

4. YES
5. YES

The most widely used machine in industry is the three-phase induction motor. Its characteristic features are:

1. Simple and rugged construction
2. Low cost and minimum maintenance
3. High reliability and sufficiently high efficiency
4. Needs no extra starting motor and need not be synchronized

Q3)

ω_{ss} is obtained when the electric drive system is stable, and stability of electric drive system occurs when $T_m = T_L$, so:

$$T_m = T_L$$
$$15 + 0.5\omega_{ss} = 5 + 0.6\omega_{ss}$$
$$\omega_{ss} = 100\,\frac{rad}{s}$$

And $w_2 = 0.95\ \omega_{ss} = 0.95 * 100\,\frac{rad}{s} = 95\,\frac{rad}{s}$

As the initial operating motor angular speed ω_o isn't mentioned in the question, so the drive system considered start its rotation from rest $w_o = 0\,\frac{rad}{s}$

$$t_{st} = \int_{w_1}^{w_2} \frac{J_\Sigma}{T - T_L}\,d\omega = \int_{w_o}^{w} \frac{J_\Sigma}{T - T_L}\,d\omega = \int_{0}^{95} \frac{1.0}{15 + 0.5\omega - (5 + 0.6\omega)}\,d\omega = \int_{0}^{95} \frac{1.0}{10 - 0.1\omega}\,d\omega = \frac{1.0}{-0.1}\ln(10 - 0.1\omega)\Big|_0^{95}$$
$$= 10\ln(10 - 0.1\omega)\Big|_{95}^{0} = 10(\ln(10 - 0.1 * 0) - \ln(10 - 0.1 * 95)) = 29.9573227355\ s \cong 29.96\ s$$

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Fundamentals of Electric Drive Mid Term Exam Second Semester 2021-2022

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Form A

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50 Minutes **30 Marks**

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1. The stability condition of electric drive system operation is:

a. Hardness of load torque-speed characteristic is less than hardness of motor torque-speed characteristic
b. Motor electro-magnetic torque is less than load torque
c. Hardness of motor torque-speed characteristic is less than Hardness of load torque-speed characteristic
d. Motor speed is more than load speed

2. The drive system rotational motion output power is calculated by multiplying:

a. Rotational speed and torque
b. Angular speed and torque
c. Voltage and current
d. Angular speed and efficiency

3. Fan load torque is:

a. Directly proportional with speed
b. Directly proportional with speed square
c. Inversely proportional with speed
d. Inversely proportional with speed square

4. Rotating mechanical load of 10 N.m connected to the motor through gearbox, which has gear ratio of 5. If the system efficiency equals 0.9, the referred load torque to motor shaft is:

a. 2.22 N.m
b. 50 N.m
c. 2 N.m
d. 1.8 N.m

5. The condition for the steady-state operation of the electric drive system is:

a. Load torque > Motor torque
b. Load torque <<<< Motor torque
c. Load torque = Motor torque
d. Load torque < Motor torque

6. The transmission unit as a component of electric drive system is used to:

a. Increase load speed and increase load torque

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Fundamentals of Electric Drive Mid Term Exam Second Semester 2021-2022

Number of Multiple Choice Questions: 20 Number of pages: 4

Form A

b. Increase load speed and decrease load torque
c. Decrease load speed and increase load torque
d. Decrease load speed and decrease load torque

7. The equivalent load torque referred to motor shaft calculated depending on:
a. Kinetic energy conservation law
b. Mechanical power conservation law
c. Electro-magnetic energy conservation law
d. Heating (electric) power conservation law

8. When varying load torque, separately excited dc motor has constant:
a. Torque
b. Flux
c. Back emf
d. Armature current

9. Separately excited dc motor with following ratings:

$$P_{2N} = 10\text{kW}, V_a = V_f = 220\text{ V}, \omega_s = 100\frac{\text{rad}}{\text{s}}, R_a = 0.4\Omega, I_{an} = 50\text{ A}, k\emptyset_N = 2\text{V.s}$$

Nominal motor torque is:
a. $50\ N.m$
b. $100\ N.m$
c. $25\ N.m$
d. $55\ N.m$

10. Separately excited dc motor with following ratings:

$$P_{2N} = 10\text{kW}, V_a = V_f = 220\text{ V}, \omega_s = 100\frac{\text{rad}}{\text{s}}, R_a = 0.4\Omega, I_{an} = 50\text{ A}, k\emptyset_N = 2\text{V.s}$$

No-load angular speed is:
a. 100 rad/s
b. 200 rad/s
c. 110 rad/s
d. 220 rad/s

11. The dc motor loading mode in fourth quadrant is called:
a. Motoring mode
b. Short circuit mode
c. Plugging mode
d. Regenerating mode

12. One of the following dc motors does not operate in no-load mode:

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a. Parallel field dc motor
b. Compound field dc motor
c. Series field dc motor
d. Separately excited dc motor

13. If motor rotational speed equals 1450 rpm, the number of poles is:
a. 1
b. 2
c. 3
d. 4

14. Slip losses are the:
a. Stator circuit losses
b. Rotor circuit losses
c. Magnetic circuit losses
d. Air gap losses

15. When the slip value is negative, the ac motor operates in:
a. First quadrant
b. Second quadrant
c. Third quadrant
d. Fourth quadrant

16. The ac slip ring motor of slip value more than one operates in:
a. Motoring mode
b. Short circuit mode
c. Regenerating mode
d. Plugging mode

17. If the synchronous motor load torque value is more than synchronizing torque during starting, the motor will:
a. Run in short circuit mode
b. Not be synchronized and operate as asynchronous motor
c. Be synchronized and operate as synchronous motor
d. Be synchronized and operate as synchronous generator

18. The temperature rise is related to ambient temperature which value is:
a. 10°C
b. 20°C
c. 30°C
d. 40°C

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Form A

19. Continuous duty cycle operates under constant load with following parameters:
a. Temperature rise during operation under load does not reach steady-state value
b. Starting losses are not neglected
c. Temperature rise during operation under load reaches steady-state value
d. Starting time is not neglected

20. Frequency of operation parameter measuring unit is:
a. Starting per minute
b. Starting per second
c. Starting per hour
d. Starting per shift

Insert the correct answer in the following table:

Ques	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans																				

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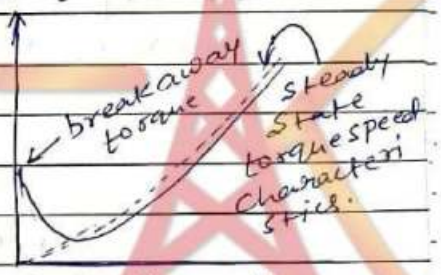
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Form A

Question Number	Correct answer symbol	Correct answer
1	c	Hardness of motor torque-speed characteristic is less than Hardness of load torque-speed characteristic
2	b	Angular speed and torque
3	b	Directly proportional with speed square Fan load torque the load torque is proportional to the square of the speed of fan. this type of loads have variable load (torque) torque characteristics. It requires very low torque at low speed than at high speeds. So load torque is low when starting it.  $T = T_{rated} \times \left(\frac{n}{n_{rated}}\right)^2$ where T_{rated} → rated torque & n_{rated} → speed of a fan load So $T \propto n^2$ option 'c' is correct. Page No.
4	a	2.22 N.m $\frac{10\ \text{N. m}}{5 * 0.9}$
5	c	Load torque = Motor torque
6	c	Decrease load speed and increase load torque

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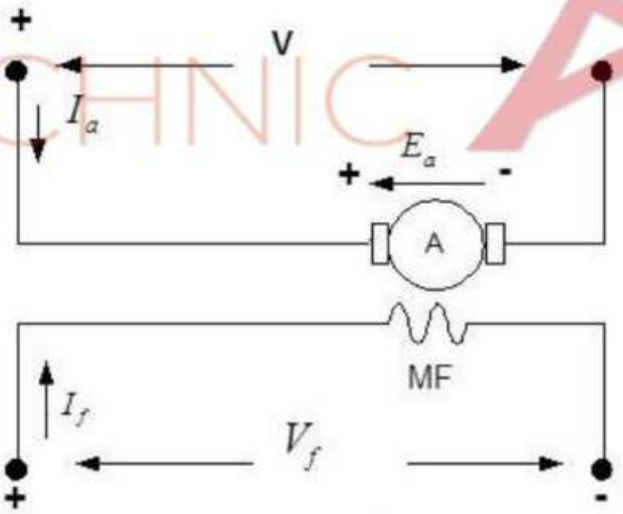
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Fundamentals of Electric Drive Mid Term Exam Second Semester 2021-2022

Number of Multiple Choice Questions: 20 Number of pages: 4

Form A

		∴ Transsimmion unit in electric drive system is usd to Decreas the load speed and increas load torque . ✓b) Decreas load speed and increas load Tora
7	b	Mechanical power conservation law
8	b	Flux Figure below shows the electrical circuit of separately excited dc motor in which the Field Windings (symbolized by MF) is separated from Armature, and it is fed from an independent source and the motor flux is independent on motor load variation, so the motor flux is constant.  **The electrical circuit of separately excited dc motor**
9	b	$100\ N.m$ $T_n = k\emptyset_N I_{an} = 2 * 50 = 100\text{N.m}$
10	c	110 rad/s $\omega_o = \frac{V_t}{K\Phi_N} = \frac{220\ V}{2\ V.s} = 110\ \text{rad/s}$

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Form A

11	c	Plugging mode
12	c	Series field dc motor
13	d	4 If the motor rotational speed equals 1450 rpm, then the nearest synchronous speed for it is 1500 rpm, and according to Synchronous Speed Formula: $$n = \frac{120 * f}{P}$$ Where: n: Synchronous Speed f: frequency P: Number of Poles If the frequency does not mention, then it's considered to be the same as the power grid frequency of your country (in Jordan the power grid frequency is 50 Hz) $$P = \frac{120 * f}{n} = \frac{120 * 50Hz}{1500\ rpm} = 4\ poles$$
14	b	Rotor circuit losses
15	b	Second quadrant
16	d	Plugging Mode
17	b	Not be synchronized and operate as asynchronous motor
18	d	40°C
19	c	Temperature rise during operation under load reaches steady-state value
20	c	Starting per hour

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Fundamentals of Electric Drive Mid Term Exam Second Semester 2021-2022

Number of Multiple Choice Questions: 20 Number of pages: 4

Form B

Faculty of Engineering Technology

Mechatronics Engineering Department

50 Minutes **30 Marks**

Student's Name: SPARK TEAM - بسواعدنا نبنيها

1. The electric drive system with following motor torque and load torque expressions is stable:

a. $T = 10 + 2\omega,\ T_L = 10$

b. $T = 25 - 3\omega,\ T_L = 10$

c. $T = 15 + 4\omega,\ T_L = 15 + \omega$

d. $T = 25 + 5\omega,\ T_L = 2 + 2\omega^2$

2. The active load torque industrial machinery is:

a. Crane

b. Pump

c. Ventilator

d. Turning machine

3. The hardness of series field dc motors torque-speed characteristic is:

a. Negative

b. Positive

c. Infinity

d. Zero

4. Drilling machine speed is:

a. Directly proportional with torque

b. Independent on torque

c. Inversely proportional with torque

d. Inversely proportional with torque square

5. What is the relationship between torque and speed in constant type loads?

a. Torque is independent on speed

b. Torque linearly increases with an increase in speed

c. Torque non-linearly increases with an increase in speed

d. Torque non-linearly decreases with an increase in speed

6. Electric drive system is:

a. Electro-mechanical system

b. Mechanical system

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Fundamentals of Electric Drive Mid Term Exam Second Semester 2021-2022

Number of Multiple Choice Questions: 20 Number of pages: 4

Form B

c. Electrical system
d. Electro-pneumatic system

7. Loading diagram is load torque as function of:
a. Speed
b. Time
c. Moment of inertia
d. Gear ratio

8. Separately excited dc motor has linear static characteristics, because:
a. Motor flux is constant
b. Load torque is variable
c. Armature current is variable
d. Motor speed is variable

9. Separately excited dc motor with following ratings:

$$P_{2N} = 10\text{kW}, V_a = V_f = 220\text{ V}, \omega_s = 100\frac{\text{rad}}{\text{s}}, R_a = 0.4\Omega, I_{an} = 50\text{ A}, k\emptyset_N = 2\text{V.s}$$

Nominal back emf is:
a. $220\ V$
b. $200\ V$
c. $240\ V$
d. $120\ V$

10. Separately excited dc motor with following ratings:

$$P_{2N} = 10\text{kW}, V_a = V_f = 220\text{ V}, \omega_s = 100\frac{\text{rad}}{\text{s}}, R_a = 0.4\Omega, I_{an} = 50\text{ A}, k\emptyset_N = 2\text{V.s}$$

Motor starting current is:
a. 50 A
b. 150 A
c. 100 A
d. 550 A

11. The dc motor loading mode in third quadrant is called:
a. Motoring mode
b. Short circuit mode
c. Plugging mode
d. Regenerating mode

12. One of the following dc motors does not operate in regenerating mode:
a. Parallel field dc motor

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b. Compound field dc motor
c. Series field dc motor
d. Separately excited dc motor

13. If ac motor rotational speed equals 1450 rpm, the no-load angular speed is:
a. 1500 rpm
b. 157.07 rad/s
c. 209.42 rad/s
d. 314 rad/s

14. The ac motor critical (maximum) torque independent on:
a. Rotor resistance
b. Stator resistance
c. Source voltage
d. Source frequency

15. Three-phase ac motor has nominal torque of 75 N.m and maximum torque of 225 N.m. The motor loading capacity (coefficient) equals:
a. 225 N.m
b. 0.333
c. 3
d. 75 N.m

16. T-form per phase equivalent open circuit describes:
a. Motoring mode
b. Regenerating mode
c. Short circuit mode
d. No-load mode

17. The three-phase synchronous motor has:
a. Five loading modes
b. Four loading modes
c. Three loading modes
d. Two loading modes

18. The insulating materials class A temperature limit is:
a. $105°C$
b. $120°C$
c. $130°C$
d. $140°C$

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Fundamentals of Electric Drive Mid Term Exam Second Semester 2021-2022

Number of Multiple Choice Questions: 20 Number of pages: 4

Form B

19. Short time duty cycle operates under constant load with following parameters:
a. Temperature rise during operation under load does not reach steady-state value
b. Starting and braking losses are not neglected
c. Temperature rise during operation under load reaches steady-state value
d. Starting time and braking time is not neglected

20. Which duty cycle has on-time and off-time period?
a. Intermittent periodic duty cycle
b. Short time duty cycle
c. Continuous duty cycle with constant load
d. Continuous duty cycle with variable load

Insert the correct answer in the following table:

Ques	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans																				

POLYTECHNIC

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Fundamentals of Electric Drive Mid Term Exam Second Semester 2021-2022

Number of Multiple Choice Questions: 20 Number of pages: 4

Form B

Question Number	Correct answer symbol	Correct answer
1	b	$T = 25 - 3\omega, \quad T_L = 10$
2	a	Crane
3	a	Negative
4	c	Inversely proportional with torque
5	a	Torque is independent on speed
6	a	Electro-mechanical system
7	b	Time
8	a	Motor flux is constant
9	b	$200\ V$ $\omega_n = \frac{V}{k\emptyset_N} - \frac{I_a R_a}{k\emptyset_N} = 100\ rad/sec$ $E_n = k\emptyset_N \omega_n = 2 * 100 = 200\ V$
10	d	550 A $I_{st} = \frac{V_a}{R_a} = \frac{220\ V}{0.4\ \Omega} = 550\ A$
11	a	Motoring mode
12	c	Series field dc motor
13	b	157.07 rad/s If the motor rotational speed equals 1450 rpm, then the nearest synchronous speed for it is 1500 rpm which is approximately equals the no-load speed, so no-load rotational speed is 1500 rpm: $\omega_o = \frac{2\pi}{60} n = \frac{2\pi}{60} * 1500\ rpm = 157.079632679\ \text{rad/s} \approx 157.07\ \text{rad/s}$
14	a	Rotor resistance

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Form B

		$T_{e\,max} = \frac{3}{2\omega_{sm}} \frac{V_{th}^2}{\left\{R_{th} + \sqrt{R_{th}^2 + (X_{th} + X_2')^2}\right\}}$ → (1) if stator impedance is neglected, then $T_{e\,max} = \frac{3}{2\omega_{sm}} \times \frac{V_1^2}{X_2'}$ V_1 = source voltage ω_s = synchronous speed from (1), T_{emax} depends on source voltage stator resistance source frequency and is independent of rotor resistance (d) rotor resistance
15	c	3
16	d	No-load mode
17	c	Three loading modes
18	a	105°C
19	a	Temperature rise during operation under load does not reach steady-state value
20	a	Intermittent periodic duty cycle

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Fundamentals of Electric Drive Mid Term Exam Second Semester 2021-2022

Number of Multiple Choice Questions: 20 Number of pages: 4

Form C

Faculty of Engineering Technology

Mechatronics Engineering Department

50 Minutes **30 Marks**

Student's Name: SPARK TEAM - بسواعدنا نبنيها

1. The drive system output power is:
a. Thermal power
b. Mechanical power
c. Electrical power
d. Electromagnetic power

2. Production machineries torque-speed characteristic is:
a. Decreasing linear characteristic
b. Increasing non-linear characteristic
c. Decreasing non-linear characteristic
d. Independent on speed

3. The hardness of torque-speed characteristic is the relationship between the changes of:
a. Speed to torque
b. Current to torque
c. Speed to current
d. Torque to speed

4. Transporting machineries load torque is:
a. Directly proportional with speed
b. Directly proportional with speed square
c. Inversely proportional with speed
d. Independent on speed

5. Elevator torque-speed characteristic is:
a. Constant torque characteristic
b. Linearly rising torque characteristics
c. Non-Linearly rising torque characteristic
d. Non-Linearly decreasing torque characteristic

6. Electric motor has moment of inertia of $1\ kg.m^2$ directly connected to motor shaft drum with moment of inertia $4\ kg.m^2$. The drive system total (equivalent) moment of inertia equals:
a. $5\ kg.m^2$

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b. $3\ kg.m^2$
c. $4\ kg.m^2$
d. $1\ kg.m^2$

7. Speed profile is a speed as function of:
a. Motor torque
b. Time
c. Moment of inertia
d. Load torque

8. Series field dc motor has non-linear static characteristics, because:
a. Motor flux is variable
b. Load torque is variable
c. Armature current is variable
d. Motor speed is variable

9. Separately excited dc motor with following ratings:

$$P_{2N} = 10\text{kW}, V_a = V_f = 220\text{ V}, \omega_s = 100\frac{\text{rad}}{\text{s}}, R_a = 0.4\Omega, I_{an} = 50\text{ A}, k\emptyset_N = 2\text{V}.s$$

Nominal rotational speed is:
a. $100\ rpm$
b. $1000\ rpm$
c. $981\ rpm$
d. $955\ rpm$

10. Separately excited dc motor with following ratings:

$$P_{2N} = 10\text{kW}, V_a = V_f = 220\text{ V}, \omega_s = 100\frac{\text{rad}}{\text{s}}, R_a = 0.4\Omega, I_{an} = 50\text{ A}, k\emptyset_N = 2\text{V}.s$$

Nominal angular speed drop is:
a. 10 rad/s
b. 20 rad/s
c. 90 rad/s
d. 80 rad/s

11. The dc motor loading mode in second quadrant is called:
a. Motoring mode
b. Short circuit mode
c. Plugging mode
d. Regenerating mode

12. No-load mode is in-efficient, but it is efficiently used in:

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a. Pump machinery
b. Drilling machine
c. Escalator
d. Elevator

13. If motor rotational speed equals 950 rpm, the slip is:
a. 0.95
b. 0.05
c. 0.005
d. 0.1

14. When the slip value more than one, the ac motor operates in:
a. First quadrant
b. Second quadrant
c. Third quadrant
d. Fourth quadrant

15. The slip value at starting point of ac motor equals:
a. Zero
b. One
c. Infinity
d. Minus two

POLYTECHNIC

16. If the synchronous motor load torque value is less than synchronizing torque during starting, the motor will:
a. Run in short circuit mode
b. Not be synchronized and operate as asynchronous motor
c. Be synchronized and operate as synchronous motor
d. Be synchronized and operate as synchronous generator

17. The three-phase squirrel cage asynchronous motor has:
a. Five loading modes
b. Four loading modes
c. Three loading modes
d. Two loading modes

18. The insulating materials class B temperature rise is:
a. $105°C$
b. $130°C$
c. $90°C$
d. $110°C$

بسواعدنا نبنيها

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Fundamentals of Electric Drive Mid Term Exam Second Semester 2021-2022

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Form C

19. Frequency of operation parameter is used to specify the following duty:
a. Intermittent periodic cycle
b. Continuous duty cycle with intermittent periodic loading
c. Intermittent periodic duty cycle with starting and braking
d. Continuous duty cycle with starting and braking

20. The thermal time constant of electric motor is:
a. The ratio of the thermal capacity and the dissipation constant
b. The ratio of the dissipation constant and the thermal capacity
c. The ratio of the temperature rise and the thermal capacity
d. The ratio of the temperature rise and the dissipation constant

Insert the correct answer in the following table:

Ques	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans																				

POLYTECHNIC

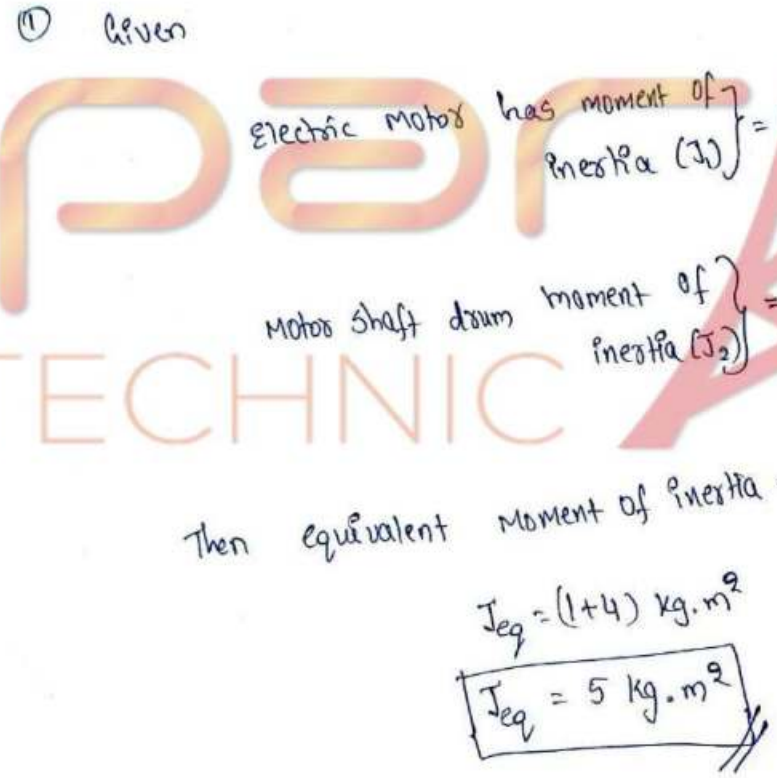
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Fundamentals of Electric Drive Mid Term Exam Second Semester 2021-2022

Number of Multiple Choice Questions: 20 Number of pages: 4

Form C

Question Number	Correct answer symbol	Correct answer
1	b	Mechanical power
2	c	Decreasing non-linear characteristic
3	d	Torque to speed
4	d	Independent on speed
5	a	Constant torque characteristic
6	a	$5\ kg.m^2$ 
7	b	Time
8	a	Motor flux is variable
9	d	$955\ rpm$ $n_N = \frac{60}{2\pi}\omega_N = \frac{60}{2\pi} * 100\ rpm = 954.92965856\ \text{rpm} \approx 955\ \text{rpm}$
10	a	10 rad/s

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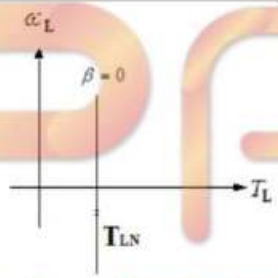
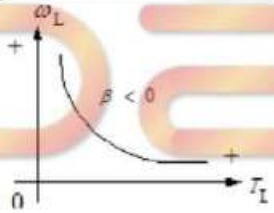
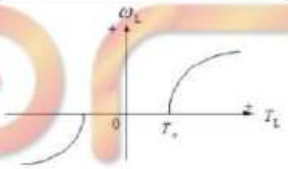
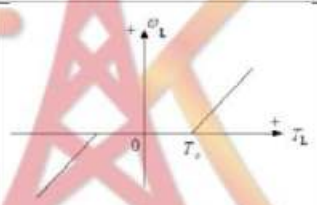
Form C

		$\Delta\omega = \frac{I_{AR} * R_a}{K\Phi_N} = \frac{50\ A * 0.4\ \Omega}{2\ V.s} = 10\ \text{rad/s}$
11	d	Regenerating mode
12	d	Elevator
13	b	0.05 $S = \frac{n_o - n}{n_o}$ Where: S: Slip value n_o: Rotational speed of stator field (synchronous speed) n: Rotational speed of rotor If the motor rotational speed equals 950 rpm, then the nearest synchronous speed for it is 1000 rpm $S = \frac{1000 - 950}{1000} = \frac{50}{1000} = 0.05$
14	d	Fourth quadrant
15	b	One
16	c	Be synchronized and operate as synchronous motor
17	b	Four loading modes Because three-phase squirrel cage asynchronous motor hasn't plugging mode
18	b	$130°C$
19	c	Intermittent periodic duty cycle with starting and braking
20	a	The ratio of the thermal capacity and the dissipation constant

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Appendix I: Comparison between different Industrial machinery types

Industrial machinery type	Transporting machineries	Production machineries	Fan-load characteristic machineries	Machineries with linear incremental characteristics
Torque-Speed Relationship	The load torque has a fixed value and does not depend on the value and direction of the speed (Speed independent torque-speed characteristic)	The load torque of machinery is inversely proportional to speed (non-linear decaying characteristics)	The load torque of machinery is directly proportional to speed squared (second order non-linear incremental characteristic)	The load torque of machinery is directly proportional to speed and increases linearly (first order linear incremental characteristic)
Torque-Speed Characteristics	 Hard Speed-Torque Characteristic	 Soft Speed-Torque Characteristic		
The hardness β value Its relationship is: $\beta = \frac{\Delta T}{\Delta \omega}$	$\beta = 0$	$\beta < 0$	$\beta > 0$	$\beta > 0$

The torque-speed characteristics of different types of industrial machineries can be derived using an experimental relationship in steady state (stable mode), this experimental relationship is:

$$T_L = T_O + (T_{Ln} - T_O)\left(\frac{\omega_L}{\omega_{Ln}}\right)^X$$

Where:

T_L: Load Torque

T_O: Initial friction torque

T_{Ln}: Nominal load torque

ω_L: Load angular speed

ω_{Ln}: Load nominal angular speed

بسواعدنا نبنيها

ملاحظة مهمة جدا: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل مُنشأ هذا العمل

References المراجع

X: A parameter that determines the relationship between torque change and velocity change according to the type of machinery.				
X parameter value	0	-1	+2	+1
Initial friction torque T_O value	=0	=0	≠0	≠0
Torque-Speed Equation	$T_L = T_{Ln}$	$T_L = T_{Ln}\left(\frac{\omega_{Ln}}{\omega_L}\right)$	$T_L = T_O + (T_{Ln} - T_O)\left(\frac{\omega_L}{\omega_{Ln}}\right)$	$T_L = T_O + (T_{Ln} - T_O)\left(\frac{\omega_L}{\omega_{Ln}}\right)^2$
Examples of this type of Industrial machineries	1. Lift (Elevator) 2. Crane 3. Escalator 4. Conveyor	which are raw materials (wood, plastic, and metal) manufacturing machines, including: 1. Drilling machines. 2. Cutting machines. 3. Milling machines. 4. Turning machines. 5. Sawing machines. and others.	Machineries such as: 1. Ventilation Machineries 2. Fan Machineries 3. Pump Machineries 4. Compressor Machineries	1. Calendaring machines 2. Eddy current brakes
Torque Type	active torque *	reactive torque **	reactive torque **	reactive torque **
Constant Parameter in the equation $P_L = T_L\omega_L$	Constant load torque T_L machineries	Constant load power P_L machineries ***	None of them	None of them

* The transporting machineries deal with transportation of masses, that have weights depending on gravity acceleration. So, the industrial machineries load torque is active.

** The manufacturing process is a result of direct contact between workpiece and tool, and a contact friction force will produce reactive torque.

*** Constant load power machineries because the speed of tool is inversely proportional with torque variation.

بسواعدنا نبنيها

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